

Action–Multiplicity Duality for Spacetime–Gauge Incidence

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Abstract

We prove an exact action–multiplicity duality for finite gravity–matter systems in which spacetime, internal typing, and matter incidence are represented on a common carrier. The complete represented action and the residual matter multiplicity determine one another as mutual commutants. The result applies to arbitrary incidence maps, including singular and rank-deficient ones, and admits intrinsic support, quiver, rigidity, and cofinal-stability formulations. On a certified structural branch, this duality reconstructs the Standard-Model internal algebra. Finite naturality selects the $S(U(3) \times U(2)) \cong [SU(3) \times SU(2) \times U(1)]/\mathbb{Z}_6$ global form, while, within the fixed tensor-type central embedding, anomaly conditions recover the same primitive abelian direction and the usual hypercharge ratios. A source-minimal fermion analysis, together with a positive endpoint-allocation criterion, yields exactly three complete charged copies while excluding future-visible charged exotic and mirror sectors. The construction is finite and operator algebraic; it does not depend on a particular microscopic realization. Renewal Geometry provides one realization of the required joint carrier. All selection statements retain explicit occurrence and compatibility hypotheses; absent source data are not supplied by algebraic completion.

Keywords. double-centralizer duality; matter multiplicity; Standard Model; finite operator algebras; gauge-group reconstruction; generations; commutant rigidity; incidence geometry; Renewal Geometry.

1 Introduction

The two central theories of fundamental physics organize structure in different ways. General relativity identifies gravity with the geometry of spacetime itself, whereas the Standard Model is formulated over an already given spacetime and introduces a separate internal gauge structure, chiral matter representations, a scalar sector, and Yukawa couplings. Under the usual assumptions on an ordinary relativistic S -matrix, the Coleman–Mandula theorem sharply constrains attempts to combine continuous spacetime and internal Lie symmetries into a larger nontrivial symmetry group [1]; the graded extension is characterized by the Haag–Łopuszański–Sohnius theorem, which admits supersymmetry [2].

This separation does not imply that spacetime action, internal symmetry, and matter multiplicity must be algebraically unrelated. They may instead occur as distinct structures on a common finite carrier and constrain one another through their joint representation. The relevant question is then not whether the Lorentz and Standard-Model gauge groups should be unified as subgroups of a larger ordinary symmetry group. It is whether the complete algebra of represented spacetime, internal, and matter-incidence operations determines the multiplicity left invisible to that action, and conversely whether this residual multiplicity determines the represented action.

The present paper formulates this question in finite operator-algebraic terms. Its central result is an intrinsic coefficient-space reconstruction and exact double-centralizer theorem for a source-complete joint packet. Once an external Clifford action, a typed internal action, and physical matter-incidence operators are represented on one finite Hilbert carrier, the algebra generated by the complete visible action and the residual typed matter multiplicity determine one another as mutual commutants. The theorem depends only on this represented finite packet and not on a particular microscopic mechanism by which the packet was constructed.

Relation to earlier geometric and algebraic approaches. Noncommutative geometry provides a natural point of comparison because it places spacetime and internal particle structure in a common operator-theoretic framework. In spectral geometry, the data $(\mathcal{A}, \mathcal{H}, D)$ encode geometric information [3, 4], while the Connes–Lott and almost-commutative constructions combine ordinary spin geometry with a finite noncommutative sector and recover gauge, Higgs, fermionic, and gravitational terms through the spectral action [5, 6, 7, 8, 9, 10]. Finite spectral triples admit a substantial classification theory [11, 12, 13], and modified spectral assumptions can lead to larger internal symmetries such as Pati–Salam models [14]. Lorentzian and twisted formulations address the corresponding indefinite geometric structures [15, 16]. Related algebraic reformulations of the finite Standard-Model data and quantum gauge-symmetry constructions are developed in [17, 18].

Particularly relevant here are constructions in which internal information is encoded by commutants or centralizers. Morita formulations of the noncommutative Standard Model relate the coordinate algebra to its spectral Clifford algebra [19], while the Hodge condition asks for exact commutant relations between spectral Clifford actions [20, 21]. More generally, classical Howe duality organizes mutually centralizing actions through their multiplicity structure [22], and reconstruction of internal gauge symmetry from algebraic or representation-theoretic data also appears in algebraic quantum field theory [23, 24, 25].

At the purely representation-theoretic level, mutual-centralizer decompositions are classical: a finite-dimensional semisimple algebra and the endomorphism algebra of its multiplicity module satisfy the usual double-centralizer theorem [26]. The new point pursued here is therefore not the abstract bicommutant implication by itself. It is the intrinsic identification of the first commutant from a physically represented, typed incidence packet, including arbitrary operator–Schmidt rank, singular and rectangular incidences, together with the source, support, quiver, and stability consequences developed below. Once the first commutant has been identified, the reverse equality is the classical finite bicommutant step.

A complementary class of approaches seeks Standard-Model structure inside larger Clifford, division, or exceptional algebras [27, 28, 29, 30, 31, 32]. These constructions typically identify a distinguished centralizer, stabilizer, or subalgebra inside a prescribed ambient algebra.

A separate literature seeks an algebraic explanation for fermion triplication. Three-generation structures have been obtained from complex-octonionic and Clifford-algebra constructions [33, 34], from exceptional-algebraic structures [35], and from extensions of noncommutative geometry [36]. More recently, categorical coherence obstructions have been proposed as a mechanism in which the order of an associator cocycle fixes the minimal family replication [37]. In the standard almost-commutative construction, by contrast, the number of generations is an input [9]. These mechanisms are logically distinct from the one used below: here three first appears as the rank of a source-minimal tetrahedral endpoint residual and becomes a physical charged family count only after a represented same-carrier ancestry occurrence, a positive allocation test, complete charged inventory, and the anomaly relations have all been imposed.

The logical structure considered here is different. The primitive datum for the main theorem is

not a larger unifying algebra and is not a preassigned finite spectral triple. It is a *joint represented packet*: an external Clifford factor, a commuting typed internal algebra, and physical incidence operators acting on the same carrier. The algebra dual to the complete action is also not the Standard-Model gauge algebra itself. It is the residual matter multiplicity that the complete represented action cannot resolve.

Finite action–multiplicity duality. Let $\mathcal{A}_{\text{Cl},X} \cong M_4(\mathbb{C})$ be the represented external Clifford factor, let $\mathcal{A}_{\text{type},X}$ be a commuting finite-dimensional kinematic typing algebra, and let $Y_{e,X}$ denote represented matter-incidence operators. The complete action algebra is

$$\boxed{\mathcal{A}_{\text{act},X} = C^*(\mathcal{A}_{\text{Cl},X}, \mathcal{A}_{\text{type},X}, Y_{e,X}, Y_{e,X}^*)} \quad (1.1)$$

Finite representation theory separates the carrier into external, internal-representation, and residual multiplicity factors. The operators on the residual factors that intertwine the intrinsic coefficient spaces of all represented incidence maps form the typed multiplicity algebra $\mathcal{M}_{\text{type},X}$.

The main finite theorem proves

$$\boxed{\mathcal{A}'_{\text{act},X} = \mathcal{M}_{\text{type},X}, \quad \mathcal{M}'_{\text{type},X} = \mathcal{A}_{\text{act},X}} \quad (1.2)$$

Thus the duality is between the complete spacetime–gauge–incidence action and residual matter multiplicity. It is exact for arbitrary operator-Schmidt rank, including rectangular, singular, rank-deficient, and zero incidence maps. Intrinsic coefficient-space quivers require no one-dimensional intertwiner assumption. Support-polar data give a support-exact formulation without introducing inverse incidence operators; noncommuting supports lead to a multiplicity quiver; and on connected invertible branches the same residual algebra is described by positive polar metrics and cycle holonomies. This is a finite double-centralizer duality, without assuming a classical reductive dual pair. Its reciprocal equality uses the classical finite bicommutant theorem [26]; the constructive content is the coefficient-space identification of the first commutant from the represented incidence packet.

The incidence family itself has a finite minimality theory. A submodular rank function, in the standard polymatroid sense [38], measures how much each represented incidence removes from the residual commutant, yielding inclusion-minimal incidence skeletons. This makes it possible to distinguish structural incidence data from redundant generators and to identify precisely when deleting one physical incidence enlarges the multiplicity algebra.

One upstream realization: Renewal Geometry. Predictive and observable-state approaches provide one route to finite structure without introducing a manifold or field configuration at the outset. Computational mechanics identifies states through equivalence of conditional future statistics [39]; finite-rank realization theory, observable operator models, predictive-state representations, and multiplicity automata encode related stochastic processes through finite operator or Hankel representations and predictions of observable future experiments [40, 41, 42, 43]. Multitime operational descriptions with memory provide a related framework for intervention-dependent quantum processes [44, 45]. One predictive realization of the finite joint-carrier input is developed through the Lorentzian and spectral Renewal Geometry constructions [46, 47], while the double-centralizer theorem itself is independent of that microscopic construction.

In this realization, operational histories are quotiented by equality of conditional future statistics. The resulting minimum persistent predictive carrier $Z_X^{\min}$, represented history algebra $\mathcal{A}_X^{\text{hist}}$, and

positive word-Gram hierarchy are organized into the *Renewal Grand Tensor*,

$$\mathbb{T}_{\text{Grand}, X} = \left(\mathcal{A}_X^{\text{hist}}, \{\mathbb{K}_{X,r}\}_{r \geq 0}, Z_X^{\text{min}} \right). \quad (1.3)$$

Here “tensor” denotes this finite predictive data structure rather than a tensor field on a pre-existing manifold. The predictive carrier, represented history algebra, and word-Gram framework underlying this object are developed in [46, 47]. The Lorentzian construction [46] extracts, on a certified branch, a tetrahedral relational sector, calibrated geometric data, and a faithful complex Clifford factor, while the spectral construction [47] supplies a compatible history-algebraic and first-order spectral realization. These results furnish one route from the Grand Tensor to the represented input of the present theory; they are not hypotheses of the finite duality theorem.

Structural Standard-Model specialization. The general double-centralizer theorem does not select the Standard Model. A second layer of the paper specializes to a tetrahedrally equivariant source-complete branch and asks which residual multiplicity structures survive a finite sequence of occurrence, connectivity, positivity, and incidence tests.

On the certified branch the landed internal algebra is

$$\mathcal{A}_{\text{int}}^{\text{word}} \cong M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}, \quad (1.4)$$

and a canonical incidence line, selected by six finite naturality identities and retained relative to a coherent neutral reference, selects

$$S(\text{U}(3) \times \text{U}(2)) \cong \frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}. \quad (1.5)$$

The determinant-source route therefore selects this $\mathbb{Z}_6$ global form at the level of represented incidence rather than from the Lie algebra alone [48, 49]. Before that naturality test is applied, a finite commutator residual certifies that the represented colour and two weak factor actions occupy mutually commuting tensor slots on one carrier. Once this factor skeleton is fixed, the naturality operator and its determinant projector are defined independently of any matter-source population; positive source mass and vanishing naturality defect are separate occurrence and purity tests. Independently, once the displayed tensor types and their two-parameter central action are fixed, anomaly cancellation determines the same primitive central embedding and, together with the faithful matter kernel, reconstructs the same global group without assuming determinant-source occurrence. It also gives the usual Standard-Model hypercharge ratios [50, 51, 52]. This uniqueness is conditional on that fixed tensor-type embedding: anomaly cancellation for more general matter inventories or abelian charge assignments need not determine a unique $\text{U}(1)$ direction [53].

At the process level, a Kraus-independent cross-sign Choi mass tests primitive grading-changing occurrence before a coherent odd representative is chosen. This active mass is unchanged by unread trace-preserving spectator extensions, so hidden spectator oddness cannot substitute for matter occurrence on the represented carrier.

The same structural branch contains a distinct rank-three endpoint residual. After finite *-cyclic saturation of the physical fermion source, canonical isotypic projectors audit the complete future-visible gauge-charged carrier. On the zero-inventory branch four linear anomaly relations force the five charged multiplicities to coincide. A finite Schur-covariance test extracts a unique multiplicity map from a represented endpoint-to-type ancestry occurrence. A positive allocation short then measures the excess of one complete typed source beyond that rank-three endpoint

image; its vanishing yields exactly three copies of every charged matter type. This already gives an abstract common generation factor; common endpoint ancestry is needed only when a particular source-determined relative generation frame is claimed. A separate positive weak-copy Gram counts actual weak-doublet copies, while a finite relation-space defect tests the allowed Yukawa and optional Majorana blocks of a specified grading-odd generator. This endpoint-allocation route should be distinguished from octonionic, Clifford, exceptional, extended-NCG, categorical, and discrete-flavour mechanisms for family replication [33, 34, 35, 36, 37, 54]. These statements are structural rather than numerical: they do not fix fermion masses or mixing parameters, gauge couplings, Higgs parameters, or renormalization-group data.

Importantly, the Standard-Model branch is reached through finite alternatives rather than by completing a desired answer. Before a multiplicity of three appears, an internal colour factor is algebraically impossible; if the word hierarchy becomes flat before that occurrence, the obstruction is terminal. Once the required multiplicity spaces exist, failure of the colour bridge, weak router, determinant incidence, or other structural tests returns the corresponding smaller algebra instead of supplying the missing structure by assumption.

Rigidity, refinement, and variational compatibility. The paper also addresses whether the finite duality is stable rather than accidental. The commutant Laplacian gives a positive finite certificate whose kernel is exactly the commutant, and the associated relative Howe Gram quantifies the gap to additional multiplicity directions. For polynomial or analytic parameter families, its determinant describes the symmetry-enhancement locus. A positive finite gap is stable under controlled perturbations.

A separate cofinal result treats regulator refinement. A cutoff-uniform commutator gap on the complement of a finitely transported protected algebra, together with transported Mosco convergence, forces both the late-cutoff and limiting commutant kernels to equal the protected multiplicity algebra; limiting kernel exactness is therefore a conclusion rather than an independent hypothesis. Asymptotic compactness then upgrades this statement to norm-resolvent, eigenvalue, and spectral-projection convergence. A common abstract algebraic identification additionally requires multiplicatively compatible transports.

Finally, the common source quotient separates internal directions from an already represented gravitational source. This is logically separate from the double-centralizer theorem and does not construct a gravity–matter action or a continuum Einstein–Standard-Model theory. The nonlinear finite-to-continuum closure problem requires additional variational, compactness, and PDE analysis.

The principal operator-algebraic result is therefore general at the level of finite joint packets. The tetrahedral Standard-Model branch and Renewal Geometry realization provide progressively more specialized structures built around that theorem rather than premises required for it.

Organization. Section 2 specifies the joint packet and separates represented input from its operational identification. Section 3 proves the exact action–multiplicity theorem, then develops its principal consequences and incidence minimality. Section 4 reconstructs the internal algebra on the specified tetrahedral branch. Section 5 reconstructs the global gauge group and matter typing algebra. Section 6 treats the charged inventory, endpoint allocation, and the resulting generation structure. Section 7 establishes finite rigidity and cofinal stability. Section 8 discusses interpretation and scope. Appendices A–F retain the supporting operator-algebraic, source-identification, representation-theoretic, occurrence, and analytic details in the same article.

2 Joint packets and the commutant obstruction

The finite duality proved below is formulated directly in terms of represented operator-algebraic data. No predictive reconstruction, spectral triple, or microscopic stochastic law is part of the theorem statement. Such structures may provide realizations of the required finite packet, but the double-centralizer argument uses only the common carrier, the external and internal actions, and the represented incidence operators specified here.

2.1 Source-complete joint packets

Definition 2.1 (Finite source-complete joint packet). At cutoff X , a finite source-complete spacetime–gauge packet consists of:

- (J1) a finite Hilbert carrier $\mathcal{H}_X$ on which a faithfully represented external Clifford algebra $\mathcal{A}_{\text{Cl},X} \cong M_4(\mathbb{C})$ acts;
- (J2) a commuting finite-dimensional kinematic typing C^* -algebra $\mathcal{A}_{\text{type},X}$;
- (J3) a central type decomposition

$$\mathcal{H}_X \cong \bigoplus_{\lambda \in \Lambda_X} \mathbb{C}^4 \otimes V_{\lambda,X} \otimes M_{\lambda,X},$$

such that

$$\mathcal{A}_{\text{type},X} = \bigoplus_{\lambda} I_4 \otimes \mathcal{B}(V_{\lambda,X}) \otimes I_{M_{\lambda,X}};$$

- (J4) a finite directed incidence graph $\Gamma_X = (\Lambda_X, E_X)$ and represented incidence operators

$$Y_{e,X} : \mathbb{C}^4 \otimes V_{s(e),X} \otimes M_{s(e),X} \longrightarrow \mathbb{C}^4 \otimes V_{t(e),X} \otimes M_{t(e),X};$$

- (J5) the mixed represented products needed to determine products and adjoints of $\mathcal{A}_{\text{Cl},X}$, $\mathcal{A}_{\text{type},X}$, and the $Y_{e,X}$ on the same carrier.

All operators entering a claimed mixed relation are required to be represented on this common carrier. No inverse incidence map is part of the definition.

Definition 2.2 (Joint typed action algebra). For a joint packet define

$$\mathcal{A}_{\text{act},X} := C^*(\mathcal{A}_{\text{Cl},X}, \mathcal{A}_{\text{type},X}, Y_{e,X}, Y_{e,X}^* : e \in E_X). \quad (2.1)$$

The notation “action” refers here to represented operator action on the common finite carrier, not to a variational action functional.

The common carrier is essential: replacing it by a direct sum of unrelated external and internal packets removes the mixed incidence relations. We first identify the freedom left by the external and typing actions alone.

2.2 External, internal, and multiplicity factors

Proposition 2.3 (Factor obstruction and typed decomposition). *Let a unital copy of $M_n(\mathbb{C})$ act on a finite Hilbert space $\mathcal{H}$. For a multiplicity space $\mathcal{K}$,*

$$\mathcal{H} \cong \mathbb{C}^n \otimes \mathcal{K}, \quad M_n(\mathbb{C})' = I_n \otimes \mathcal{B}(\mathcal{K}). \quad (2.2)$$

Thus a reducible internal algebra with nontrivial centre cannot be the full commutant of the external factor. If $\mathcal{A}_{\text{type}}$ is a commuting unital finite-dimensional C^ -algebra, there are spaces V_λ, M_λ with*

$$\mathcal{H} \cong \bigoplus_{\lambda} \mathbb{C}^n \otimes V_\lambda \otimes M_\lambda, \quad (2.3)$$

$$\mathcal{A}_{\text{type}} = \bigoplus_{\lambda} I_n \otimes \mathcal{B}(V_\lambda) \otimes I_{M_\lambda}. \quad (2.4)$$

Writing $\mathcal{C} = C^*(\mathcal{A}_{\text{Cl}}, \mathcal{A}_{\text{type}})$, one has

$$\mathcal{C} = \bigoplus_{\lambda} \mathcal{B}(\mathbb{C}^n \otimes V_\lambda) \otimes I_{M_\lambda}, \quad (2.5)$$

$$\mathcal{C}' = \bigoplus_{\lambda} I_{\mathbb{C}^n \otimes V_\lambda} \otimes \mathcal{B}(M_\lambda). \quad (2.6)$$

The residual spaces M_λ , not the internal representation spaces V_λ , carry flavour and generation multiplicity.

Proof. Every finite unital representation of $M_n(\mathbb{C})$ is a direct sum of its defining representation, giving $\mathcal{H} = \mathbb{C}^n \otimes \mathcal{K}$ and the first commutant identity by matrix units. The commuting typing algebra acts on $\mathcal{K}$. Its finite-dimensional representation decomposition gives $\mathcal{K} = \bigoplus_{\lambda} V_\lambda \otimes M_\lambda$ and (2.4). On each central block the two full matrix factors generate $\mathcal{B}(\mathbb{C}^n \otimes V_\lambda)$; Schur's lemma gives (2.6). $\square$

The preceding decomposition identifies everything invisible to the bare external–typing action. The remaining question is how the represented incidences constrain the spaces M_λ . Section 3 answers it without assuming that an incidence factors as a single tensor.

2.3 Represented input and source identification

The joint packet is the input to the finite theorem, not its output. A microscopic realization must identify the operations acting on the system carrier, including their mixed products. Classical records alone need not suffice: Proposition B.1 constructs two normalized finite instruments with the same resolved records at every word length, one-step effects, coefficient Gram, and nonzero coarse Choi spectrum, but different system algebras and commutants.

There is also a positive reconstruction route. For an isometry $V = \sum_{\nu} C_{\nu} \otimes |\nu\rangle$, the complementary Heisenberg pullback satisfies $\mathcal{P}^*(|z\rangle\langle w|) = C(z)^* C(w)$. If occurring coefficients include a nonzero identity pivot and nonzero multiples of a system-algebra generating family, these pullbacks generate exactly that system algebra (Proposition B.2). Appendix B proves both statements and gives the finite coherent reconstruction panel.

Renewal Geometry is one source of such represented data. Its Lorentzian branch supplies, under the hypotheses of the cited construction,

$$\mathcal{A}_{\text{Cl},X} = C^*(\gamma^0, \gamma^1, \gamma^2, \gamma^3) \cong M_4(\mathbb{C}), \quad (2.7)$$

while its spectral branch gives a compatible history-algebraic realization [46, 47]. Neither construction is a premise of the finite theorem. For historical realizations, Appendix B.3 additionally distinguishes executable system operations from invariant endomorphisms of a regular word space, and specifies when mixed source Grams are computable from existing word moments. These identification questions do not interrupt the algebraic argument once the joint packet is supplied.

3 Exact action–multiplicity duality

Throughout this section the represented joint packet is fixed and the cutoff index is suppressed. The proof proceeds in three steps: extract the full multiplicity coefficient space of each incidence, identify its intertwining algebra, and apply finite bicommutant closure.

3.1 Intrinsic coefficient spaces and the reconstruction theorem

Put $K_\lambda = \mathbb{C}^4 \otimes V_\lambda$ and $\mathcal{M} = \bigoplus_\lambda M_\lambda$. Let Z_λ be the vertex projections on $\mathcal{M}$, and let

$$\iota((R_\lambda)_\lambda) = \bigoplus_\lambda I_{K_\lambda} \otimes R_\lambda$$

be the faithful representation of the block-diagonal multiplicity operators on $\mathcal{H}$. Commutants on $\mathcal{H}$ and on $\mathcal{M}$ are distinguished below; when the carrier is unambiguous, the representation ι is suppressed.

Definition 3.1 (Intrinsic incidence coefficient space). For an arbitrary represented edge $Y_e : K_\lambda \otimes M_\lambda \rightarrow K_\mu \otimes M_\mu$, define

$$\mathfrak{B}_e = \{(\varphi \otimes \text{id})(Y_e) : \varphi \in \mathcal{B}(K_\lambda, K_\mu)^*\} \subseteq \mathcal{B}(M_\lambda, M_\mu). \quad (3.1)$$

This is a linear operator space, not a choice of a single residue. Equivalently, for a Hilbert–Schmidt orthonormal basis $(D_{e,a})_a$ of $\mathcal{B}(K_\lambda, K_\mu)$, write

$$Y_e = \sum_a D_{e,a} \otimes F_{e,a}, \quad \mathfrak{B}_e = \text{span}\{F_{e,a} : a\}. \quad (3.2)$$

The coefficient-space quiver has vertices M_λ , forward arrow spaces $\mathfrak{B}_e$, and the corresponding adjoint arrow spaces.

Definition 3.2 (Typed multiplicity algebra). The endomorphism algebra of this adjoint-closed coefficient-space quiver is

$$\mathcal{M}_{\text{type}} = \{(R_\lambda)_\lambda : R_\mu B = B R_\lambda, \quad R_\lambda B^* = B^* R_\mu \\ \text{for every } B \in \mathfrak{B}_e, \quad e : \lambda \rightarrow \mu\}. \quad (3.3)$$

It is a unital C^* -algebra on $\mathcal{M}$, represented on the full carrier by ι . Define also the multiplicity coefficient action algebra

$$\mathcal{O}_Y = C^*(Z_\lambda, B, B^* : B \in \mathfrak{B}_e, \quad e \in E) \subseteq \mathcal{B}(\mathcal{M}). \quad (3.4)$$

Commutation with the external and typing algebras has already reduced an operator to the multiplicity factors. The coefficient spaces now retain exactly the additional information needed to commute with every physical incidence and its adjoint.

Theorem 3.3 (Arbitrary-incidence action–multiplicity reconstruction). *For every finite source-complete joint packet of Definition 2.1, with no restriction on the operator-Schmidt rank of its incidences,*

$$\mathcal{A}'_{\text{act}} \mathcal{H} = \iota(\mathcal{M}_{\text{type}}), \quad \iota(\mathcal{M}_{\text{type}})' \mathcal{H} = \mathcal{A}_{\text{act}}. \quad (3.5)$$

On the total multiplicity carrier,

$$\mathcal{O}'_{Y, \mathcal{M}} = \mathcal{M}_{\text{type}} = \text{End } \mathfrak{Q}_Y, \quad \mathcal{M}'_{\text{type}, \mathcal{M}} = \mathcal{O}_Y, \quad (3.6)$$

where $\mathfrak{Q}_Y$ is the adjoint-closed coefficient-space quiver. Both algebras are independent of all coefficient bases and of the chosen Wedderburn coordinates up to the canonical multiplicity unitaries of Corollary A.1. These conclusions include rectangular, singular, rank-deficient, and zero incidences.

Proof. By Proposition 2.3, every operator commuting with $\mathcal{A}_{\text{Cl}}$ and $\mathcal{A}_{\text{type}}$ is uniquely $X = \iota((R_\lambda)_\lambda)$. On an edge $e : \lambda \rightarrow \mu$,

$$XY_e - Y_e X = \sum_a D_{e,a} \otimes (R_\mu F_{e,a} - F_{e,a} R_\lambda).$$

Linear independence of the $D_{e,a}$ makes this commutator vanish exactly when the forward coefficient equations hold. Applying the same argument to Y_e^* gives the adjoint equations in (3.3). This proves the first identity in (3.5); finite bicommutant closure proves the second.

On $\mathcal{M}$, commutation with every Z_λ is equivalent to block diagonality. Commutation with all coefficient arrows and their adjoints is then exactly (3.3), proving the first identity in (3.6). Finite bicommutant closure again proves the reverse identity. A change of carrier basis changes the coefficient list by an invertible linear transformation and leaves its span, hence every defining equation, unchanged. The same is true of a change of spanning family within an arrow space. None of these arguments uses an inverse coefficient or a positive-rank assumption. $\square$

This is the algebraic core of the paper. The constructive content is the intrinsic coefficient-space description of the first commutant; the reverse identity follows from finite-dimensional bicommutant closure. The result recovers a represented algebra, not a preferred incidence list, Clifford marking, or numerical Yukawa matrix.

The following example shows why retaining all coefficient directions matters. A rank-one replacement can enlarge the commutant even when it preserves a nonzero part of the same physical incidence.

Example 3.4 (Two coefficients impose an additional multiplicity relation). *Take two types with $K_1 = K_2 = \mathbb{C}^4$ and $M_1 = M_2 = \mathbb{C}^2$, and the base algebra $M_4(\mathbb{C}) \oplus M_4(\mathbb{C})$ acting trivially on the multiplicities. On the single physical edge put*

$$Y = E_{11} \otimes I_2 + E_{22} \otimes D, \quad D = \text{diag}(1, 2).$$

Its intrinsic coefficient space is $\text{span}\{I_2, D\}$. The identity coefficient forces $R_2 = R_1$, and the second coefficient forces $[R_1, D] = 0$. Hence the residual algebra is the common diagonal $\mathbb{C}^2$. Keeping only the identity coefficient would instead leave a common $M_2(\mathbb{C})$ commutant. This example is a source-complete joint packet: the external Clifford factor is the same M_4 on both types, and the type projections separate them. It shows why the unrestricted theorem must retain an entire coefficient space rather than assume rank-one factorization of every physical edge.

3.2 Canonicity and reciprocal structure

Corollary 3.5 (Canonical coefficient action and reciprocal blocks). *The coefficient-action algebra $\mathcal{O}_Y$ is the smallest unital C^* -algebra on $\mathcal{M}$ containing the vertex projections and all multiplicity slices. For any unital finite-dimensional algebra $\mathcal{C} \subseteq \mathcal{B}(\mathcal{M})$,*

$$\mathcal{C}' = \mathcal{M}_{\text{type}} \iff \mathcal{C} = \mathcal{O}_Y. \quad (3.7)$$

A change of Wedderburn coordinates acts by multiplicity unitaries W_λ , with $\tilde{\mathfrak{B}}_e = W_{t(e)} \mathfrak{B}_e W_{s(e)}^$. Thus the dual pair is intrinsic up to these unitary identifications.*

On the full carrier there is a simultaneous decomposition

$$\mathcal{H}_X \cong \bigoplus_{\alpha} H_{\alpha,X}^{\text{act}} \otimes H_{\alpha,X}^{\text{mult}}, \quad (3.8)$$

with

$$\mathcal{A}_{\text{act},X} \cong \bigoplus_{\alpha} \mathcal{B}(H_{\alpha,X}^{\text{act}}) \otimes I, \quad (3.9)$$

$$\mathcal{M}_{\text{type},X} \cong \bigoplus_{\alpha} I \otimes \mathcal{B}(H_{\alpha,X}^{\text{mult}}). \quad (3.10)$$

Consequently

$$Z(\mathcal{A}_{\text{act},X}) = Z(\mathcal{M}_{\text{type},X}) = \mathcal{A}_{\text{act},X} \cap \mathcal{M}_{\text{type},X}. \quad (3.11)$$

The common central projections label the same sectors on both sides.

Proof. Slicing any realization of Y_e through a coefficient subspace shows that it contains $\mathfrak{B}_e$; expansion in a carrier basis gives the converse. The algebraic minimality follows by generation, and uniqueness from the commutant follows from Theorem 3.3. Coordinate changes are absorbed by the full family of carrier linear functionals and by multiplicity unitaries, as detailed in Corollary A.1. The finite-dimensional structure theorem applied to $\mathcal{A}_{\text{act},X}$ gives the simultaneous blocks and their common centre. $\square$

The coefficient spaces also have a canonical positive realization. This is useful later because it separates exact algebraic information from the stability of its numerical presentation. The operator–Schmidt terminology used below is standard [55].

Proposition 3.6 (Slice Gram and operator–Schmidt spectrum). *For one represented incidence Y , write $\mathfrak{B}_Y$ for its intrinsic coefficient space. Thus, for $Y \in \mathcal{B}(K_s, K_t) \otimes \mathcal{B}(M_s, M_t)$, equip both operator spaces with their Hilbert–Schmidt products. For $D \in \mathcal{B}(K_s, K_t)$ put $\varphi_D(A) = \text{Tr}(D^* A)$ and define*

$$\mathcal{S}_Y(\overline{D}) := (\varphi_D \otimes \text{id})(Y), \quad H_Y := \mathcal{S}_Y \mathcal{S}_Y^* \succeq 0.$$

Then

$$\text{Ran } H_Y = \text{Ran } \mathcal{S}_Y = \mathfrak{B}_Y. \quad (3.12)$$

If $Y = \sum_{k=1}^r \sigma_k D_k \otimes F_k$ is an operator–Schmidt decomposition with orthonormal coefficient families, then

$$H_Y F_k = \sigma_k^2 F_k, \quad \text{rank } H_Y = r, \quad \text{Tr } H_Y = \|Y\|_{\text{HS}}^2. \quad (3.13)$$

Thus the intrinsic coefficient space is the support of a canonical positive operator and its positive spectrum is the squared operator–Schmidt spectrum.

Proof. In any Hilbert–Schmidt orthonormal carrier basis (D_a) write $Y = \sum_a D_a \otimes F_a$. Then $\mathcal{S}_Y(\overline{D_a}) = F_a$, so its range is exactly the intrinsic slice space. Finite-dimensional range equality $\text{Ran}(SS^*) = \text{Ran } S$ gives the first identity. In a Schmidt basis, $\mathcal{S}_Y(\overline{D_k}) = \sigma_k F_k$, which proves the spectral and rank statements. Finally $\|\mathcal{S}_Y\|_{\text{HS}}^2 = \sum_a \|F_a\|_{\text{HS}}^2 = \|Y\|_{\text{HS}}^2$. $\square$

For an orthonormal expansion $Y_e = \sum_a D_{e,a} \otimes F_{e,a}$, the same calculation as in the main proof gives

$$\|[\iota(R), Y_e]\|_{\text{HS}}^2 = \sum_a \|R_{t(e)} F_{e,a} - F_{e,a} R_{s(e)}\|_{\text{HS}}^2, \quad (3.14)$$

and $\|\iota(R)\|_{\text{HS}}^2 = \sum_\lambda (\dim K_\lambda) \|R_\lambda\|_{\text{HS}}^2$. The adjoint incidence adds the adjoint constraints. These vertex weights enter the spectral problem, although they do not change its kernel.

3.3 Support-polar and quiver formulations

Choose a finite bank (F_e) spanning the intrinsic arrow spaces, with adjoints. This is a presentation choice: coefficient recombination can change individual support projections, but not the generated coefficient algebra. For $F_e = U_e P_e$, retain the polar partial isometry and the positive metric rather than an inverse of the incidence.

Definition 3.7 (Support-polar action algebra). Define

$$\mathcal{O}_F^{\text{sup}} = C^*(Z_\lambda, F_e^* F_e, U_e, U_e^* : \lambda, e) \subseteq \mathcal{B}(\mathcal{M}). \quad (3.15)$$

Proposition 3.8 (Exact support-polar duality). *For arbitrary rectangular, singular, rank-deficient, or zero incidence maps,*

$$\mathcal{M}_{\text{type}} = (\mathcal{O}_F^{\text{sup}})', \quad \mathcal{O}_F^{\text{sup}} = \mathcal{M}'_{\text{type}}. \quad (3.16)$$

Moreover,

$$C^*(Z_\lambda, F_e, F_e^* : \lambda, e) = \mathcal{O}_F^{\text{sup}}. \quad (3.17)$$

Thus a zero singular value remains a genuine support boundary and is never replaced by an inverse residue. Proposition A.5 quantifies leakage across this boundary, and Proposition A.6 gives a finite support certificate when a source-side rank ceiling is available.

Proof. Commutation with the vertex projections makes an operator block diagonal. Lemma A.4 shows that commutation with $F_e^* F_e, U_e, U_e^*$ is exactly the pair of incidence intertwining equations in Definition 3.2. This gives the first equality, and finite bicommutant closure gives the second.

For (3.17), $F_e^* F_e$ belongs to the edge algebra. Since its spectrum is finite, interpolate the function $g(0) = 0$, $g(t) = t^{-1/2}$ for $t > 0$ on that spectrum by a polynomial. Then

$$U_e = F_e g(F_e^* F_e)$$

belongs to the incidence algebra. Conversely $F_e = U_e (F_e^* F_e)^{1/2}$. $\square$

The singular-edge equations and their support-leakage energy are proved in Appendix A.3. When supports do not commute, their isotypic decomposition gives a refined operator-space quiver with

$$C^*(\mathcal{S}_F, \mathbf{T})' \cong \text{End } \mathfrak{Q}_F(\mathbf{T}). \quad (3.18)$$

Its full definition and commutator form are in Proposition A.8. On a connected bank of invertible arrows, Proposition A.9 instead transports all multiplicities to a root fibre: the residual algebra is the commutant of the transported positive polar metrics and cycle holonomies. These are formulations of the same duality, not additional conditions on the general theorem.

3.4 Incidence essentiality and minimal skeletons

Fix the non-incidence action $\mathcal{A}_0 = C^*(\mathcal{A}_{\text{Cl}}, \mathcal{A}_{\text{type}})$ and a finite incidence bank E . For $S \subseteq E$ set

$$\mathcal{M}(S) = C^*(\mathcal{A}_0, Y_e, Y_e^* : e \in S)', \quad \mathcal{M} = \mathcal{M}(E),$$

and define the residual test space $V = \mathcal{A}'_0 \cap \mathcal{M}^\perp$, with $n = \dim_{\mathbb{C}} V$. For $X \in V$ let

$$D_e X = 2^{-1/2}([Y_e, X], [Y_e^*, X]), \quad G_e = D_e^* D_e, \quad G_S = \sum_{e \in S} G_e. \quad (3.19)$$

The full bank is exact precisely when $G_E \succ 0$ on V . The set function below is a representable polymatroid rank function [38].

Proposition 3.9 (Incidence rank polymatroid). *Let $R_e = \text{Ran } D_e^* \subseteq V$ and*

$$f(S) = \dim_{\mathbb{C}} \left(\sum_{e \in S} R_e \right). \quad (3.20)$$

Then

$$f(S) = \text{rank } G_S = n - \dim_{\mathbb{C}}(\mathcal{M}(S) \cap V), \quad \mathcal{M}(S) = \mathcal{M} \oplus \text{Ker } G_S \quad (3.21)$$

as orthogonal vector spaces. The function f is normalized, monotone and submodular:

$$f(S) + f(T) \geq f(S \cup T) + f(S \cap T). \quad (3.22)$$

An incidence $e \in E$ is essential for the full duality exactly when

$$f(E \setminus \{e\}) < n \quad (3.23)$$

or, equivalently, when deleting e strictly enlarges the commutant.

Proof. The quadratic form of G_S is $\sum_{e \in S} \|D_e X\|^2$, hence

$$\text{Ker } G_S = \bigcap_{e \in S} \text{Ker } D_e = \left(\sum_{e \in S} \text{Ran } D_e^* \right)^\perp.$$

Every element of $\mathcal{M}(S)$ already lies in $\mathcal{A}'_0$. Orthogonal projection onto $\mathcal{M}$ therefore leaves a residual vector in V , and the incidence constraints on that residual are exactly $G_S X = 0$. This proves (3.21). If $R(S) = \sum_{e \in S} R_e$, then $R(S) + R(T) = R(S \cup T)$ and $R(S \cap T) \subseteq R(S) \cap R(T)$. The dimension identity for two subspaces yields submodularity. The deletion criterion is the kernel formula with $S = E \setminus \{e\}$. $\square$

Corollary 3.10 (Finite source-minimal incidence skeleton). *Every finite exact incidence bank contains an inclusion-minimal subset $S \subseteq E$ with $\mathcal{M}(S) = \mathcal{M}(E)$. Such a subset is obtained by repeatedly deleting nonessential incidences and has at most n members. In general these minimal subsets need not be unique and need not have equal cardinality; the correct grouped rank object is the polymatroid f , not a matroid on the incidence labels.*

Proof. Finite reverse deletion terminates. In an inclusion-minimal sufficient bank, adding its members one by one strictly increases the span of the corresponding constraint ranges, so at most n additions are possible. The remaining claims follow from the grouped nature of the subspaces R_e ; an explicit unequal-cardinality example is given in Appendix A.6. $\square$

The grouped rank is a polymatroid rather than, in general, a matroid on the physical incidence labels. Appendix A.6 gives the unequal-cardinality example, a positive determinant refinement, and exact simultaneous-deletion margins. The ordering and typing-refinement consequences of the duality are collected in Appendix A.1.

The general theory is now complete. We next impose additional finite source and representation hypotheses to identify a particular internal algebra; those hypotheses are not required for Theorem 3.3.

4 Reconstruction of the structural internal algebra

The general duality determines the multiplicity left by any represented joint packet. It does not select a gauge theory. We now consider the specified tetrahedrally equivariant branch and ask when its internal word algebra saturates a particular relative commutant. The argument separates three questions: which source directions survive removal of geometry, which multiplicity spaces occur on the physical carrier, and which represented operations connect those spaces.

4.1 Geometry shorting and the endpoint residual

The oriented A_3 root carrier used in this specialization is the tetrahedral relational carrier reconstructed in [46]. The representation decomposition and source-shortening analysis below are carried out here on the resulting finite joint packet.

Let Φ be the twelve oriented A_3 roots associated with the tetrahedral cell and let

$$P = \mathbb{R}^\Phi, \quad P_0 = \left\{ f \in P : \sum_{\alpha \in \Phi} f(\alpha) = 0 \right\}.$$

Let $\mathcal{T} = \{c, \ell\}$ be two accepted categorical types and set

$$t_0 = 2^{-1/2}(\mathbf{e}_c + \mathbf{e}_\ell), \quad t_1 = 2^{-1/2}(\mathbf{e}_c - \mathbf{e}_\ell).$$

After removal of the overall accepted mean, define

$$E_{\text{acc}}^0 = (P_0 \otimes t_0) \oplus (\mathbb{R}\mathbf{1}_\Phi \otimes t_1) \oplus (P_0 \otimes t_1). \quad (4.1)$$

The three summands have dimensions 11, 1, 11.

Write

$$E_{\text{ST}}^{\text{cat}} := P_0 \otimes t_0, \quad E_{\text{int}}^{\text{cat}} := (\mathbb{R}\mathbf{1}_\Phi \otimes t_1) \oplus (P_0 \otimes t_1).$$

Let

$$S_{\text{ST}} : E_{\text{ST}} \rightarrow \mathcal{H}_{\text{acc}}, \quad S_{\text{int}} : E_{\text{int}}^{\text{cat}} \rightarrow \mathcal{H}_{\text{acc}}$$

denote respectively the complete represented spacetime source and the accepted internal categorical source on one common carrier. The spacetime source may contain, in addition to $E_{\text{ST}}^{\text{cat}}$, independently reconstructed waiting-time, calibration, bracket, shift, or geometric-circulation directions. Put

$$G_{\text{ST}} = S_{\text{ST}}^* S_{\text{ST}}, \quad G_{\text{int}} = S_{\text{int}}^* S_{\text{int}}, \quad C = S_{\text{int}}^* S_{\text{ST}},$$

and let P_{ST} be the orthogonal projection onto $\text{Ran } S_{\text{ST}}$. The positive residual below is the finite-dimensional shorted-operator/Schur- complement construction in the sense of [56, 57].

Proposition 4.1 (Universal geometry short and active residual). *For every source-complete common realization, shorting the accepted internal source by the complete represented spacetime source gives*

$$G_{\text{int}|\text{ST}} = G_{\text{int}} - CG_{\text{ST}}^\dagger C^* = S_{\text{int}}^*(I - P_{\text{ST}})S_{\text{int}} \succeq 0. \quad (4.2)$$

The source-minimal internal categorical carrier is therefore

$$E_{\text{int}|\text{ST}}^{\min} = E_{\text{int}}^{\text{cat}} / \text{Ker } G_{\text{int}|\text{ST}}, \quad \dim E_{\text{int}|\text{ST}}^{\min} = \text{rank } G_{\text{int}|\text{ST}}. \quad (4.3)$$

On the explicit orthogonal active realization, the accepted contrast synthesis has scalar Gram

$$S_{\text{acc}}^* S_{\text{acc}} = \vartheta I_{E_{\text{acc}}^0}, \quad \vartheta > 0,$$

and the additional spacetime scores are orthogonal to the accepted categorical source. Hence the mixed block vanishes on $E_{\text{int}}^{\text{cat}}$ and

$$G_{\text{int}|\text{ST}} = \vartheta I_{12}. \quad (4.4)$$

On the tetrahedrally equivariant branch, the categorical residual carries the multiplicity-free representation

$$E_{\text{int}}^{\text{cat}} \otimes \mathbb{C} \cong \mathbb{C}_{\text{type}}^+ \oplus W_+ \oplus (V_2)_+ \oplus W_- \oplus (W \otimes \text{sgn})_-. \quad (4.5)$$

Here $\dim W = 3$ and $\dim V_2 = 2$. The odd standard block W_- is the internal endpoint module, while the odd sign-twisted block is the distinct loop module. If the restriction of $G_{\text{int}|\text{ST}}$ to the endpoint block is strictly positive, that block survives the source-minimal quotient at complex rank three.

Proof. The short is the Gram of $(I - P_{\text{ST}})S_{\text{int}}$, so positivity and the quotient/rank identity follow. The stated orthogonal realization makes the mixed block zero and leaves ϑI_{12} . The reversal-even oriented-root sector is $\mathbf{1} \oplus W \oplus V_2$, and the reversal-odd sector is $W \oplus (W \otimes \text{sgn})$; Appendix C.1 proves this decomposition. Equivariance makes the inequivalent endpoint and loop summands reducing for the shorted Gram. Strict positivity on the endpoint block therefore preserves its three dimensions in the source quotient. $\square$

The endpoint residual is defined after the complete represented spacetime source has been removed. It is not a second name for the spatial carrier. Its later interpretation as a generation source will require a represented map into an actual charged multiplicity and an allocation test.

4.2 Compatibility with the Clifford factor

The algebra $\mathcal{A}_{\text{Cl}} \cong M_4(\mathbb{C})$ and the tetrahedral relabeling algebra $\mathcal{A}_{\text{tet},r} = C^*(\rho_r(S_4))$ are distinct represented objects. The latter acts on a reduced physical carrier and need not be a Clifford factor. Its commutant can supply internal operations for a joint packet only after compatibility with the represented Clifford action has been established. The following finite criterion specifies that handoff.

Proposition 4.2 (Clifford–tetrahedral compatibility and internal lift). *Let $(e_{ij})_{1 \leq i,j \leq 4}$ be matrix units for a unital represented $\mathcal{A}_{\text{Cl}}$ on $\mathcal{H}$, and let P_r be a reducing carrier projection: $[P_r, e_{ij}] = 0$. Suppose on $P_r \mathcal{H}$ there are a genuine unitary representation $v_r : S_4 \rightarrow \text{U}(P_r \mathcal{H})$ and a finite historical operator bank (b_j) such that*

$$[v_r(g), e_{ij}|_{P_r \mathcal{H}}] = 0, \quad [b_j, e_{ik}|_{P_r \mathcal{H}}] = [b_j, v_r(g)] = 0. \quad (4.6)$$

Put $\mathcal{V}_r = e_{11}P_r \mathcal{H}$. There is a unitary

$$\mathcal{U}_r : \mathbb{C}^4 \otimes \mathcal{V}_r \longrightarrow P_r \mathcal{H}, \quad \mathcal{U}_r(e_i \otimes \xi) = e_{i1}\xi, \quad (4.7)$$

under which

$$\begin{aligned}\mathcal{A}_{\text{Cl}}|_{P_r\mathcal{H}} &= M_4(\mathbb{C}) \otimes I, \\ v_r(g) &= I_4 \otimes \rho_r(g), \quad b_j = I_4 \otimes \tilde{b}_j, \\ C^*(\tilde{b}_j) &\subseteq \mathcal{A}'_{\text{tet},r}.\end{aligned}\tag{4.8}$$

If the reduced bank satisfies the hypotheses of Theorem 4.5, its lift is a represented copy of $M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}$ in $\mathcal{A}'_{\text{Cl}} \cap v_r(S_4)'$. Conversely, any such tensor-factor realization satisfies (4.6).

Proof. The matrix-unit identities give

$$\langle e_{i1}\xi, e_{j1}\eta \rangle = \delta_{ij}\langle \xi, \eta \rangle, \quad \sum_i e_{i1}e_{1i} = I.$$

Thus (4.7) is unitary. An operator commuting with all e_{ij} is $I_4 \otimes \tilde{b}$ in this factorization, with $\tilde{b}$ its restriction to $e_{11}P_r\mathcal{H}$. Apply this to $v_r(g)$ and b_j . Their remaining commutators become $[\tilde{b}_j, \rho_r(g)] = 0$, and the genuine group law descends to ρ_r . Relative-commutant saturation on $\mathcal{V}_r$ gives the asserted internal lift. The converse is immediate in tensor coordinates. $\square$

Finite verification and nontrivial spin action. All commutation conditions above are finite identities. For example $\sum_{i,k} \| [b, e_{ik}] \|^2_{\text{HS}} = 0$ certifies Clifford commutation exactly. Commutation with tetrahedral relabeling alone does not imply it: a trivial tetrahedral action commutes with every b , including a nonscalar Clifford matrix unit. If an original relabeling $u_r(g)$ normalizes rather than commutes with the Clifford factor, a specified coherent spin implementation $s_r(g) \in \mathcal{A}_{\text{Cl}}$ must first be removed: $v_r(g) = s_r(g)^* u_r(g)$. The commuting residual must satisfy the genuine S_4 group law to use the ordinary multiplicity census below. A projective residual instead requires its corresponding projective representation census. No such compatibility is inferred from a dimension count.

The word-generated internal seed is also distinct from the kinematic typing algebra. The seed supplies matrix-factor actions; their tensorial matter representation and type projections generate $\mathcal{A}_{\text{type}}$ on the matter carrier. Proposition 5.8 makes this handoff explicit.

4.3 Multiplicity birth and terminating obstructions

The external tetrahedral action is inherited from the Lorentzian relational carrier of [46], while the internal gauge action reconstructed below must be kept on the opposite side of the centralizer. This tetrahedral S_4 is not a postulated S_4 flavour symmetry of the type used in discrete-flavour model building [54]: it acts first as an external relabeling symmetry of the relational carrier, and the endpoint triplet acquires a generation interpretation only after the separate same-carrier ancestry and allocation tests. Let $\mathcal{V}_r$ be the future-minimal reachable–observable residual *physical carrier* spanned by represented source words of length at most r , after the complete spacetime and physical-null shorts. The source letters are required to descend to this fixed quotient, its word spans are nested, and its tetrahedral action preserves the spans. Compatibility with the Clifford carrier is the separate finite condition of Proposition 4.2. This is not the regular word Hilbert space $L^2(\mathcal{A}, \tau_{\text{reg}})$: by Corollary B.10, large invariant multiplicities of the latter do not by themselves create physical internal factors. Decompose the external S_4 action as

$$\mathcal{V}_r \cong \bigoplus_{\lambda \in \widehat{S_4}} V_\lambda \otimes M_{\lambda,r}, \quad m_{\lambda,r} := \dim_{\mathbb{C}} M_{\lambda,r}.\tag{4.9}$$

Since S_4 has five irreducible complex representations, we record the *complete* multiplicity census in the fixed order

$$\mathbf{m}_r := (m_{\mathbf{1},r}, m_{\text{sgn},r}, m_{W,r}, m_{W^{\text{sgn}},r}, m_{V_2,r}), \quad W^{\text{sgn}} := W \otimes \text{sgn}. \quad (4.10)$$

No unlisted isotypic sector is understood to have zero multiplicity unless

Proposition 4.3 (Complete census, colour eligibility, and flat termination). *At every finite physical word depth,*

$$\mathcal{A}_{\text{tet},r} = C^*(\rho_r(S_4)) \cong \bigoplus_{\lambda} M_{\dim V_{\lambda}}(\mathbb{C}) \otimes I_{M_{\lambda,r}}, \quad (4.11)$$

$$\mathcal{A}'_{\text{tet},r} \cong \bigoplus_{\lambda} I_{V_{\lambda}} \otimes M_{m_{\lambda,r}}(\mathbb{C}). \quad (4.12)$$

An internal simple factor $M_n(\mathbb{C})$ on the λ -multiplicity space requires $m_{\lambda,r} \geq n$. The categorical degree-one census is $(1, 0, 2, 1, 1)$, with

$$\mathcal{A}'_{\text{tet},1} \cong \mathbb{C} \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}. \quad (4.13)$$

One additional external-trivial private line changes the census to $(2, 0, 2, 1, 1)$ and the commutant to

$$M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}. \quad (4.14)$$

Neither carrier supports a commuting internal colour $M_3(\mathbb{C})$. Classical matrix-algebra length theory gives quantitative bounds on the word length needed to span a generated matrix algebra [58]; the flatness test below has a different role, namely to certify stabilization of the particular source-generated physical carrier.

For the proposed assignment of the colour, weak, and scalar blocks to the trivial, standard, sign-twisted-standard, and two-dimensional sectors, respectively, exact exhaustion of the relative commutant requires precisely

$$\mathbf{m}_r = (3, 0, 2, 1, 1). \quad (4.15)$$

In particular, a nonzero sign-isotypic entry produces an additional summand and cannot be omitted.

Let K_r be the labelled physical source-word Gram. If

$$\text{rank } K_r = \text{rank } K_{r+1}, \quad (4.16)$$

then $\mathcal{V}_s = \mathcal{V}_r$ for all $s \geq r$, and every isotypic multiplicity is frozen. Writing

$$m_{0,r} = \dim_{\mathbb{C}} \text{Hom}_{S_4}(\mathbf{1}, \mathcal{V}_r), \quad r_{\text{col}}^{\text{birth}} = \min\{r : m_{0,r} \geq 3\}, \quad (4.17)$$

with the value ∞ when no birth occurs before flatness, a flat depth with $m_{0,r} \leq 2$ permanently excludes an external-trivial internal colour factor. More generally, failure of $m_{\lambda,r} \geq n$ at a flat depth is terminal for an internal $M_n(\mathbb{C})$ on that multiplicity space.

Proof. The isotypic decomposition, Burnside's theorem on each irreducible external factor, and Schur's lemma give the two algebras. A nonzero representation of $M_n(\mathbb{C})$ has dimension at least n , giving the multiplicity obstruction. The degree-one representation is $\mathbb{C}_{\text{type}} \oplus (W \otimes \mathbb{C}^2) \oplus V_2 \oplus W^{\text{sgn}}$, which gives the two displayed censuses. Independent isotypic blocks give (4.15) for the specified assignment. Finally, $\mathcal{V}_{r+1}$ is spanned by $\mathcal{V}_r$ and its one-letter images. Equality of Gram ranks makes those spaces equal, so every source letter preserves $\mathcal{V}_r$. Induction proves stabilization and the terminal multiplicity obstructions. $\square$

Multiplicity three only creates a colour-eligible carrier. Full matrix operations appear only when the represented word algebra connects it. The next tests separate this algebra-generation question from occurrence.

4.4 Internal generation and relative-commutant saturation

Proposition 4.4 (Colour bridge, weak router, and central separation). *On an external-trivial three-space $T \oplus H_{\text{priv}}$, with $\dim T = 1$ and $\dim H_{\text{priv}} = 2$, assume the represented private support words generate $M_2(H_{\text{priv}})$. For an actual word u , put*

$$\omega_{\text{col}}(u) = \|p_H u p_T\|_{\text{HS}}^2 + \|p_T u p_H\|_{\text{HS}}^2. \quad (4.18)$$

Then

$$C^*(\mathbb{C}p_T \oplus M_2(H_{\text{priv}}), u) = M_3(\mathbb{C}) \iff \omega_{\text{col}}(u) > 0. \quad (4.19)$$

At zero bridge the algebra is exactly $\mathbb{C} \oplus M_2(\mathbb{C})$. On the standard-isotypic multiplicity plane, let p_t, p_h project onto unit vectors t, h . Then

$$0 < |\langle t, h \rangle| < 1 \iff C^*(p_t, p_h) = M_2(\mathbb{C}). \quad (4.20)$$

A single rank-one projection generates only $\mathbb{C}^2$.

Finally, a unital subalgebra of $M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}$ surjecting onto each factor is either the full 15-dimensional product or the 14-dimensional algebra

$$\{(A, B, z, z) : A \in M_3(\mathbb{C}), B \in M_2(\mathbb{C}), z \in \mathbb{C}\}. \quad (4.21)$$

For a generating bank (b_j) and the two scalar characters χ_1, χ_2 , the full product occurs exactly when

$$\eta_{\text{cen}} = \sum_j |\chi_1(b_j) - \chi_2(b_j)|^2 > 0. \quad (4.22)$$

Independent represented scalar central supports suffice.

Proof. A nonzero line–plane corner, its adjoint, and the private matrix units generate all rectangular matrix units, hence M_3 ; absent both corners the two blocks reduce the algebra. Two distinct nonorthogonal rank-one projections in dimension two have no common reducing line, so Burnside’s theorem gives M_2 . Finally, a subdirect product of simple matrix algebras can identify only isomorphic factors (Proposition C.4). Only the two scalar factors can therefore be locked, and a difference of their characters on a generator is exactly the condition excluding that identification. $\square$

A bridge can itself be a composite historical word. Appendix C.4 proves the finite reciprocal-return criterion: nonorthogonality of the two mediator cyclic multiplicity spaces is equivalent to a positive return mass, while zero mass gives a reducing projection excluding every longer excursion in the declared route bank. This alternative supplies a bridge rather than assuming an extra direct one.

Define $\mathcal{A}_{\text{int},r}^{\text{word}}$ to be the C^* -algebra generated by all represented physical word compressions at depth r that commute with the external tetrahedral action.

Theorem 4.5 (Internal-algebra reconstruction by relative-commutant saturation). *Suppose that at one finite word depth the complete external multiplicity census is*

$$\mathbf{m}_r = (3, 0, 2, 1, 1) \quad (4.23)$$

in the ordering (4.10), that the external-trivial three-space is coherently realized as $T \oplus H_{\text{priv}}$, and that:

- (I1) the private support words generate $M_2(H_{\text{priv}})$ and either one represented word has $\omega_{\text{col}} > 0$, or an admitted reciprocal mediator bank satisfies $\mathbf{m}_{\text{ret}} > 0$ in Proposition C.5;
- (I2) the standard multiplicity plane contains two distinct nonorthogonal represented rank-one projections;
- (I3) the V_2 and W^{sgn} external blocks have multiplicity one;
- (I4) the two scalar sectors are centrally separated, equivalently the generated algebra contains their independent central supports or passes $\eta_{\text{cen}} > 0$ in Proposition 4.4.

Then

$$\mathcal{A}_{\text{int},r}^{\text{word}} = \mathcal{A}'_{\text{tet},r} \cong M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}. \quad (4.24)$$

Its complex dimension is 15, its centre has dimension four, and the two nonabelian simple factors are intrinsically distinguished by their matrix sizes. Hence the labels “colour” and “weak” are canonical up to inner automorphism once the multiplicity packet has landed.

Proof. Condition (I1) and Proposition 4.4 generate the M_3 block; when the reciprocal route alternative is used, Proposition C.5 supplies the required represented bridge. Condition (I2) and Proposition 4.4 generate the M_2 block. The multiplicity-one sectors give surjective scalar compressions, and condition (I4) excludes their diagonal identification by Proposition 4.4. The zero sign entry in (4.23), together with Proposition 4.3, excludes every additional sign-isotypic commutant block. Thus the generated internal algebra is the full product and exhausts the complete relative commutant. $\square$

The saturation theorem is conditional on the complete census and on the occurrence of the stated connecting words. At that census it also gives an explicit classification of the smaller branches rather than a completion of missing structure.

Proposition 4.6 (Exact internal-algebra deficit alternatives). *On a coherent carrier with complete census $\mathbf{m}_r = (3, 0, 2, 1, 1)$, assume the colour three-space initially carries $M_2(\mathbb{C}) \oplus \mathbb{C}$, the weak multiplicity plane carries one rank-one projection, and the displayed colour, weak and two scalar central supports occur independently in the represented internal algebra. Then exactly the following dimensions occur:*

represented internal words	$\dim_{\mathbb{C}} \mathcal{A}_{\text{int}}^{\text{word}}$
neither colour bridge nor weak router	9
weak router but no colour bridge	11
colour bridge but no weak router	13
both bridge and router	15

The terminal value 15 is equivalent to saturation of the complete relative commutant in Theorem 4.5. Each failure is therefore an exact finite source-word rank deficit rather than a discarded branch. If independent central supports have not yet been established, Proposition 4.4 must be applied before componentwise dimensions are added; in particular, the terminal blockwise-surjective branch may still be the 14-dimensional scalar-locked algebra (4.21).

Proof. Without the bridge the colour block is $M_2 \oplus \mathbb{C}$, of dimension five; with the bridge it is M_3 , of dimension nine. One weak rank-one projection generates $\mathbb{C}^2$, of dimension two; a noncommuting second projection generates M_2 , of dimension four. The two multiplicity-one sectors add two scalar dimensions, giving 9, 11, 13, 15. $\square$

For fixed multiplicity carriers and diagonal minimal projections, nonabelian router saturation is open and dense in the affine router class; its failure is a finite union of disconnected-support linear subspaces (Proposition C.8). This genericity is relative to that admitted class and asserts no microscopic occurrence law.

The algebra just reconstructed is the complex relative commutant of the reduced tetrahedral action, lifted through Clifford compatibility. It is neither the commutant of the complete matter-incidence action nor the real finite coordinate algebra of an almost-commutative spectral triple. The next section determines the physical gauge group from represented incidence or anomaly data, and then its actual matter typing algebra.

5 Gauge group and matter typing

Landing the internal matrix factors does not yet fix their relative tensor position, the central gauge subgroup, or the matter inventory. We treat these questions in that order. The determinant-source and anomaly routes to the global group are logically independent; the charged-copy count then requires additional source-completeness and endpoint-allocation hypotheses.

5.1 Tensor alignment and finite naturality

The global form of the Standard-Model gauge group is not fixed by its Lie algebra alone [48, 49]; the construction below selects the $S(U(3) \times U(2))$ or $\mathbb{Z}_6$ quotient on the stated branch. We first identify a determinant incidence by finite equations on the landed matrix factors, then distinguish its operator ray from the phase-sensitive datum that selects a subgroup. Throughout, $C \cong \mathbb{C}^3$ and $W_2 \cong \mathbb{C}^2$ are the actual internal multiplicity carriers after Theorem 4.5. Their actions on the indicated source and target legs must be represented on the same identified typed carrier. They are not external tetrahedral Burnside blocks.

The relative orientation of these factor actions is additional represented data. Let $\mathcal{K}_{\text{det}}$ be the carrier supporting the colour input and the two retained weak input slots, and let $\mathcal{A}_C \cong M_3(\mathbb{C})$ and $\mathcal{A}_{W,1}, \mathcal{A}_{W,2} \cong M_2(\mathbb{C})$ be their represented unital factor algebras on $\mathcal{K}_{\text{det}}$. Choose Hermitian spanning families (C_α) , $(W_\beta^{(1)})$, and $(W_\gamma^{(2)})$ and put

$$\begin{aligned} \Delta_\otimes := & \sum_{\alpha,\beta} \| [C_\alpha, W_\beta^{(1)}] \|_{\text{HS}}^2 + \sum_{\alpha,\gamma} \| [C_\alpha, W_\gamma^{(2)}] \|_{\text{HS}}^2 \\ & + \sum_{\beta,\gamma} \| [W_\beta^{(1)}, W_\gamma^{(2)}] \|_{\text{HS}}^2. \end{aligned} \tag{5.1}$$

Proposition 5.1 (Same-carrier tensor-factor alignment). *The condition*

$$\Delta_\otimes = 0 \tag{5.2}$$

holds if and only if the three represented factor algebras commute pairwise. On this branch multiplication induces an injective unital $$ -homomorphism*

$$M_3(\mathbb{C}) \otimes M_2(\mathbb{C}) \otimes M_2(\mathbb{C}) \hookrightarrow \mathcal{B}(\mathcal{K}_{\text{det}}), \tag{5.3}$$

and the generated algebra is unitarily equivalent to $M_{12}(\mathbb{C}) \otimes I_M$ for a finite multiplicity space M . Thus the intrinsic tensor action $C \otimes W_2 \otimes W_2$ is represented on the same carrier, while any residual multiplicity remains a spectator for this tensor factorization.

Proof. Vanishing of (5.1) is equivalent to commutation of all elements of the chosen spanning families and hence, by linearity, adjoints and multiplication, to pairwise commutation of the generated factor algebras. Multiplication then defines a unital $*$ -homomorphism from their finite tensor product, which is the simple algebra $M_{12}(\mathbb{C})$. A unital homomorphism from a simple full matrix algebra has zero kernel, so the product representation is faithful. The finite-dimensional representation theorem then gives the displayed spatial form. $\square$

A sufficient reconstruction of these commuting tensor actions from mixed pair and triple source occurrences is proved in Proposition D.1. Separate factor marginals are not sufficient (Proposition D.2). Once alignment is established, the matrix units below are coordinates of already represented actions, not additional physical controls.

Set

$$\mathcal{U} = \text{Hom}(C \otimes W_2 \otimes W_2, \Lambda^2 C^*), \quad \dim_{\mathbb{C}} \mathcal{U} = 36. \quad (5.4)$$

Source and target typing alone do not determine an incidence line: Appendix D computes the normalization kernel of the unrestricted rectangular envelope. The required additional information is supplied by finite identities for the represented colour and weak actions.

For $X \in \text{End}(C)$ and $Y \in \text{End}(W_2)$ define

$$\begin{aligned} d\rho_C^{\text{in}}(X) &= X \otimes I \otimes I, \\ d\rho_C^{\text{out}}(X)(\varphi \wedge \psi) &= -(\varphi \circ X) \wedge \psi - \varphi \wedge (\psi \circ X), \\ d\rho_W^{\text{in}}(Y) &= I_C \otimes Y \otimes I + I_C \otimes I \otimes Y, \end{aligned} \quad (5.5)$$

and the linear defect maps on $\mathcal{U}$

$$\mathcal{R}_X(T) = d\rho_C^{\text{out}}(X)T - Td\rho_C^{\text{in}}(X), \quad \mathcal{S}_Y(T) = -Td\rho_W^{\text{in}}(Y). \quad (5.6)$$

Choose matrix-unit systems for the represented factors and put

$$\mathcal{G}_C = \{E_{12}, E_{21}, E_{23}, E_{32}\}, \quad \mathcal{G}_W = \{F_{12}, F_{21}\}. \quad (5.7)$$

No central generator or tensor-slot interchange is included in this bank.

Theorem 5.2 (Determinant-line reconstruction and positive naturality certificate). *The common kernel*

$$\mathcal{S}_{\text{nat}} = \bigcap_{X \in \mathcal{G}_C} \text{Ker } \mathcal{R}_X \cap \bigcap_{Y \in \mathcal{G}_W} \text{Ker } \mathcal{S}_Y \quad (5.8)$$

is one-dimensional and canonically isomorphic to $(\det C \otimes \det W_2)^$. Its nonzero elements are $\text{SU}(C) \times \text{SU}(W_2)$ -equivariant, vanish on $C \otimes \text{Sym}^2 W_2$, and obey*

$$T(I_C \otimes \tau_W) = -T, \quad T \mapsto e^{-i(3\alpha+2\beta)}T \quad (5.9)$$

under weak-slot interchange and central phases. The colour equations alone have kernel dimension four, and the weak equations alone dimension nine.

Choose T_0 on this line with $T_0^ T_0 = P_{\text{alt}}$, where P_{alt} projects onto $C \otimes \Lambda^2 W_2$. Then $\|T_0\|_{\text{HS}}^2 = 3$ and $P_{\text{det}} = |T_0\rangle\langle T_0|/3$ is its orthogonal line projector. For standard normalized matrix units and orthonormal wedge bases, the positive defect operator*

$$H_{\text{nat}} = \sum_{X \in \mathcal{G}_C} \mathcal{R}_X^* \mathcal{R}_X + \sum_{Y \in \mathcal{G}_W} \mathcal{S}_Y^* \mathcal{S}_Y \quad (5.10)$$

satisfies

$$\text{Ker } H_{\text{nat}} = S_{\text{nat}}, \quad H_{\text{nat}} \succeq 2(I - P_{\text{det}}), \quad \lambda_{\min}^+(H_{\text{nat}}) = 2. \quad (5.11)$$

For any positive Choi operator J_D on this port, without a Kraus-rank assumption, put $d_{\text{nat}} = \text{Tr}(H_{\text{nat}} J_D)$. Then

$$\text{Tr}[(I - P_{\text{det}})J_D] \leq \frac{1}{2}d_{\text{nat}}, \quad (5.12)$$

and

$$d_{\text{nat}} = 0 \iff J_D = x|T_0\rangle\rangle\langle\langle T_0|, \quad x \geq 0. \quad (5.13)$$

The additional condition $\text{Tr } J_D > 0$ is equivalent to $x > 0$. The spectrum, kernel, and simultaneously transported certificates are unchanged by unitary changes of matrix-unit systems.

Proof. The four colour units generate $\mathfrak{sl}_3(\mathbb{C})$ and the two weak units $\mathfrak{sl}_2(\mathbb{C})$ under brackets and linear combinations. Intertwining the generators therefore intertwines these Lie algebras. The weak decomposition $W_2 \otimes W_2 = \text{Sym}^2 W_2 \oplus \Lambda^2 W_2$ has only one trivial summand, the alternating line. Since the target is weak-trivial, the weak identities leave $\text{Hom}(C, \Lambda^2 C^*)$, of dimension nine. The volume identification makes its two colour modules irreducible and isomorphic for $\mathfrak{sl}_3$, so Schur's lemma leaves one line. Colour identities alone leave four weak covector coefficients. Connectedness exponentiates the identities to the compact special-unitary groups. The target weight -2α and inverse source weight $-\alpha - 2\beta$ give (5.9).

The canonical line identification is given by

$$(\Theta_\tau(c \otimes w_1 \otimes w_2))(c_1, c_2) = \tau(c \wedge c_1 \wedge c_2 \otimes w_1 \wedge w_2).$$

It is nonzero and hence an isomorphism between the two lines. The sum-of-squares definition gives the kernel of H_{nat} ; Appendix D.3 diagonalizes its colour and weak summands and proves the sharp gap. Tracing this bound against $J_D \succeq 0$ gives (5.12). Finally, $\text{Tr}(H_{\text{nat}} J_D) = \|H_{\text{nat}}^{1/2} J_D^{1/2}\|_{\text{HS}}^2$ vanishes exactly when J_D is supported on the one-dimensional kernel. Positivity gives the stated ray and its trace $3x$. Unitary changes of coordinates conjugate the representations and the defect operator simultaneously. $\square$

The factor actions determine the selector before a source populates its port. Zero defect tests purity, whereas positive mass tests occurrence; small positive defect bounds leakage but does not imply exact selection. For positive branch refinements, zero total defect forces zero defect on each branch. Proposition D.8 gives the two-sided spectral bounds and branchwise occurrence estimates.

5.2 The global group: determinant and anomaly routes

Definition 5.3 (Phase-sensitive determinant seed). A determinant seed is a fixed nonzero functional, normalized to a unitary isomorphism,

$$\tau : \det C \otimes \det W_2 \xrightarrow{\sim} \mathbb{C}, \quad (5.14)$$

where the target scalar line is fixed pointwise. Such a tensor can be supplied directly as a represented incidence or recovered relative to a marked coherent neutral reference. Changing its overall normalization phase does not change its stabilizer. Retaining only the operator ray or its Choi projector is a weaker datum and is not a phase-sensitive seed.

Theorem 5.4 (Gauge-group reconstruction and the phase-sensitive criterion). *The unitary stabilizer of a phase-sensitive determinant seed is*

$$\text{Aut}_U(C, W_2, \tau) = S(U(3) \times U(2)) \cong \frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}. \quad (5.15)$$

Equivalently, let $\chi(A, B) = \det A \det B$ and let the group act by 1 on a marked neutral amplitude space $\mathcal{V}_0$ and by χ^{-1} on $\mathcal{S}_{\text{nat}}$. For a positive packet

$$J = \begin{pmatrix} J_{00} & J_{0D} \\ J_{0D}^* & J_{DD} \end{pmatrix}, \quad h_{\text{det}} = \|J_{0D}\|_{\text{HS}}^2,$$

one has

$$h_{\text{det}} > 0 \iff \text{Stab}_{U(3) \times U(2)}(J) = \ker \chi. \quad (5.16)$$

If $h_{\text{det}} = 0$, this marked packet has the full stabilizer $U(3) \times U(2)$, even when J_{DD} is a nonzero exact naturality ray. For $\chi(g) = e^{i\phi}$,

$$\|\mathcal{R}(g)J\mathcal{R}(g)^* - J\|_{\text{HS}}^2 = 8h_{\text{det}} \sin^2(\phi/2). \quad (5.17)$$

A larger marked packet has this full stabilizer only when its other retained components are $\ker \chi$ -invariant; otherwise their stabilizers must be intersected.

Proof. A pair (A, B) preserves τ exactly when $\det A \det B = 1$. The homomorphism $\Phi(g_3, g_2, z) = (z^{-2}g_3, z^3g_2)$ maps onto this subgroup: for (A, B) in it, choose $z^6 = \det B$, and put $g_3 = z^2A$, $g_2 = z^{-3}B$. Its kernel is $\{(z^2I_3, z^{-3}I_2, z) : z^6 = 1\} \cong \mathbb{Z}_6$. On the marked amplitude packet, the diagonal blocks are invariant and conjugation multiplies J_{0D} by χ . A nonzero cross block is fixed exactly when $\chi = 1$. The two orthogonal cross blocks contribute $2|e^{i\phi} - 1|^2\|J_{0D}\|_{\text{HS}}^2$, proving the quantitative formula. Additional marked data restrict a stabilizer by intersection. $\square$

A rank-one Choi packet is phase blind. Neither its normalization nor its positive mass creates the coherent neutral-incidence cross block. The normalization and minimal-arity statements are retained in Appendix D.5. We next obtain the same group by a different route, using a specified chiral tensor packet and anomaly freedom rather than determinant-source occurrence.

Let the infinitesimal central action on $C \oplus W_2$ be

$$iaI_C \oplus ibI_{W_2}.$$

Define the tensor types

$$Q = C \otimes W_2, \quad u = \Lambda^2 C^*, \quad d = C, \quad L = W_2^*, \quad e = \Lambda^2 W_2^*, \quad H = W_2. \quad (5.18)$$

Their central weights are

$$(a + b, -2a, a, -b, -2b, b). \quad (5.19)$$

Proposition 5.5 (Anomaly-forced central weights). *The following statement is made within the fixed tensor types (5.18) and the two-parameter central action above; broader abelian charge problems can possess additional anomaly-free solutions [53]. Write right-handed fermions as left-handed conjugates when computing chiral anomalies. The local $\text{SU}(3)^2\text{U}(1)$, $\text{SU}(2)^2\text{U}(1)$, gravitational- $\text{U}(1)$, and cubic $\text{U}(1)$ anomalies vanish simultaneously if and only if*

$$3a + 2b = 0. \quad (5.20)$$

Up to overall reversal of the $U(1)$ orientation, the primitive integral solution is

$$(a, b) = (-2, 3).$$

With $y = 6Y$, the resulting physical weights are

$$y(Q, u, d, L, e, H) = (1, 4, -2, -3, -6, 3). \quad (5.21)$$

If a neutral right-handed singlet is retained, $y(\nu) = 0$.

Proof. After charge conjugation of the right-handed fields,

$$\begin{aligned} \mathcal{A}_{\text{SU}(3)^2 Y} &= \mathcal{A}_{\text{SU}(2)^2 Y} = \frac{1}{2}(3a + 2b), \\ \mathcal{A}_{\text{grav}^2 Y} &= 3(3a + 2b), \\ \mathcal{A}_{Y^3} &= 3(3a + 2b)(3a^2 + 2b^2). \end{aligned}$$

The pure colour anomaly cancels as $2 - 1 - 1$, and the weak doublet count is even. For a nontrivial abelian action $3a^2 + 2b^2 > 0$, so simultaneous local anomaly cancellation is equivalent to (5.20). The primitive coprime integral solution is $(-2, 3)$ up to sign; substitution gives (5.21). $\square$

For the global quotient one must also determine the common kernel of the represented matter action. We display it in left-handed convention so that the group kernel and the chiral anomaly convention can be compared directly.

Proposition 5.6 (Faithful matter realization of the determinant quotient). *Normalize the abelian charge by $y = 6Y$ and write all fermions as left-handed Weyl fields. On one generation the represented matter packet is*

$$\begin{aligned} Q : (\mathbf{3}, \mathbf{2})_1, \quad u^c : (\overline{\mathbf{3}}, \mathbf{1})_{-4}, \quad d^c : (\overline{\mathbf{3}}, \mathbf{1})_2, \\ L : (\mathbf{1}, \mathbf{2})_{-3}, \quad e^c : (\mathbf{1}, \mathbf{1})_6, \quad H : (\mathbf{1}, \mathbf{2})_3, \quad [\nu^c : (\mathbf{1}, \mathbf{1})_0]. \end{aligned}$$

The common kernel of these representations of $\tilde{G} = \text{SU}(3) \times \text{SU}(2) \times U(1)_y$ is exactly

$$K = \left\langle \left(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3} \right) \right\rangle \cong \mathbb{Z}_6. \quad (5.22)$$

Consequently the represented matter packet is faithful precisely for $\tilde{G}/K$, which is the same global group selected intrinsically by the determinant seed in Theorem 5.4.

Proof. A common-kernel element must be central, so write it as $(\omega_3^a I_3, (-1)^b I_2, e^{i\theta})$. The charged singlet e^c gives $\theta = m\pi/3$ modulo 2π ; the lepton doublet gives $b \equiv m \pmod{2}$; and the colour antitriplet d^c gives $a \equiv m \pmod{3}$. Substitution in the remaining representations shows that precisely the six powers of the generator in (5.22) act trivially. The optional neutral singlet is trivial under all factors and adds no further restriction. $\square$

Corollary 5.7 (Faithful global group from anomaly freedom). *For the represented chiral tensor packet, primitive local anomaly freedom fixes $(a, b) = (-2, 3)$ up to simultaneous sign. Together with the faithful matter kernel of Proposition 5.6, this gives*

$$G_{\text{phys}} \cong S(U(3) \times U(2)) \cong (\text{SU}(3) \times \text{SU}(2) \times U(1))/\mathbb{Z}_6.$$

This reconstruction does not assume occurrence or phase coherence of a historical determinant-incidence source. When such a source is represented, Theorem 5.4 gives the independent fixed-tensor and marked-source stabilizer statements.

Proof. Proposition 5.5 fixes the primitive embedding $(a, b) = (-2, 3)$ up to simultaneous sign, and Proposition 5.6 gives its matter kernel $\mathbb{Z}_6$. The homomorphism Φ in the proof of Theorem 5.4 identifies the quotient with $S(U(3) \times U(2))$ as a group-theoretic identity; using that identity requires no determinant-source occurrence. $\square$

Proposition 5.8 (Kinematic typing algebra on the represented matter carrier). *Suppose the fermion modules of Proposition 5.6, with multiplicity spaces N_f , occur on one Clifford-compatible carrier*

$$\mathcal{H}_{\text{mat}} \cong \mathbb{C}^4 \otimes \bigoplus_{f \in \mathcal{F}} V_f \otimes N_f, \quad \mathcal{F} = \{Q, u^c, d^c, L, e^c\} \quad [\text{and } \nu^c].$$

Let p_f be the actual type projections and $r(g)$ the represented G_{det} action. Then

$$\mathcal{A}_{\text{type}} := C^*(r(g), p_f : g \in G_{\text{det}}, f \in \mathcal{F}) = \bigoplus_f I_4 \otimes \mathcal{B}(V_f) \otimes I_{N_f}. \quad (5.23)$$

It commutes with $\mathcal{A}_{\text{Cl}}$ and supplies the kinematic algebra in Definition 2.1. Without the neutral singlet, its abstract block sizes are $(6, 3, 3, 2, 1)$, rather than the seed sizes $(3, 2, 1, 1)$. Any represented incidence family on these same sectors, including non-factorizing incidences, then gives the joint packet of Theorem 3.3.

Proof. Every displayed matter representation is irreducible. On each type block, Schur's lemma gives the scalar commutant on V_f , and finite bicommutant closure gives $\mathcal{B}(V_f)$. The type projections separate the blocks, while the represented multiplicity factors are untouched. Clifford compatibility makes the spinor factor the identity for this algebra. The stated block dimensions are those of the tensor representations. The incidence conclusion is precisely the unrestricted coefficient-space construction; the hypothesis that these types occur on the carrier is not inferred from the abstract seed algebra alone. $\square$

Thus the seed algebra, its physical gauge group, and the matter typing algebra have different roles. In particular the two scalar seed summands are not two additional physical $U(1)$ groups, and the matter block sizes are $(6, 3, 3, 2, 1)$ rather than $(3, 2, 1, 1)$. With this represented action fixed, we can now ask which charged sectors are actually sourced or reached.

6 Charged inventory and generations

The preceding section fixes the represented gauge action and its matter types. It does not yet determine the reached charged inventory or the number of copies. We first remove unsourced ambient directions, then combine the chiral inventory and linear anomaly tests with a faithful endpoint occurrence and its positive allocation residual.

6.1 Source-minimal charged inventory

All counts below refer to the complete retained physical fermion source and its finite cyclic closure, not to an arbitrary ambient completion. The external Clifford factor is suppressed, so its dimension is not included in the charged multiplicities. Completeness is a premise of the source packet; when an independently complete atlas is available, the positive residual of Proposition E.1 tests it explicitly.

Let $\mathcal{H}_F^{\text{amb}}$ be a finite represented carrier and let

$$\mathcal{A}_F = C^*(c_1, \dots, c_m) \subseteq \mathcal{B}(\mathcal{H}_F^{\text{amb}}) \quad (6.1)$$

be the C^* -algebra generated by the represented gauge action, grading, full-cell generator, matter incidences, and every other physical letter used by the structural claim. Let $B_F : E_F \rightarrow \mathcal{H}_F^{\text{amb}}$ synthesize the complete retained physical fermion source after the declared null short. Define

$$\mathcal{K}_0 = \text{Ran } B_F, \quad \mathcal{K}_{r+1} = \mathcal{K}_r + \sum_{j=1}^m (c_j \mathcal{K}_r + c_j^* \mathcal{K}_r). \quad (6.2)$$

Proposition 6.1 (Terminating source-minimal fermion saturation). *The filtration (6.2) stabilizes after finitely many strict increases at*

$$\mathcal{K}_F^{\min} = \mathcal{A}_F \text{Ran } B_F. \quad (6.3)$$

At the first equality $\mathcal{K}_{r+1} = \mathcal{K}_r$, the stabilized space reduces $\mathcal{A}_F$. If S_r synthesizes $\mathcal{K}_r$ and N_r synthesizes all columns $c_j S_r, c_j^* S_r$, write

$$\begin{pmatrix} G_r & C_r \\ C_r^* & D_r \end{pmatrix} = \begin{pmatrix} S_r^* \\ N_r^* \end{pmatrix} \begin{pmatrix} S_r & N_r \end{pmatrix}$$

and put

$$\mathbb{L}_r = D_r - C_r^* G_r^\dagger C_r = N_r^* (I - P_r) N_r \succeq 0. \quad (6.4)$$

Then

$$\text{rank } \mathbb{L}_r = \dim \mathcal{K}_{r+1} - \dim \mathcal{K}_r, \quad (6.5)$$

so $\mathbb{L}_r = 0$ is an exact terminating saturation test. Compression to $\mathcal{K}_F^{\min}$ preserves every loaded source-word Gram, while its orthogonal complement is invisible to all such source correlations.

Proof. The spaces are nested and every strict inclusion increases finite dimension. At the first equality all c_j and c_j^* preserve the stabilized space, which is therefore reducing and is the smallest reducing subspace containing $\text{Ran } B_F$. The Schur complement in (6.4) is the Gram of the component of the next columns orthogonal to $\mathcal{K}_r$; its rank is therefore exactly the new cyclic dimension. Every loaded word applied to $\text{Ran } B_F$ lies in the cyclic space, proving the final assertion. $\square$

Fix on $\mathcal{K}_F^{\min}$ the represented gauge action, its self-adjoint central generator Y normalized by $y = 6Y$, and a fermionic chirality grading $\Gamma_F = \Gamma_F^* = \Gamma_F^{-1}$ commuting with the gauge action. Choose a finite family of represented nonabelian infinitesimal generators $X_1, \dots, X_s$ which, together with Y , generate the finite gauge-action algebra. For a target irreducible gauge type $f \in \{Q, u, d, L, e\}$ from (5.18), with central weight y_f and physical chirality $\epsilon_f \in \{\pm 1\}$, define on $\text{Hom}(V_f, \mathcal{K}_F^{\min})$

$$\mathbb{T}_{f,\epsilon} = \sum_{j=1}^s \mathcal{R}_{f,j}^* \mathcal{R}_{f,j} + \mathcal{Y}_f^* \mathcal{Y}_f + \mathcal{G}_{f,\epsilon}^* \mathcal{G}_{f,\epsilon}, \quad (6.6)$$

where

$$\mathcal{R}_{f,j}(T) = X_j T - T \rho_f(X_j), \quad \mathcal{Y}_f(T) = Y T - y_f T, \quad \mathcal{G}_{f,\epsilon}(T) = \Gamma_F T - \epsilon T.$$

Let $\mathcal{I}_{f,\epsilon} = \text{Ker } \mathbb{T}_{f,\epsilon}$ and equip it with $\langle S, T \rangle_{\text{mult}} = d_f^{-1} \text{Tr}(S^* T)$, $d_f = \dim V_f$.

Proposition 6.2 (Canonical chiral projectors and the charged inventory). *For an orthonormal basis $T_1, \dots, T_n$ of $\mathcal{I}_{f,\epsilon}$,*

$$T_a^* T_b = \delta_{ab} I_{V_f}, \quad P_{f,\epsilon} = \sum_a T_a T_a^*, \quad n_{f,\epsilon} = \dim \mathcal{I}_{f,\epsilon} = \frac{\text{rank } P_{f,\epsilon}}{d_f}. \quad (6.7)$$

The projection is basis independent and its range is the complete physical (f, ϵ) -isotypic subspace. In particular the evaluation map $\text{ev}_f : V_f \otimes \mathcal{I}_{f, \epsilon_f} \rightarrow P_{f, \epsilon_f} \mathcal{K}_F^{\min}$, $v \otimes T \mapsto Tv$, is unitary.

Let P_0^F project onto the fully gauge-trivial subspace, set $P_{\text{gch}}^F = P_F - P_0^F$, and define

$$\begin{aligned} P_{\text{SM}}^F &= \sum_f P_{f, \epsilon_f}, & P_{\text{mir}}^F &= \sum_f P_{f, -\epsilon_f}, \\ P_{\text{ex}}^F &= P_{\text{gch}}^F - \sum_f (P_{f, +} + P_{f, -}). \end{aligned} \quad (6.8)$$

These are orthogonal projections with

$$P_{\text{gch}}^F = P_{\text{SM}}^F \oplus P_{\text{mir}}^F \oplus P_{\text{ex}}^F, \quad (6.9)$$

and hence

$$\Delta_{\text{inv}}^F := \text{Tr}(P_{\text{gch}}^F - P_{\text{SM}}^F) = \text{rank } P_{\text{mir}}^F + \text{rank } P_{\text{ex}}^F. \quad (6.10)$$

Thus $\Delta_{\text{inv}}^F = 0$ exactly when the complete future-visible nontrivial gauge carrier consists of the target chiral types. A nonzero residual is at least one, and its projection has operator norm one.

Proof. The tester is a sum of positive squared defects, so its kernel is the intertwiner space for the stated gauge type, charge, and chirality. Schur's lemma gives $T_a^* T_b = c_{ab} I$; the normalized multiplicity inner product gives $c_{ab} = \delta_{ab}$. The resulting isometries have orthogonal ranges and exhaust the isotypic component, proving both the projector formula and unitary evaluation. The gauge action and commuting chirality grading give an orthogonal decomposition into these isotypic spaces. The remaining nontrivial gauge part is P_{ex}^F , and the trace of each finite projection is its rank. Cyclic saturation excludes decoupled ambient spectators but includes every sourced or reached gauge submodule together with its gauge orbit. $\square$

The fully gauge-trivial sector, rather than the entire zero-hypercharge sector, is removed. A nonabelian multiplet of zero hypercharge is still gauge charged; Appendix D.5 gives an explicit exotic pair that would be missed by a hypercharge-only audit.

Write

$$n_f := n_{f, \epsilon_f}, \quad f \in \{Q, u, d, L, e\}, \quad (6.11)$$

and, with the determinant-normalized integer charge table of (5.21), define the four normalized linear anomaly rows

$$\begin{aligned} r_3 &= 2n_Q - n_u - n_d, \\ r_{3Y} &= n_Q - 2n_u + n_d, \\ r_{2Y} &= n_Q - n_L, \\ r_{gY} &= n_Q - 2n_u + n_d - n_L + n_e, \end{aligned} \quad (6.12)$$

and

$$\Delta_{\text{lin}} = r_3^2 + r_{3Y}^2 + r_{2Y}^2 + r_{gY}^2. \quad (6.13)$$

Proposition 6.3 (Linear anomaly rigidity of the charged multiplicities). *For nonnegative integer multiplicities with $n_Q > 0$,*

$$\Delta_{\text{lin}} = 0 \iff n_Q = n_u = n_d = n_L = n_e =: g > 0. \quad (6.14)$$

On this branch the cubic $U(1)^3$ anomaly vanishes automatically and the number of left-handed weak doublets is $4g$, hence even. Moreover

$$\Delta_{\text{lin}} \neq 0 \implies \Delta_{\text{lin}} \geq 1. \quad (6.15)$$

Thus, after the central charge table is fixed, the cubic anomaly is not an independent multiplicity constraint. If also $\Delta_{\text{inv}}^F = 0$ and one charged multiplicity is independently certified as three, the complete charged carrier is $3Q \oplus 3u \oplus 3d \oplus 3L \oplus 3e$, with no future-visible charged mirror or exotic sector. Gauge-trivial singlets remain unconstrained.

Proof. The equations $r_3 = r_{3Y} = 0$ give $n_d = 2n_Q - n_u$ and then $3(n_Q - n_u) = 0$, so $n_u = n_d = n_Q$. The equation $r_{2Y} = 0$ gives $n_L = n_Q$, and $r_{gY} = 0$ then gives $n_e = n_Q$. The converse is immediate. For a common value g , the cubic anomaly is g times the already vanishing one-generation cubic anomaly of Proposition 5.5; the weak-doublet count is $3g + g = 4g$. Finally a nonzero sum of squares of integers is at least one. $\square$

The anomaly relations make all charged multiplicities equal, but do not select the common integer. The endpoint source supplies a possible rank-three anchor. We must still verify its occurrence in a charged type and test whether its image exhausts that type's complete multiplicity.

6.2 Endpoint ancestry and the charged-copy count

On the positive endpoint branch of Proposition 4.1, denote the supported complex endpoint space by G_{gen} . Then

$$\dim_{\mathbb{C}} G_{\text{gen}} = 3. \quad (6.16)$$

The odd sign-twisted loop triplet is a different source sector and is not included. Its allocation to matter would require separate ancestry data. For a populated type set $N_f = \mathcal{I}_{f, \epsilon_f}$, with the unitary evaluation map above. Lemma E.2 obtains a surjective synthesis $S_f : E_f \rightarrow N_f$ by contracting a terminal fermion source-word bank against V_f . Redundant source columns are allowed.

Proposition 6.4 (Gauge-covariant endpoint-ancestry compiler). *Fix a populated charged type f , its canonical multiplicity space $N_f = \mathcal{I}_{f, \epsilon_f}$, and the supported endpoint carrier G_{gen} of Proposition 4.1. Let*

$$\Xi_f : V_f \otimes G_{\text{gen}} \longrightarrow P_{f, \epsilon_f} \mathcal{K}_F^{\min} \quad (6.17)$$

be a represented same-carrier endpoint-ancestry occurrence, and put

$$\widehat{\Xi}_f := \text{ev}_f^* \Xi_f : V_f \otimes G_{\text{gen}} \longrightarrow V_f \otimes N_f. \quad (6.18)$$

Using the finite gauge generators $X_1, \dots, X_s$, the central generator Y , and the chirality grading from Proposition 6.2, define the positive covariance defect

$$\begin{aligned} \Delta_f^{\text{anc}}(\Xi_f) := & \sum_{j=1}^s \|X_j \Xi_f - \Xi_f(\rho_f(X_j) \otimes I_{G_{\text{gen}}})\|_{\text{HS}}^2 \\ & + \|Y \Xi_f - y_f \Xi_f\|_{\text{HS}}^2 + \|\Gamma_F \Xi_f - \epsilon_f \Xi_f\|_{\text{HS}}^2. \end{aligned} \quad (6.19)$$

Then

$$\Delta_f^{\text{anc}}(\Xi_f) = 0 \iff \widehat{\Xi}_f = I_{V_f} \otimes T_f \text{ for a unique } T_f : G_{\text{gen}} \rightarrow N_f. \quad (6.20)$$

On the zero-defect branch the multiplicity map is recovered canonically by

$$T_f = \frac{1}{d_f} \text{Tr}_{V_f}(\widehat{\Xi}_f), \quad d_f = \dim V_f, \quad (6.21)$$

where Tr_{V_f} denotes contraction of the input and output V_f legs, and obeys

$$\Xi_f^* \Xi_f = I_{V_f} \otimes T_f^* T_f, \quad \|\Xi_f\|_{\text{HS}}^2 = d_f \|T_f\|_{\text{HS}}^2. \quad (6.22)$$

Consequently endpoint faithfulness is the finite positive condition $T_f^* T_f \succ 0$, equivalently strict positivity of $\Xi_f^* \Xi_f$ on $V_f \otimes G_{\text{gen}}$. Thus, once the represented cross-source occurrence (6.17) is supplied, the endpoint map used in the allocation theorem below is not an arbitrary rank-three embedding: it is a unique coefficient extracted from the physical same-carrier operator.

Proof. Zero defect means that Ξ_f intertwines the represented finite gauge-action algebra with the stated charge and chirality. Transport by the unitary evaluation map gives an intertwiner from $V_f \otimes G_{\text{gen}}$ to $V_f \otimes N_f$. Irreducibility of V_f and Schur's lemma give the unique coefficient T_f ; conversely any $I_{V_f} \otimes T_f$ satisfies these equations. Contracting the V_f legs gives the partial-trace formula, and taking the Gram and trace gives the remaining identities and the equivalence of strict positivity. $\square$

This compiler removes a coordinate choice, not an occurrence hypothesis: Ξ_f is a represented cross-source operator. Old word moments cannot supply it when it is absent from the historical algebra (Proposition B.14). Once it occurs faithfully, the following theorem tests surjectivity without assuming it.

Theorem 6.5 (Endpoint allocation and the exact charged-copy count). *Fix one populated charged type f and a complete represented typed synthesis $S_f : E_f \rightarrow N_f$, for example (E.4). Let $T_f : G_{\text{gen}} \rightarrow N_f$ be the canonical coefficient of a represented ancestry occurrence supplied by Proposition 6.4, or more generally a specified represented endpoint subspace, satisfying*

$$A_f := T_f^* T_f \succ 0, \quad P_{fG} := T_f A_f^{-1} T_f^*. \quad (6.23)$$

No surjectivity of T_f is assumed. Put

$$G_f = S_f^* S_f, \quad C_f = T_f^* S_f, \quad \mathbb{A}_{f|G} := G_f - C_f^* A_f^{-1} C_f. \quad (6.24)$$

Then

$$\mathbb{A}_{f|G} = S_f^* (I - P_{fG}) S_f \succeq 0, \quad a_{f|G} := \text{Tr } \mathbb{A}_{f|G} = \|(I - P_{fG}) S_f\|_{\text{HS}}^2, \quad (6.25)$$

and

$$n_f = \dim(E_f / \text{Ker } G_f) = \text{rank } G_f = 3 + \text{rank } \mathbb{A}_{f|G}. \quad (6.26)$$

In particular,

$$a_{f|G} = 0 \iff \mathbb{A}_{f|G} = 0 \iff \text{Ran } T_f = N_f \iff n_f = 3. \quad (6.27)$$

The quotient carries the inner product induced by G_f , and S_f descends to a unitary from that quotient onto N_f . If, in addition, $\Delta_{\text{inv}}^F = \Delta_{\text{lin}} = 0$, then

$$n_Q = n_u = n_d = n_L = n_e = 3 + \text{rank } \mathbb{A}_{f|G}. \quad (6.28)$$

Consequently $a_{f|G} = 0$ gives exactly three copies of every charged target type and no future-visible charged exotic or mirror sector. Fully gauge-trivial singlets require separate occurrence and multiplicity data.

Proof. Faithfulness of T_f and $\dim G_{\text{gen}} = 3$ make P_{fG} the orthogonal projection onto a three-dimensional subspace of N_f . Substitution proves (6.25). Since S_f is onto,

$$N_f = \text{Ran } T_f \oplus (I - P_{fG})N_f, \quad \text{Ran}((I - P_{fG})S_f) = (I - P_{fG})N_f.$$

For $V = (I - P_{fG})S_f$, the identity $\text{rank}(V^*V) = \text{rank } V$ therefore gives $\text{rank } \mathbb{A}_{f|G} = n_f - 3$. Also $\text{Ker } G_f = \text{Ker } S_f$, which proves the quotient and rank statements. Positivity makes zero trace equivalent to the zero residual. Its vanishing is equivalent to $\text{Ran } S_f \subseteq \text{Ran } T_f$; completeness of S_f and $\text{rank } T_f = 3$ give the remaining equivalences. Finally, $\langle e, G_f e' \rangle = \langle S_f e, S_f e' \rangle$ proves the assertion about the quotient inner product. The four linear anomaly rows force every charged multiplicity to equal $n_f = 3 + \text{rank } \mathbb{A}_{f|G}$ by Proposition 6.3, while the zero-inventory test excludes every other nontrivial gauge-charged sector. $\square$

A small positive allocation trace is a leakage bound, not an exact three-copy conclusion. Appendix E.4 proves the rank-separating threshold under an independently controlled source frame bound and gives counterexamples without such control. The same appendix records the source-native and same-source coefficient formulas, retaining their occurrence hypotheses.

6.3 Generation factors and relative frames

Proposition 6.6 (Abstract replication and source-determined generation frames). *Suppose all populated charged multiplicities N_f have dimension g . For any g -space G and arbitrary unitaries $W_f : G \rightarrow N_f$,*

$$\bigoplus_f V_f \otimes N_f \cong \left(\bigoplus_f V_f \right) \otimes G, \quad (6.29)$$

and a coefficient arrow $B : N_f \rightarrow N_h$ becomes $W_h^ B W_f \in M_g(\mathbb{C})$. Thus the three-copy conclusion already supplies an abstract common $\mathbb{C}^3$ factor, without common endpoint ancestry.*

For each connected incidence component C , suppose instead that a specified rank-three endpoint source G_C occurs with maps $S_f : G_C \rightarrow N_f$ satisfying $S_f^ S_f \succ 0$ and $\text{Ran } S_f = N_f$. Their polar maps $U_f = S_f(S_f^* S_f)^{-1/2}$ are unitaries and give*

$$\mathcal{H}_{\text{mat}} \cong \bigoplus_{C \in \pi_0(\Gamma)} \left(\bigoplus_{f \in C} V_f \right) \otimes G_C, \quad \dim G_C = 3. \quad (6.30)$$

The gauge and typing algebras act trivially on G_C , and every arrow inside C becomes $U_h^ B U_f$. For the usual quark and lepton components this gives separate G_q and G_ℓ , when both component sources occur.*

If a single represented endpoint source G_{gen} supplies all types, these polar maps determine

$$\mathcal{H}_{\text{mat}} \cong \mathcal{H}_F^{(1)} \otimes G_{\text{gen}}, \quad \mathcal{H}_F^{(1)} = \bigoplus_f V_f, \quad \dim G_{\text{gen}} = 3. \quad (6.31)$$

An arrow preserves the full common generation algebra exactly when

$$U_h^* B U_f \in \mathbb{C} I_{G_{\text{gen}}}, \quad (6.32)$$

equivalently when

$$\sum_{i,j=1}^3 \|B U_f E_{ij} U_f^* - U_h E_{ij} U_h^* B\|_{\text{HS}}^2 = 0. \quad (6.33)$$

Arbitrary Yukawa coefficients remain ordinary matrices in this frame; their surviving commutant is determined by Theorem 3.3.

More generally, for equal-dimensional component fibres, if every retained gauge, typing, source, and incidence operator is block diagonal across components, the unselected relative coordinate frames form

$$U(g)^{|\pi_0(\Gamma)|}/U(g)_{\text{diag}}. \quad (6.34)$$

For the quark and lepton components with $g = 3$, this is one relative $U(3)$. No datum internal to the disconnected packet selects a point of this space.

Proof. The direct sum of $I_{V_f} \otimes W_f^*$ gives the first factorization and transports all arrows. Faithful surjective endpoint maps have unitary polar factors, giving the componentwise and common-source identifications. In common coordinates an arrow intertwines every generation matrix exactly when it commutes with all matrix units, hence is scalar; the sum of squared defects tests that condition. To compare disconnected components, choose $J_C : G \rightarrow G_C$. Replacing J_C by $J_C U_C$ conjugates the complete represented tuple by

$$\mathcal{U}_U = \bigoplus_C I_{V_C} \otimes U_C^*. \quad (6.35)$$

Words, Grams, and commutants transform covariantly, while spectra, singular values, trace polynomials, and scalar occurrence residuals are unchanged. A common right multiplication changes only the basis of the reference space, leaving the quotient (6.34). There is no represented intercomponent relation to select a relative frame. $\square$

6.4 Grading change, weak copies, and finite Dirac relations

These remaining tests concern occurrence and the structure of a specified operator, not the charged-copy count. For a protected grading $J = P_+ - P_-$ and an active completely positive map Φ , the odd-Choi mass is

$$\mu_J(\Phi) = \text{Tr}[P_- \Phi(P_+)] + \text{Tr}[P_+ \Phi(P_-)].$$

It is Kraus independent and vanishes exactly when the Choi support consists of grading-even operators. Appendix E.5 proves this criterion, its finite-alphabet stopping alternative, and its invariance under unread trace-preserving spectator lifts. Positivity certifies active grading change; it does not by itself select a weak-doublet copy.

Proposition 6.7 (Positive weak-copy census). *Let $W_2 \cong \mathbb{C}^2$ be the represented weak defining module, let M_H be a finite multiplicity space, and let $J_H \succeq 0$ be the complete actual positive weak-scalar occurrence packet on the retained sector $W_2 \otimes M_H$. Define*

$$B_H := \text{Tr}_{W_2} J_H \succeq 0, \quad (6.36)$$

and let its gauge-cyclic occurrence carrier be

$$\mathcal{K}_H^{\text{occ}} := \text{span}\{(g \otimes I)v : g \in \text{SU}(W_2), v \in \text{Ran } J_H\}. \quad (6.37)$$

For normalized Haar measure,

$$\overline{J}_H := \int_{\text{SU}(W_2)} (g \otimes I) J_H (g^* \otimes I) dg = \frac{1}{2} I_{W_2} \otimes B_H, \quad (6.38)$$

and

$$\mathcal{K}_H^{\text{occ}} = \text{Ran } \overline{J}_H = W_2 \otimes \text{Ran } B_H. \quad (6.39)$$

Thus the source-minimal gauge-cyclic number of reached weak-doublet copies is

$$g_H^{\text{occ}} = \text{rank } B_H. \quad (6.40)$$

In particular, there is exactly one such doublet if and only if

$$m_H := \text{Tr } B_H > 0, \quad \mathfrak{p}_H := m_H^2 - \text{Tr}(B_H^2) = 0. \quad (6.41)$$

The criterion is independent of Kraus coordinates, phases, and the choice of basis of M_H .

Proof. Schur averaging gives $\bar{J}_H = I_{W_2} \otimes B_H/2$. The kernel of a positive group average is the intersection of the kernels of its conjugates, so its range is precisely the gauge-cyclic occurrence carrier. Hence the copy count is $\text{rank } B_H$. If $\lambda_i \geq 0$ are the eigenvalues of B_H , then $m_H^2 - \text{Tr}(B_H^2) = 2 \sum_{i < j} \lambda_i \lambda_j$; together with $m_H > 0$, its vanishing is equivalent to rank one. All constructions are independent of coefficient coordinates. The kernel-of-the-average argument is written out in Appendix E.7. $\square$

The count concerns the complete retained packet's gauge-cyclic support, not an unused ambient multiplicity. The rank of J_H itself need not be the copy count: the same appendix gives a rank-one packet with two weak copies.

Let A be a source-saturated self-adjoint full-cell generator on the common carrier and let

$$\tilde{D}_F = P_- A P_+ + P_+ A P_- = \frac{1}{2}(A - J A J) \quad (6.42)$$

be its grading-odd component.

Definition 6.8 (Finite-Dirac relation space). Let $\mathfrak{D}_F$ be the finite-dimensional $*$ -closed linear subspace of grading-odd operators whose typed blocks are supported only on the represented fermionic Yukawa incidence relations determined by the matter packet; when the neutral and charge-conjugate sectors are represented, the declared Majorana relation is included as well. Let Π_F^D be the Hilbert–Schmidt orthogonal projection onto $\mathfrak{D}_F$ and define

$$D_F^{\text{can}} = \Pi_F^D \tilde{D}_F, \quad \Delta_F^{\text{rel}} = \|(I - \Pi_F^D) \tilde{D}_F\|_{\text{HS}}^2. \quad (6.43)$$

Proposition 6.9 (Fixed-generator finite-Dirac closure). *For the specified self-adjoint generator A ,*

$$\tilde{D}_F = D_F^{\text{can}} \iff \Delta_F^{\text{rel}} = 0. \quad (6.44)$$

Moreover

$$\text{dist}_{\text{HS}}(\tilde{D}_F, \mathfrak{D}_F) = \sqrt{\Delta_F^{\text{rel}}}. \quad (6.45)$$

If $\Delta_F^{\text{rel}} = 0$ and $\|D_F^{\text{can}}\|_{\text{HS}} > 0$, the complete odd component of this generator is a nonzero admitted finite-Dirac operator. In any generation factorization of Proposition 6.6, its typed coefficient blocks are ordinary generation matrices. If the stronger common-ancestry conditions of Proposition 6.6 hold, those matrices are expressed in the corresponding source-determined common frame. Odd-provenance completeness is not required for these fixed-generator conclusions; when imposed, it additionally excludes independent future-visible odd source directions outside the declared odd synthesis.

Proof. Orthogonal decomposition relative to $\mathfrak{D}_F$ gives

$$\tilde{D}_F = \Pi_F^D \tilde{D}_F + (I - \Pi_F^D) \tilde{D}_F.$$

The second summand has squared Hilbert–Schmidt norm Δ_F^{rel} , proving both displayed statements. The remaining claims follow from the typed decomposition and the generation-coordinate identifications already established above. $\square$

Odd-provenance completeness, when imposed, means that every future-visible grading-odd action of the saturated carrier lies in its represented odd source span. This stronger exhaustion condition is not needed for the fixed-generator projection test above.

Corollary 6.10 (Structural synthesis). *Suppose a same-source, Clifford-compatible packet satisfies the census and internal-landing hypotheses of Theorem 4.5. Suppose also that its global group is obtained either from the aligned, phase-sensitive determinant data of Theorem 5.4, or from the represented chiral tensor packet, primitive anomaly freedom, and faithful kernel of Corollary 5.7. On its complete source-minimal fermion carrier, assume the faithful endpoint occurrence and complete typed synthesis of Theorem 6.5, with $\Delta_{\text{inv}}^F = \Delta_{\text{lin}} = a_{f_0|G} = 0$ for one charged type f_0 . Then*

$$\begin{aligned} \mathcal{A}_{\text{int}}^{\text{word}} &\cong M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}, \\ G_{\text{phys}} &\cong S(\text{U}(3) \times \text{U}(2)) \cong (\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6, \\ \mathcal{K}_{F,\text{gch}}^{\text{min}} &\cong 3Q \oplus 3u \oplus 3d \oplus 3L \oplus 3e. \end{aligned} \tag{6.46}$$

The canonical matter action supplies the typing algebra, so all its represented incidences, of arbitrary coefficient rank, satisfy the general duality. The independent weak-copy and fixed-generator tests above give respectively one reached weak doublet when $m_H > 0$, $\mathfrak{p}_H = 0$, and an admitted nonzero finite-Dirac operator when $\Delta_F^{\text{rel}} = 0$, $\|D_F^{\text{can}}\|_{\text{HS}} > 0$.

Proof. Combine the internal saturation and Clifford lift, the applicable global-group route, the endpoint-allocation count, and the two independent occurrence and operator tests. Proposition 5.8 then supplies the typing algebra for Theorem 3.3. $\square$

The synthesis retains the distinction between classification and occurrence: a failed test returns its smaller algebra, missing source direction, or nonzero residual. It determines no numerical Yukawa or Majorana entries, flavour hierarchy, CKM or PMNS parameters, gauge or Higgs couplings, renormalization-group data, or fermionic measure phase.

7 Rigidity and refinement

The preceding sections determine exact finite algebras. We now ask how that exactness can be certified and maintained. One positive commutator operator governs three distinct statements: recognition at a fixed packet, perturbative stability with a protected algebra, and cofinal stability on varying screened spaces. None of these statements supplies a missing physical generator or a missing source occurrence.

7.1 A positive commutant certificate

Theorem 7.1 (Finite commutant certificate and coercivity). *Let $c_1, \dots, c_s$ span a unital $*$ -algebra $\mathcal{C} \subseteq \mathcal{B}(\mathcal{H})$. On Hilbert–Schmidt operator space, define the commutant Laplacian*

$$\mathcal{L}_{\mathcal{C}} = \sum_j \text{ad}_{c_j}^* \text{ad}_{c_j}, \quad \text{ad}_c(X) = [c, X]. \tag{7.1}$$

Then $\mathcal{L}_C \succeq 0$ and

$$\text{Ker } \mathcal{L}_C = \mathcal{C}'. \quad (7.2)$$

If $\lambda_C > 0$ is its least positive eigenvalue,

$$\|X - P_{\mathcal{C}'} X\|_{\text{HS}}^2 \leq \lambda_C^{-1} \sum_j \|[c_j, X]\|_{\text{HS}}^2. \quad (7.3)$$

For a proposed protected algebra $\mathcal{M} \subseteq \mathcal{C}'$ and an orthonormal basis (E_a) of $\mathcal{M}^\perp$, define the relative Howe Gram

$$\mathbb{G}_{\text{Howe}}(\mathbf{c} \mid \mathcal{M})_{ab} = \sum_j \langle [c_j, E_a], [c_j, E_b] \rangle_{\text{HS}}. \quad (7.4)$$

Then

$$\mathbb{G}_{\text{Howe}}(\mathbf{c} \mid \mathcal{M}) \succ 0 \iff \mathcal{C}' = \mathcal{M}. \quad (7.5)$$

On this branch its least eigenvalue, when the complement is nonzero, is the first positive eigenvalue of $\mathcal{L}_C$.

Proof. The Hilbert–Schmidt adjoint of ad_c is ad_c^* , so the quadratic form is $\sum_j \|[c_j, X]\|_{\text{HS}}^2$. Its zero set is exactly the simultaneous commutant. The spectral theorem gives coercivity on the orthogonal complement of that kernel. The relative Howe Gram is precisely the compression to $\mathcal{M}^\perp$; since $\mathcal{M}$ is already in the kernel, strict positivity there is equivalent to absence of every additional commutant direction. $\square$

For typed incidences, the type-central part of this certificate is a weighted graph Laplacian. Its connected-component kernel and its Schur separation from noncentral flavour directions are proved in Appendix F.1. The full test remains an operator-algebraic one, not a dimensional match.

7.2 Enhancement loci and finite perturbations

The positive certificate also determines the symmetry-enhancement locus of a parameter family. Let $\Theta \subset \mathbb{R}^p$ be an open parameter domain, keep a fixed represented protected algebra $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H})$, and assume $[c_j(\theta), \mathcal{M}] = 0$ for every j and θ . On $\mathcal{K} := \mathcal{M}^\perp$ define the stacked commutator map

$$D(\theta)X = ([c_1(\theta), X], \dots, [c_s(\theta), X]), \quad G(\theta) = D(\theta)^* D(\theta).$$

In an orthonormal basis of $\mathcal{K}$, $G(\theta)$ is the relative Howe Gram. Put $r = \dim_{\mathbb{C}} \mathcal{K}$ and

$$\Delta_{\mathcal{M}}(\theta) = \det G(\theta). \quad (7.6)$$

Proposition 7.2 (Howe enhancement discriminant and generic rigidity). *Assume every matrix entry of every $c_j(\theta)$ is a complex polynomial in the real coordinates θ , of degree at most d . Then $\Delta_{\mathcal{M}}$ is a nonnegative real polynomial of degree at most $2dr$ and admits the Cauchy–Binet representation*

$$\Delta_{\mathcal{M}}(\theta) = \sum_{|I|=r} |\det D_I(\theta)|^2. \quad (7.7)$$

Moreover,

$$\{\theta : \Delta_{\mathcal{M}}(\theta) = 0\} = \{\theta : C^*(\mathbf{c}(\theta))' \supsetneq \mathcal{M}\}. \quad (7.8)$$

If the proposed commutant is exact at one parameter value, then $\Delta_{\mathcal{M}}$ is not the zero polynomial. Hence its enhancement locus has empty interior and Lebesgue measure zero, while exact duality holds on an open dense full-measure subset of every open admissible parameter region. The same open-dense conclusion holds for a real-analytic family on each connected analytic component containing one exact point.

Proof. The entries of $D(\theta)$ are polynomial of degree at most d . Cauchy–Binet applied to D^*D gives (7.7); each squared modulus is a sum of squares of real polynomials, proving nonnegativity and the degree bound. Since $G \succeq 0$, its determinant is positive exactly when $G \succ 0$, which is equivalent to $C^*(\mathbf{c}(\theta))' = \mathcal{M}$ by Theorem 7.1. A nonzero real polynomial has a zero set of Lebesgue measure zero and empty interior; continuity makes the complement open. For a connected real-analytic family, the identity theorem gives the corresponding empty-interior conclusion once the determinant is nonzero at one point. $\square$

Corollary 7.3 (Determinantal nullity filtration). *For $1 \leq k \leq r$, the locus on which the commutant acquires at least k additional complex dimensions is the common zero set of all $(r-k+1) \times (r-k+1)$ minors of $D(\theta)$. Additional commutant dimension is therefore upper semicontinuous in every continuous family.*

The genericity is relative to a specified admissible family with a protected algebra; it does not imply that a microscopic source realizes that family. A finite quantitative statement is available around any exact anchor.

Proposition 7.4 (Finite-anchor stability and generator-scale control). *Transport a family of tuples $\mathbf{c}_X = (c_{1,X}, \dots, c_{s,X})$ to one common finite screened carrier, and suppose a fixed algebra $\mathcal{M}$ commutes with every tuple. Set*

$$C_X = \left(\sum_j \|c_{j,X}\|_{\text{op}}^2 \right)^{1/2}, \quad \varepsilon_{X,Y} = \left(\sum_j \|c_{j,X} - c_{j,Y}\|_{\text{op}}^2 \right)^{1/2}.$$

Assume $\mathbb{G}_{\text{Howe}}(\mathbf{c}_{X_0} \mid \mathcal{M}) \succeq \gamma_0 I$, $\gamma_0 > 0$. At every later X for which

$$4(C_X + C_{X_0})\varepsilon_{X,X_0} < \gamma_0, \tag{7.9}$$

one has

$$C^*(\mathbf{c}_X)' = \mathcal{M}, \quad \lambda_{\min}^+(\mathcal{L}_X) \geq \gamma_0 - 4(C_X + C_{X_0})\varepsilon_{X,X_0}. \tag{7.10}$$

Alternatively, the condition $2\varepsilon_{X,X_0} < \sqrt{\gamma_0}$ gives

$$C^*(\mathbf{c}_X)' = \mathcal{M}, \quad \lambda_{\min}^+(\mathcal{L}_X) \geq (\sqrt{\gamma_0} - 2\varepsilon_{X,X_0})^2, \tag{7.11}$$

without an upper bound on the full generator norm. If the corresponding condition holds for every later X , the conclusion holds throughout that tail.

Proof. For the stacked commutator maps, $\|\partial_{\mathbf{c}_X} - \partial_{\mathbf{c}_Y}\|_{2 \rightarrow 2} \leq 2\varepsilon_{X,Y}$ and $\|\partial_{\mathbf{c}_X}\|_{2 \rightarrow 2} \leq 2C_X$. Thus $\|\mathcal{L}_X - \mathcal{L}_Y\| \leq 4(C_X + C_Y)\varepsilon_{X,Y}$, giving the first bound on $\mathcal{M}^\perp$. Directly on that complement, the triangle inequality gives $\|\partial_{\mathbf{c}_X} A\| \geq (\sqrt{\gamma_0} - 2\varepsilon_{X,X_0})\|A\|$, giving the second. The protected algebra stays in the kernel, so either positive lower bound makes that kernel exact. $\square$

Coefficient-space and singular-support stability require a separate singular-value gap. Appendix F.2 retains the operator–Schmidt cluster estimate and the one-sided rank certificate, and Appendix A.3 gives support leakage. In particular, small measured tails do not certify exact absence without an independent upper-rank premise.

7.3 Cofinal kernel exactness

A cofinal family need not remain in a norm neighbourhood on a fixed carrier. We therefore formulate a second stability mechanism on transported screened Hilbert spaces. Kernel exactness will use Mosco convergence, finite protected transport, and a uniform complement gap; compactness will be added only for the stronger spectral conclusion.

At a finite cutoff the screened operator space is the full Hilbert–Schmidt space of the finite physical screen on which the displayed generators act. If instead a proper invariant operator test space is used, all kernel statements are relative to that space and do not identify an unseen commutant on a larger carrier.

Let $\mathcal{H}$ be a separable Hilbert space and let $J_X : \mathcal{H}_X \rightarrow \mathcal{H}$ be isometric embeddings of the screened Hilbert–Schmidt operator spaces, with $P_X = J_X J_X^*$ the corresponding orthogonal projections. The limiting space is embedded by J_∞ , and we assume the transport is compatible in the sense that $P_X \rightarrow P_\infty$ strongly on $\mathcal{H}$. For a densely defined closed nonnegative form q_X on $\mathcal{H}_X$, denote its self-adjoint operator by $\mathcal{L}_X$ and its transported resolvent by

$$R_X := J_X(\mathcal{L}_X + I)^{-1} J_X^* \in \mathcal{B}(\mathcal{H}). \quad (7.12)$$

Definition 7.5 (Transported Mosco convergence). The forms q_X converge to q_∞ in the transported Mosco sense when:

- (M1) whenever $J_X A_X \rightarrow B$ weakly in $\mathcal{H}$, the strong convergence $P_X \rightarrow P_\infty$ gives $B \in J_\infty \mathcal{H}_\infty$; writing $B = J_\infty A$,

$$q_\infty(A) \leq \liminf_X q_X(A_X);$$

- (M2) for every $A \in D(q_\infty)$ there are $A_X \in D(q_X)$ such that $J_X A_X \rightarrow J_\infty A$ strongly and

$$q_X(A_X) \rightarrow q_\infty(A).$$

Lemma 7.6 (Finite protected basis transport). *Let $\mathcal{M}_X \subseteq \mathcal{H}_X$ and $\mathcal{M}_\infty \subseteq \mathcal{H}_\infty$ have the same finite complex dimension m . Choose Hilbert–Schmidt orthonormal bases $e_{1,X}, \dots, e_{m,X}$ and $e_{1,\infty}, \dots, e_{m,\infty}$ and suppose*

$$\eta_X := \max_a \|J_X e_{a,X} - J_\infty e_{a,\infty}\| \rightarrow 0. \quad (7.13)$$

Then the transported orthogonal projections $\hat{P}_X = J_X P_{\mathcal{M}_X} J_X^$ satisfy*

$$\|\hat{P}_X - \hat{P}_\infty\|_{\text{op}} \leq 2m\eta_X \rightarrow 0. \quad (7.14)$$

If the $\mathcal{M}_X$ are images of one fixed finite C^ -algebra under specified unital $*$ -isomorphisms, the same data also identify multiplication and adjoints across cutoffs.*

Proof. Write the transported projections as sums of rank-one projections onto the transported orthonormal basis vectors. For unit vectors u, v , $\| |u\rangle\langle u| - |v\rangle\langle v| \|_{\text{op}} \leq 2\|u - v\|$. Summing over m basis vectors proves the estimate. The final assertion is immediate from the specified $*$ -isomorphisms. $\square$

Theorem 7.7 (Derived-kernel cofinal action–multiplicity duality). *Let*

$$q_X(A) = \sum_j \| [c_{j,X}, A] \|_{\text{HS}}^2 \quad (7.15)$$

be the closed commutator forms of a cofinal family of screened support-polar or multiplicity-quiver generators, with associated commutant Laplacians $\mathcal{L}_X$. Assume:

(D1) $q_X \rightarrow q_\infty$ in transported Mosco sense;

(D2) for every X there is a protected multiplicity algebra $\mathcal{M}_X \subseteq \text{Ker } \mathcal{L}_X$ of the same finite complex dimension m , together with a specified m -dimensional $\mathcal{M}_\infty$ satisfying the basis transport condition (7.13);

(D3) for all sufficiently late cutoffs there is one constant $\gamma_* > 0$ such that

$$q_X(A) \geq \gamma_* \|(I - P_{\mathcal{M}_X})A\|_{\text{HS}}^2 \quad (A \in D(q_X)). \quad (7.16)$$

Then for every sufficiently late cutoff

$$\text{Ker } \mathcal{L}_X = \mathcal{M}_X, \quad C^*(\mathbf{c}_X)' = \mathcal{M}_X, \quad (7.17)$$

where the commutant identity is understood on the complete represented Hilbert–Schmidt test space. Moreover the limiting form satisfies

$$q_\infty(A) \geq \gamma_* \|(I - P_{\mathcal{M}_\infty})A\|_{\text{HS}}^2, \quad (7.18)$$

and therefore

$$\text{Ker } \mathcal{L}_\infty = \mathcal{M}_\infty. \quad (7.19)$$

Thus limiting kernel exactness is derived from finite protected transport and a uniform late-cutoff gap; it is not an independent cofinal hypothesis. Neither asymptotic form compactness nor compact resolvent is required for these kernel statements.

Proof. At a finite cutoff, $A \in \text{Ker } \mathcal{L}_X$ gives $q_X(A) = 0$; the gap (7.16) forces $A \in \mathcal{M}_X$, and the reverse inclusion is assumed. The commutant identity is then Theorem 7.1.

Lemma 7.6 gives norm convergence of the transported protected projections. For every transported basis vector, $e_{a,X} \in \mathcal{M}_X \subseteq \text{Ker } \mathcal{L}_X$ gives $q_X(e_{a,X}) = 0$; the Mosco liminf inequality and the strong convergence in (7.13) therefore imply $q_\infty(e_{a,\infty}) = 0$. Hence $\mathcal{M}_\infty \subseteq \text{Ker } \mathcal{L}_\infty$.

For $A \in D(q_\infty)$ choose a Mosco recovery sequence A_X with $J_X A_X \rightarrow J_\infty A$ strongly and $q_X(A_X) \rightarrow q_\infty(A)$. Projection convergence implies

$$J_X P_{\mathcal{M}_X} A_X \longrightarrow J_\infty P_{\mathcal{M}_\infty} A,$$

and hence

$$\|(I - P_{\mathcal{M}_X})A_X\| \longrightarrow \|(I - P_{\mathcal{M}_\infty})A\|.$$

Passing to the limit in (7.16) proves (7.18). Its zero set is exactly $\mathcal{M}_\infty$, proving (7.19). $\square$

7.4 Compact spectral upgrade

Definition 7.8 (Asymptotic form compactness). The family (q_X) is asymptotically compact if every sequence $A_X \in D(q_X)$ satisfying

$$\sup_X (\|A_X\|^2 + q_X(A_X)) < \infty \quad (7.20)$$

has a subsequence for which $J_X A_X$ converges strongly in $\mathcal{H}$. The same requirement is imposed on the limiting form. In particular, $(\mathcal{L}_\infty + I)^{-1}$ is compact on the limiting screened space.

Corollary 7.9 (Compact spectral upgrade). *Assume Theorem 7.7 and, in addition, asymptotic form compactness in the sense of Definition 7.8. If the limiting screened space has a nonzero complement to $\mathcal{M}_\infty$, then*

$$\gamma_\infty := \lambda_{m+1}(\mathcal{L}_\infty) \geq \gamma_* > 0, \quad (7.21)$$

the transported resolvents converge in operator norm,

$$\|R_X - R_\infty\|_{\text{op}} \longrightarrow 0, \quad (7.22)$$

and

$$\lambda_{m+1}(\mathcal{L}_X) \longrightarrow \gamma_\infty, \quad \|\Pi_X - \Pi_\infty\|_{\text{op}} \longrightarrow 0. \quad (7.23)$$

Here Π_X is the transported kernel projection. Consequently, for every $0 < \gamma_0 < \gamma_\infty$ and all sufficiently late X ,

$$q_X(A) \geq \gamma_0 \|(I - P_{\mathcal{M}_X})A\|_{\text{HS}}^2. \quad (7.24)$$

Proof. Theorem 7.7 supplies exact limiting kernel dimension m and the positive lower bound γ_* . Proposition F.5 gives norm-resolvent convergence. Compact self-adjoint spectral convergence then gives convergence of the first positive eigenvalue and of the isolated zero-cluster projection. The final estimate follows from the spectral theorem. $\square$

Appendix F.3 proves the compact-Mosco resolvent mechanism used in the corollary, including the varying-space bookkeeping [59]. It also records the distinct spectral locking criterion that assumes an exact limiting kernel, and an escaping-mode counterexample showing why protected transport cannot be replaced by a dimension count.

The two stability mechanisms address different hypotheses. The finite anchor controls a norm perturbation of generators; the cofinal theorem controls varying domains through forms. Neither constructs an interacting continuum field theory. Identifying the protected spaces as one fixed abstract C^* -algebra additionally requires multiplicatively compatible transports, for example specified unital $*$ -isomorphisms. A particular spectral screen construction may supply the analytic hypotheses, but is not itself a premise of the theorem [47].

8 Interpretation and discussion

The main result of this paper is a finite representation-theoretic statement about the relation between what acts on a matter carrier and what remains invisible to that action. Once an external Clifford action, a typed internal action, and physical matter-incidence operators are represented on a common finite carrier, the residual matter multiplicity is no longer merely a count of repeated irreducible representations. It becomes an intrinsic algebraic object: the commutant of the complete represented action. Conversely, finite bicommutant closure reconstructs that action from the multiplicity algebra.

This observation separates two levels that are often conflated. The double-centralizer theorem is general at the level of finite joint packets and does not single out any particular gauge theory. The Standard Model appears only after additional structural conditions are imposed on a particular finite branch. Likewise, the predictive construction discussed below gives one origin for the required joint packet, but is not part of the algebraic proof. The discussion is organized according to these three levels: the general action–multiplicity duality, its Standard-Model specialization, and the relation between the abstract finite result and possible microscopic or continuum realizations.

8.1 What the duality determines

The complete represented action and its residual multiplicity have the same central decomposition and act on opposite matrix factors within each sector. Incidence is the information that restricts the otherwise large commutant of the external–typing action. The reconstruction is exact even at singular incidence maps: support-polar data retain the support boundary, and quiver and root metric–holonomy descriptions give alternative presentations of the same algebra.

The commutant does not recover a preferred physical generator bank or its numerical markings. Incidence minimality is therefore relative to the specified bank: the rank polymatroid identifies sufficient subsets and which deletions enlarge the commutant. Likewise, stability concerns a protected algebra within an admitted family. Its hypotheses do not provide occurrence of the generators whose commutant is being studied.

8.2 The structural branch and generation multiplicity

The Standard-Model specialization has three distinct algebraic stages. The tetrahedral multiplicity census and actual internal words determine a complex seed algebra; its represented determinant structure or its anomaly-free chiral tensor action determines the physical global group; that group’s matter representation generates the kinematic typing algebra on the common Clifford-compatible carrier. The external tetrahedral irreducible spaces are not identified with colour and weak multiplicity spaces, and the seed is not the commutant of the full matter-incidence action.

This also distinguishes the construction from the usual almost-commutative spectral geometry, where spacetime spin geometry and a finite internal geometry enter a spectral triple and its spectral action [6, 7, 8, 9, 13]. The comparison is between resulting structures, not an identification of the underlying finite geometries. In particular, the standard almost-commutative construction reproduces the one-generation representation structure while retaining the number of generations as an input [9]; the endpoint-allocation argument here addresses that residual multiplicity on the stated structural branch.

The determinant and anomaly routes agree on the global quotient but make different source assumptions. A determinant ray is phase blind; a fixed tensor or a nonzero neutral–incidence cross block imposes its stabilizer. The anomaly route instead assumes the displayed chiral tensor packet and primitive charge normalization. The kernel of its matter representation then determines the faithful quotient. Neither route infers a missing matter representation from the seed algebra alone.

For generations, endpoint rank, charged replication, and frame synchronization are separate conclusions. A positive rank-three endpoint residual must first occur faithfully in a complete charged multiplicity source. Its allocation short counts the excess beyond that image, while the inventory and anomaly tests control all charged types. Zero excess then proves three complete charged copies. Equal multiplicities already give an abstract common factor; common ancestry additionally supplies a physical relative frame. Without a cross-component datum, the quark–lepton relative frame is a presentation freedom, not an extra observable. The mechanism is therefore different from family triplication obtained from prescribed octonionic, Clifford, exceptional, extended-NCG, categorical, or discrete-flavour structures [33, 34, 35, 36, 37, 54]. None of these statements fixes numerical masses, mixing parameters, or couplings.

8.3 Source identification and continuum scope

The operational inverse problem precedes the commutant problem. A classical record may leave the acted-on algebra undetermined, whereas an appropriate operator-valued complementary pullback reconstructs it. Historical word moments determine mixed source data only when the relevant

operations are already represented and word native. The module-recognition criterion prevents invariant multiplicities of a regular word space from being mistaken for executable internal factors on the physical carrier. Similarly, unread spectator dynamics cannot supply active matter occurrence.

Renewal Geometry provides one upstream route through its predictive carrier, history algebra, and word-Gram hierarchy [46, 47]. The finite theorems remain applicable to unrelated realizations that supply the same represented input. The common-source Schur quotient in Appendix E.3 separates internal variation directions from an already represented gravitational bank, but it is not a variational action. Nonlinear matter stress convergence, control of physical variations and constraints, and classical or quantum continuum closure are additional analytic problems.

Scope and outlook. The results established here are finite and structural. The double-centralizer theorem applies to general source-complete joint packets, whereas the Standard-Model classification is obtained only on the tetrahedrally equivariant branch considered above. The finite criteria identify multiplicity birth, incidence connectivity, determinant structure, generation support, and rigidity of the resulting commutant, but they do not imply that a broad class of microscopic dynamics must realize these conditions. Nor do they determine gauge couplings, Higgs parameters, Yukawa eigenvalues, or renormalization-group data.

A separate analytic problem is required to pass from the stabilized finite gravity–matter structure to a coupled continuum theory. The cofinal commutant theorem controls the protected multiplicity sector under refinement, but does not itself supply the nonlinear compactness or measure-theoretic control required for a classical or quantum continuum limit. Renewal Geometry [46, 47] provides one predictive upstream realization of the finite data considered here, while the action–multiplicity duality remains independent of that realization.

The central conclusion is therefore not that gauge symmetry is spacetime in disguise, nor that the Standard Model is the commutant of a bare four-dimensional Clifford algebra. Rather, once spacetime action, typed internal action, and physical matter incidence are represented on a common finite carrier, the complete visible action and the matter multiplicity that it cannot resolve become mutually determining operator-algebraic structures. The structural Standard-Model branch gives one distinguished realization of this general dual pair.

Data and formalization availability

Selected finite algebraic components of the construction have been formalized in `Lean`. Source code is available at <https://github.com/AI-MathPhys/renewal-geometry>.

A Finite operator algebras, support-polar forms, and incidence minimality

This appendix supplies the coordinate, support, and deletion details for Section 3. The finite bicommutant input is the reciprocal matrix decomposition in Corollary 3.5: taking the commutant twice returns the original unital finite-dimensional C^* -algebra.

A.1 Coordinates, order reversal, and typing refinement

Corollary A.1 (Canonicity under Wedderburn coordinates). *Fix the represented pair $(\mathcal{A}_{\text{Cl}}, \mathcal{A}_{\text{type}})$ and let z_λ be the minimal central projections of $C^*(\mathcal{A}_{\text{Cl}}, \mathcal{A}_{\text{type}})$. Any two decompositions of the form*

$$z_\lambda \mathcal{H} \cong K_\lambda \otimes M_\lambda, \quad z_\lambda \mathcal{H} \cong \tilde{K}_\lambda \otimes \tilde{M}_\lambda$$

which realize the same represented external-typing algebra differ, after relabeling by the same central projections, by unitary multiplicity identifications $W_\lambda : M_\lambda \rightarrow \tilde{M}_\lambda$. For every represented incidence edge $e : \lambda \rightarrow \mu$,

$$\tilde{\mathfrak{B}}_e = W_\mu \mathfrak{B}_e W_\lambda^*,$$

and, with $W = \bigoplus_\lambda W_\lambda$,

$$\tilde{\mathcal{O}}_Y = W \mathcal{O}_Y W^*, \quad \tilde{\mathcal{M}}_{\text{type}} = W \mathcal{M}_{\text{type}} W^*. \quad (\text{A.1})$$

Thus the typed dual pair is independent of matrix units, carrier bases and Wedderburn coordinates. The physical typing algebra itself, however, remains represented input.

Proof. The projections z_λ are intrinsic minimal central projections of the represented algebra. On each central block the commutant is a full matrix algebra, and finite-dimensional uniqueness of its defining representation gives the multiplicity unitary W_λ . A change of carrier coordinates is absorbed by the full family of carrier linear functionals used in the definition of the slice space, while the multiplicity coordinates transform by left-right unitary transport. This gives the slice-space identity and hence the two generated-algebra identities. $\square$

Proposition A.2 (Order reversal and represented typing refinement). *For two incidence banks on the same typed multiplicity carrier,*

$$\mathcal{O}_Y \subseteq \mathcal{O}_{\tilde{Y}} \iff \mathcal{M}_{\text{type}}(\tilde{Y}) \subseteq \mathcal{M}_{\text{type}}(Y). \quad (\text{A.2})$$

Thus equality of the closed coefficient-action algebras is equivalent to equality of the exact multiplicity algebras. For a fixed external factor and incidence bank Y , let $\mathcal{A}_{\text{type}}^{(1)} \subseteq \mathcal{A}_{\text{type}}^{(2)} \subseteq \mathcal{A}'_{\text{Cl}}$, and put $\mathcal{A}_i = C^(\mathcal{A}_{\text{Cl}}, \mathcal{A}_{\text{type}}^{(i)}, Y, Y^*)$, $\mathcal{M}_i = \mathcal{A}'_i$. Then*

$$\mathcal{A}_1 \subseteq \mathcal{A}_2, \quad \mathcal{M}_2 \subseteq \mathcal{M}_1, \quad (\text{A.3})$$

and

$$\mathcal{M}_1 = \mathcal{M}_2 \iff \mathcal{A}_1 = \mathcal{A}_2 \iff \mathcal{A}_{\text{type}}^{(2)} \subseteq \mathcal{A}_1.$$

A typing refinement is therefore multiplicity-neutral exactly when its added operations already lie in the generated action.

Proof. Taking commutants reverses inclusions, and finite bicommutant closure reverses them back. The typing statement additionally uses $\mathcal{A}_2 = C^*(\mathcal{A}_1, \mathcal{A}_{\text{type}}^{(2)})$. $\square$

A.2 Coefficient banks and support refinements

Lemma A.3 (Multiplicity-one incidence factorization). *If the prescribed carrier coefficient space of an edge is a line $\mathbb{C}D_e$, with $D_e \neq 0$, then the edge has a unique factorization, relative to this choice of D_e ,*

$$Y_e = D_e \otimes F_e, \quad F_e : M_\lambda \rightarrow M_\mu. \quad (\text{A.4})$$

In particular this holds when a declared equivariance condition has a one-dimensional carrier intertwiner space and acts trivially on the multiplicity factors. On this branch the equations in (3.3) reduce to $R_\mu F_e = F_e R_\lambda$ and $R_\lambda F_e^ = F_e^* R_\mu$.*

Proof. The admitted edge space is $\mathbb{C}D_e \otimes \mathcal{B}(M_\lambda, M_\mu)$; contraction by a linear functional taking value one on D_e determines F_e . The general coefficient-space theorem then gives the two displayed equations. $\square$

Support refinements and canonical data. The intrinsic coefficient spaces, their adjoint-closed quiver, and $\mathcal{O}_Y$ are independent of coefficient bases. The support algebra of a chosen coefficient list can change when that list is linearly recombined. Its isotypic decomposition consequently gives a presentation of the same commutant, not an additional basis-independent physical decomposition. All such support-polar refinements recover the same $\mathcal{O}_Y$ because they generate exactly the original coefficient algebra. When a support bank is part of the represented data, its refined quiver is intrinsic to that bank up to the usual unitary multiplicity identifications.

A.3 Singular polar edges, leakage, and support certification

Let $F : E \rightarrow H$ have polar decomposition

$$F = UP, \quad P = (F^*F)^{1/2}, \quad p = U^*U = \text{supp } P, \quad q = UU^* = \text{supp}(FF^*).$$

Lemma A.4 (Singular polar-edge equivalence). *For $R_E \in \mathcal{B}(E)$ and $R_H \in \mathcal{B}(H)$, the equations*

$$R_H F = F R_E, \quad R_E F^* = F^* R_H$$

are equivalent to

$$[R_E, P^2] = 0, \quad R_H U = U R_E, \quad R_E U^* = U^* R_H. \quad (\text{A.5})$$

No invertibility assumption is required.

Proof. The two incidence equations imply $F^* R_H F = F^* F R_E = R_E F^* F$, hence $[R_E, P^2] = 0$. Functional calculus gives commutation with P and its support. Multiplication by the Moore–Penrose inverse of P on its support gives the partial-isometry transport equations. The converse follows by multiplying the transport equations by P and using $[R_E, P] = 0$. $\square$

Proposition A.5 (Exact singular polar-edge energy and support leakage). *For a nonzero coefficient map $F : E \rightarrow H$ with polar decomposition $F = UP$, support projections $p = U^*U$, $q = UU^*$, and $\mu = \lambda_{\min}(P|_{pE}) > 0$, define*

$$\mathcal{E}_F(R_E, R_H) = \|R_H F - F R_E\|_{\text{HS}}^2 + \|R_E F^* - F^* R_H\|_{\text{HS}}^2.$$

Put

$$A = p R_E p, \quad B = U^* q R_H q U, \quad S = \frac{A + B}{2}, \quad \Delta = B - A.$$

Then

$$\begin{aligned} \mathcal{E}_F(R_E, R_H) &= 2\|[S, P]\|_{\text{HS}}^2 + \frac{1}{2}\|\{\Delta, P\}\|_{\text{HS}}^2 \\ &\quad + \|P p R_E (1 - p)\|_{\text{HS}}^2 + \|(1 - p) R_E p P\|_{\text{HS}}^2 \\ &\quad + \|P U^* q R_H (1 - q)\|_{\text{HS}}^2 + \|(1 - q) R_H q U P\|_{\text{HS}}^2. \end{aligned} \quad (\text{A.6})$$

Consequently

$$\mathcal{E}_F(R_E, R_H) \geq \mu^2 (\|[p, R_E]\|_{\text{HS}}^2 + \|[q, R_H]\|_{\text{HS}}^2 + 2\|U^* q R_H q U - p R_E p\|_{\text{HS}}^2). \quad (\text{A.7})$$

The energy vanishes exactly when R_E and R_H reduce the source and target supports, their supported blocks are transported by U , and the supported source block commutes with P . The blocks on $\ker F$ and $\ker F^$ remain independent.*

Proof. Relative to $E = pE \oplus (1 - p)E$ and $H = qH \oplus (1 - q)H$, the map F has only the block $UP : pE \rightarrow qH$. Direct block multiplication gives the four support-mixing terms above and two supported terms which, after transport by U , are $BP - PA$ and $AP - PB$. Since

$$BP - PA = [S, P] + \frac{1}{2}\{\Delta, P\}, \quad AP - PB = [S, P] - \frac{1}{2}\{\Delta, P\},$$

the parallelogram identity gives (A.6). In an eigenbasis of $P|_{pE}$, $\{\Delta, P\}_{ij} = (\lambda_i + \lambda_j)\Delta_{ij}$, so its contribution is at least $2\mu^2\|\Delta\|_{\text{HS}}^2$. Each support-mixing term is bounded below by μ^2 times the corresponding bare block norm, yielding (A.7). Equality to zero forces all four mixing blocks to vanish; positivity of $P|_{pE}$ makes the anticommutator map invertible, so $\Delta = 0$, and the remaining term gives $[A, P] = 0$. $\square$

Proposition A.6 (Certified support stratum from a singular-value gap). *Let $F : E \rightarrow H$ be one represented coefficient map and let $\hat{F}$ satisfy $\|\hat{F} - F\|_{\text{op}} \leq \varepsilon$. Suppose an independent source or structural premise gives $\text{rank } F \leq r$ and the r th observed singular value obeys*

$$\hat{\sigma}_r > 2\varepsilon.$$

Then $\text{rank } F = r$. If

$$p = \text{supp}(F^*F), \quad q = \text{supp}(FF^*)$$

and $\hat{p}_r, \hat{q}_r$ are the right and left singular projectors of $\hat{F}$ for its r largest singular values, then

$$\|p - \hat{p}_r\|_{\text{op}}, \|q - \hat{q}_r\|_{\text{op}} \leq \frac{\varepsilon}{\hat{\sigma}_r - 2\varepsilon}. \quad (\text{A.8})$$

The same statement holds under the stronger-looking but often more directly available hypothesis $\|\hat{F} - F\|_{\text{HS}} \leq \varepsilon$.

Proof. Singular-value perturbation gives $\sigma_r(F) \geq \hat{\sigma}_r - \varepsilon > \varepsilon$, so the upper-rank premise forces rank exactly r . Put $s = \sigma_r(F)$. If $y \in \ker F$, then $\|\hat{F}y\| \leq \varepsilon\|y\|$. Orthogonality of the left-singular images associated with $\hat{p}_r$ and $I - \hat{p}_r$ gives

$$(s - \varepsilon)\|\hat{p}_r y\| \leq \|\hat{F}\hat{p}_r y\| \leq \|\hat{F}y\|.$$

Hence $\|\hat{p}_r(I - p)\| \leq \varepsilon/(s - \varepsilon)$. The projectors p and $\hat{p}_r$ have equal finite rank, so this is their operator-norm projector distance. Since $s - \varepsilon \geq \hat{\sigma}_r - 2\varepsilon$, the source-support bound follows; apply the same argument to F^* for the target support. Finally $\|X\|_{\text{op}} \leq \|X\|_{\text{HS}}$ gives the HS-error version. $\square$

Support certification is one-sided without a rank premise. An observed singular value above the error floor certifies a genuine support direction, but a small observed tail cannot certify exact absence. The upper rank/source premise in Proposition A.6 is therefore essential; it is the support-level analogue of the one-sided coefficient-rank firewall in Corollary F.4.

A.4 Noncommuting supports and the refined quiver

Support projections themselves can fail to commute. The correct residual description is then a finite operator-space quiver, in the standard representation-theoretic sense of a quiver with linear arrow spaces [60].

Definition A.7 (Multiplicity quiver). Let

$$\mathcal{S}_F = C^*(Z_\lambda, \text{supp}(F_e^* F_e), \text{supp}(F_e F_e^*) : \lambda, e)$$

and write its canonical finite isotypic decomposition as

$$\mathcal{M} \cong \bigoplus_{a \in \mathfrak{A}} V_a \otimes N_a, \quad \mathcal{S}_F = \bigoplus_a \mathcal{B}(V_a) \otimes I_{N_a}.$$

Let $\mathbf{T}$ be any finite $*$ -closed family of further reconstructed support metrics, polar arrows, cycle products, or action generators. For a block $T_{ba} : V_a \otimes N_a \rightarrow V_b \otimes N_b$, define

$$\mathfrak{B}_{ba}(\mathbf{T}) = \{(\varphi \otimes \text{id})(T_{ba}) : \varphi \in \mathcal{B}(V_a, V_b)^*\} \subseteq \mathcal{B}(N_a, N_b).$$

The multiplicity quiver $\mathfrak{Q}_F(\mathbf{T})$ has vertex spaces N_a and arrow spaces

$$\mathfrak{B}_{ba} = \text{span}_{T \in \mathbf{T}} \mathfrak{B}_{ba}(T).$$

For this fixed support bank its endomorphism algebra is

$$\text{End } \mathfrak{Q}_F = \{(X_a)_a : X_b B = B X_a \text{ for every } B \in \mathfrak{B}_{ba}\}.$$

Proposition A.8 (Noncommuting-support quiver theorem). *With the preceding notation,*

$$C^*(\mathcal{S}_F, \mathbf{T})' \cong \text{End } \mathfrak{Q}_F(\mathbf{T}). \quad (\text{A.9})$$

If

$$(T_j)_{ba} = \sum_{\alpha} E_{\alpha}^{ba} \otimes B_{j,\alpha}^{ba}$$

in Hilbert–Schmidt orthonormal bases of $\mathcal{B}(V_a, V_b)$, then the ambient commutator form restricts to

$$q_{\mathbf{T}}[(X_a)] = \sum_{j,b,a,\alpha} \|B_{j,\alpha}^{ba} X_a - X_b B_{j,\alpha}^{ba}\|_{\text{HS}}^2, \quad (\text{A.10})$$

whose kernel is exactly $\text{End } \mathfrak{Q}_F$.

Proof. An operator commuting with $\mathcal{S}_F$ has the form $\bigoplus_a I_{V_a} \otimes X_a$. Expanding its commutator with every T_j in the Hilbert–Schmidt basis gives precisely the quiver intertwining relations. Orthogonality gives (A.10). $\square$

For a fixed support algebra, changing Hilbert–Schmidt carrier bases unitarily recombines the coefficient families and leaves each arrow space unchanged. The refined quiver is consequently independent of those bases, up to multiplicity unitaries and block permutations. Replacing the support bank can change the refinement, but not the coefficient-action algebra or its commutant.

A.5 The connected invertible branch

When a chosen coefficient bank has invertible arrows, the duality has a particularly geometric form. For a multiplicity-one physical edge bank these are exactly its residue maps; general physical incidences may contribute several coefficient arrows.

Proposition A.9 (Polar metric–holonomy commutant). *Assume Γ is connected, all multiplicity spaces have common dimension g , and every edge map is invertible:*

$$F_e = U_e P_e, \quad U_e \in \mathbf{U}(g), \quad P_e \succ 0.$$

Choose a root o and a spanning tree. Let $Q_\lambda : M_o \rightarrow M_\lambda$ be the ordered product of the U_e along the tree path. For $e : \lambda \rightarrow \mu$, define on the root fibre

$$K_e = Q_\lambda^* P_e^2 Q_\lambda, \quad W_e = Q_\mu^* U_e Q_\lambda,$$

and put

$$\mathcal{O}_F = C^*(K_e, W_e : e \in E) \subseteq M_g(\mathbb{C}). \quad (\text{A.11})$$

Then

$$\mathcal{M}_{\text{type}}^{(o)} = \mathcal{O}'_F, \quad \mathcal{O}_F = (\mathcal{M}_{\text{type}}^{(o)})'. \quad (\text{A.12})$$

The multiplicity family associated with $R \in \mathcal{O}'_F$ is

$$R_\lambda = Q_\lambda R Q_\lambda^*.$$

Proof. The polar-edge equations give $[R_\lambda, P_e^2] = 0$ and $R_\mu = U_e R_\lambda U_e^*$. Along the spanning tree these force $R_\lambda = Q_\lambda R Q_\lambda^*$ from the root value $R = R_o$. The metric condition becomes $[R, K_e] = 0$, while a non-tree edge gives the cycle-consistency condition $[R, W_e] = 0$. Thus the residual family is equivalent to one root operator commuting with the algebra (A.11). Finite bicommutant closure gives the reverse equality. $\square$

Corollary A.10 (Flat and irreducible extremes). *Under Proposition A.9:*

- (i) $\mathcal{M}_{\text{type}}^{(o)} = M_g(\mathbb{C})$ if and only if every K_e and W_e is scalar;
- (ii) $\mathcal{M}_{\text{type}}^{(o)} = \mathbb{C}I$ if and only if $\mathcal{O}_F = M_g(\mathbb{C})$;
- (iii) if every F_e is unitary, then $K_e = I$ and the obstruction reduces to relative cycle holonomy;
- (iv) even on a tree, a non-scalar positive polar metric can reduce the residual multiplicity algebra.

A.6 Determinant supports and simultaneous deletion

The rank theorem in Section 3.4 has a determinant refinement useful for finite audits. Keep the notation there and introduce nonnegative incidence weights $w = (w_e)_{e \in E}$:

$$G(w) = \sum_{e \in E} w_e G_e, \quad P_E(w) = \det G(w). \quad (\text{A.13})$$

Proposition A.11 (Positive incidence determinant and sufficient supports). *The polynomial P_E is homogeneous of degree n and has nonnegative real coefficients. For any $S \subseteq E$, if $w_e > 0$ on S and $w_e = 0$ off S , then*

$$P_E(w) > 0 \iff f(S) = n \iff \mathcal{M}(S) = \mathcal{M}(E). \quad (\text{A.14})$$

If $\kappa_e = n - f(E \setminus \{e\})$, then the smallest power of w_e occurring with nonzero coefficient in P_E is exactly κ_e . Thus e is essential if and only if w_e divides P_E .

Proof. Stack the row blocks $\sqrt{w_e}D_e$. Cauchy–Binet expresses $\det G(w)$ as a sum of squared absolute values of $n \times n$ row minors, each multiplied by the monomial recording how many rows were selected from each edge block. All coefficients are therefore nonnegative. A point on a coordinate face has positive determinant exactly when the row spaces of the retained blocks span V , which is $f(S) = n$. For the last assertion, any row basis uses at most $f(E \setminus \{e\})$ rows from the other blocks and hence at least κ_e rows from block e ; extending a basis of the other-block span using rows from e attains this number. $\square$

For quantitative deletion, put $G = G_E \succ 0$ and

$$B_e = G^{-1/2}G_eG^{-1/2}, \quad \sum_e B_e = I, \quad B_R = \sum_{e \in R} B_e.$$

Proposition A.12 (Exact simultaneous-deletion margin). *For every $R \subseteq E$,*

$$G_{E \setminus R} = G^{1/2}(I - B_R)G^{1/2}, \tag{A.15}$$

and therefore

$$\mathcal{M}(E \setminus R) = \mathcal{M}(E) \iff \|B_R\|_{\text{op}} < 1. \tag{A.16}$$

If $\beta_R = \|B_R\|_{\text{op}} < 1$, then

$$G_{E \setminus R} \succeq (1 - \beta_R)G, \quad \lambda_{\min}(G_{E \setminus R}) \geq (1 - \beta_R)\lambda_{\min}(G).$$

Proof. The factorization follows from the definitions. Since $0 \preceq B_R \preceq I$, $I - B_R$ is positive definite exactly when the largest eigenvalue of B_R is smaller than one. Congruence by $G^{1/2}$ preserves nullity and positivity, giving both the commutant criterion and the lower bound. $\square$

Example A.13 (Minimal incidence skeletons need not form matroid bases). *On $\mathbb{C}^3$ with scalar base algebra let*

$$Y_a = E_{12} + E_{23}, \quad Y_b = E_{12}, \quad Y_c = E_{23}.$$

The singleton $\{a\}$ and the pair $\{b, c\}$ both generate $M_3(\mathbb{C})$ after adjoining adjoints, and hence both have scalar commutant. Each is inclusion-minimal: neither Y_b nor Y_c alone has scalar commutant. Thus minimal sufficient incidence sets can have unequal cardinality, so the incidence labels do not carry a matroid-basis structure in general.

B Source identification and historical compilation

The finite duality presupposes represented operators on the acted-on carrier. This appendix distinguishes that input from coarser record data, gives a coherent reconstruction route, and identifies historical operations that can be evaluated through existing word moments.

B.1 System-algebra identification

The source-complete packet is a represented system object. It cannot in general be reconstructed from a classical outcome record or from the algebra of an auxiliary environment alone. The following finite statement separates what such coarse data determine from what is needed for the duality.

Proposition B.1 (Classical record data do not determine the system algebra). *There exist two finite normalized instruments on the same input and output Hilbert spaces with the same resolved classical record channel at every word length, the same one-step effects, the same one-letter Hilbert–Schmidt Gram, and the same nonzero coarse Choi spectrum, but with nonisomorphic represented system coefficient algebras and commutants. Consequently no theorem using only those record/environment invariants can identify the joint action algebra of Definition 2.2.*

Proof. The explicit pair in Proposition B.5 supplies the witness. One coefficient family is simultaneously diagonal and generates $\mathbb{C}^{20}$; the other generates $M_4(\mathbb{C}) \otimes I_5$. Each resolved branch is a common scalar multiple of a unitary, so every uninterrupted classical word has the same effect in the two realizations, while the coefficient algebras and their commutants are different. The displayed Gram and Choi identities in Proposition B.5 complete the finite witness. $\square$

We use the standard Stinespring, Kraus, and Choi representations of completely positive maps throughout this reconstruction [61, 62, 63]. Let $V : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{E}$ be an isometry and choose an orthonormal environment basis $(|\nu\rangle)$. Write

$$V = \sum_{\nu} C_{\nu} \otimes |\nu\rangle, \quad C(z) = (I \otimes \langle z|)V,$$

and let $\mathcal{P}_*$ be the complementary channel. Its Heisenberg pullback obeys

$$\mathcal{P}^*(|z\rangle\langle w|) = C(z)^*C(w). \quad (\text{B.1})$$

Proposition B.2 (Identity-pivot reconstruction of the system action). *Suppose the occurring coefficient algebra satisfies $\mathcal{A}_{\text{sys}} = C^*(C_{\nu} : \nu) = C^*(A_1, \dots, A_s)$ and there are environment vectors $z_0, z_1, \dots, z_s$ with*

$$C(z_0) = \alpha I, \quad C(z_j) = \beta_j A_j, \quad \alpha\beta_j \neq 0.$$

Then

$$C^*(\mathcal{P}^*(\mathcal{B}(\mathcal{E}))) = \mathcal{A}_{\text{sys}}, \quad [\mathcal{P}^*(\mathcal{B}(\mathcal{E}))]' = \mathcal{A}'_{\text{sys}}. \quad (\text{B.2})$$

In particular, a complete operator-valued complementary pullback recovers the represented system action whenever the physical source contains such an identity pivot; no fitted remainder algebra is required.

Proof. Every pullback $C(z)^*C(w)$ belongs to $C^*(C_{\nu})$ and hence to $\mathcal{A}_{\text{sys}}$. Conversely,

$$\mathcal{P}^*(|z_0\rangle\langle z_j|) = \bar{\alpha}\beta_j A_j,$$

so every generator A_j lies in the pullback algebra. Equality of the commutants follows immediately. $\square$

Corollary B.3 (Finite coherent system export). *For the normalized six-edge source class of Appendix B, ten fixed Hermitian environment quadratures suffice to reconstruct all six edge coefficients on the acted-on carrier. With Hilbert–Schmidt coefficient error e_K as defined in (B.13), the associated star-closed commutator map changes in operator norm by at most $2\sqrt{2}e_K$. Hence an independently verified protected commutant complement with measured least singular value $\hat{s} > 2\sqrt{2}e_K$ has no additional commutant direction.*

Logical role of source identification. Proposition B.1 is a stopping theorem, not a weakening of the double-centralizer result. Once the represented system operators are supplied or reconstructed, Theorem 3.3 is exact. What cannot be inferred from coarse record data is which system operators occurred on the common carrier in the first place.

B.2 A finite nonidentifiability pair and coherent reconstruction

Let H_E be a finite edge-system space, let Q_a ($1 \leq a \leq 4$) be nonzero orthogonal projections on a record factor H_R with $\sum_a Q_a = I$, and let $K_1, \dots, K_6 \in \mathcal{B}(H_E)$ satisfy

$$\sum_{e=1}^6 K_e^* K_e = I, \quad \sum_{e=1}^6 K_e = \sqrt{2} I, \quad K_1, \dots, K_6 \text{ linearly independent.} \quad (\text{B.3})$$

For $0 < \vartheta < 1$ set

$$A_{e,a} = K_e \otimes Q_a, \quad L_0 = \sqrt{1 - \vartheta} I, \quad L_{e,a} = \sqrt{\vartheta} A_{e,a}.$$

These operators define a normalized resolved instrument.

Choose a real 6×5 matrix O with

$$O^\top O = I_5, \quad O^\top \mathbf{1} = 0, \quad O O^\top = I_6 - \frac{1}{6} \mathbf{1} \mathbf{1}^\top. \quad (\text{B.4})$$

Proposition B.4 (Five-Kraus freedom behind the six-edge normalization). *Equation (B.3) holds if and only if there are five matrices $B_1, \dots, B_5$ with $\sum_\mu B_\mu^* B_\mu = I$ and $I, B_1, \dots, B_5$ linearly independent such that*

$$K_e = \frac{1}{\sqrt{18}} I + \sqrt{\frac{2}{3}} \sum_{\mu=1}^5 O_{e\mu} B_\mu. \quad (\text{B.5})$$

Consequently the coarse edge channel has the form

$$\mathcal{E}_* = \frac{1}{3} \text{id} + \frac{2}{3} \Psi_*, \quad \Psi_*(\rho) = \sum_\mu B_\mu \rho B_\mu^*,$$

and $C^*(K_e : e) = C^*(B_\mu : \mu)$.

Proof. Apply the orthogonal coefficient change whose first column is $\mathbf{1}/\sqrt{6}$ and whose remaining columns are O . The coherent-sum constraint fixes the first transformed coefficient to $I/\sqrt{3}$; the other five, $D_\mu = \sum_e O_{e\mu} K_e$, satisfy $\sum_\mu D_\mu^* D_\mu = 2I/3$. Put $B_\mu = \sqrt{3/2} D_\mu$ and invert the orthogonal change. $\square$

B.2.1 An exact nonidentifiability pair

For this witness take $H_R = \mathbb{C}^4$ and $Q_a = |a\rangle\langle a|$. Let $\mathcal{S}$ be the twenty three-element subsets of $\{1, \dots, 6\}$. On $\mathbb{C}^{\mathcal{S}}$ define commuting Hermitian involutions

$$H_e^c |S\rangle = (2\mathbf{1}_{e \in S} - 1) |S\rangle.$$

Let $\Gamma_1, \dots, \Gamma_5$ be five anticommuting Hermitian Clifford matrices on $\mathbb{C}^4$, and on $\mathbb{C}^4 \otimes \mathbb{C}^5$ set

$$H_e^q = \sqrt{\frac{6}{5}} \sum_{\mu=1}^5 O_{e\mu} \Gamma_\mu \otimes I_5.$$

For $b \in \{c, q\}$ put

$$K_e^b = \frac{1}{\sqrt{18}} I_{20} + \frac{i}{3} H_e^b. \quad (\text{B.6})$$

Proposition B.5 (Record-complete nonidentifiability witness). *Both lists in (B.6) satisfy (B.3), and*

$$(K_e^b)^* K_e^b = I_{20}/6, \quad \text{Tr}[(K_e^b)^* K_f^b] = \begin{cases} 10/3, & e = f, \\ 2/3, & e \neq f. \end{cases} \quad (\text{B.7})$$

Yet

$$C^*(K_e^c) = \mathbb{C}^{20}, \quad C^*(K_e^q) = M_4(\mathbb{C}) \otimes I_5. \quad (\text{B.8})$$

For the resolved instrument above, every uninterrupted word in the original classical alphabet has the same effect in the two realizations. A word with $m \geq 1$ accepted letters, n_0 no-response letters and one common accepted record type a has effect

$$(1 - \vartheta)^{n_0} (\vartheta/6)^m I_{20} \otimes Q_a; \quad (\text{B.9})$$

if accepted record types disagree the effect is zero. Moreover the two one-step opportunity channels have the same nonzero Choi spectrum,

$$\text{spec}_+(J_{\text{opp}}) = \left\{ \left[\frac{20}{3}(12 - 11\vartheta) \right]^{(1)}, \left(\frac{20\vartheta}{3} \right)^{(3)}, \left(\frac{8\vartheta}{3} \right)^{(20)} \right\}. \quad (\text{B.10})$$

Thus the complete classical record channel at every word length is identical although the system algebras differ.

Proof. For the commuting family, counting three-subsets gives zero mean and pair correlation $-1/5$. For the Clifford family the row vectors of $\sqrt{6/5}O$ are unit vectors with the same pair correlations, and Clifford anticommutation gives $(H_e^q)^2 = I$. Hence both families satisfy the same normalization and Gram relations, and each $\sqrt{6}K_e^b$ is unitary. The twenty sign patterns distinguish all basis vectors in the commuting realization, giving $\mathbb{C}^{20}$. The five Clifford generators span and generate $M_4(\mathbb{C}) \otimes I_5$. Finally, every classical word effect is $L_w^* L_w$; because each accepted system factor is $6^{-1/2}$ times a unitary and the record projectors are orthogonal, this gives (B.9) in both realizations.

For the one-step Choi spectrum, the Gram of the twenty-four accepted Choi vectors is $G_{24} = G_6 \otimes I_4$ with $G_6 = (8/3)I_6 + (2/3)\mathbf{11}^\top$. It has eigenvalue $20/3$ on the uniform edge direction with four record directions and $8/3$ on the twenty edge-contrast directions. The coefficient matrix $M_\vartheta = \vartheta I + (1 - \vartheta)cc^*$ acts by $12 - 11\vartheta$ on the global uniform direction and by ϑ on its orthogonal complement. The nonzero Choi eigenvalues are therefore those of $M_\vartheta^{1/2} G_{24} M_\vartheta^{1/2}$, giving (B.10). $\square$

B.2.2 Ten coherent pullbacks reconstruct the six coefficients

Enumerate the twenty-four accepted operators $A_{e,a}$ as A_j and let $\mathbf{B}$ synthesize their Choi vectors $|A_j\rangle\rangle$. In $\mathbb{C}^{24}$ put

$$c = \mathbf{1}/\sqrt{2}, \quad M_\vartheta = \vartheta I + (1 - \vartheta)cc^*.$$

Let C_ν be the matrices synthesized by the columns of $\mathbf{B}M_\vartheta^{1/2}$ and define the minimal Stinespring isometry

$$V_\vartheta = \sum_{\nu=1}^{24} C_\nu \otimes |\nu\rangle, \quad \mathcal{P}_{\vartheta,*}(\rho) = \text{Tr}_{\text{sys}}(V_\vartheta \rho V_\vartheta^*).$$

For $z \in \mathbb{C}^{24}$ write $C(z) = (I \otimes \langle z|)V_\vartheta$. Then $\mathcal{P}_\vartheta^*(|z\rangle\langle w|) = C(z)^* C(w)$.

For the twenty-four coefficient labels (e, a) define

$$u_{e,a} = 1/\sqrt{24}, \quad (r_\mu)_{e,a} = O_{e\mu}/2, \quad \langle u, r_\mu \rangle = 0,$$

and the fixed Hermitian environment quadratures

$$X_\mu = |u\rangle\langle r_\mu| + |r_\mu\rangle\langle u|, \quad Y_\mu = -i|u\rangle\langle r_\mu| + i|r_\mu\rangle\langle u|. \quad (\text{B.11})$$

Let $\mathcal{P}_\vartheta^*$ denote the Heisenberg complementary pullback of the normalized coefficient frame and write $\mathbf{X}_\mu = \mathcal{P}_\vartheta^*(X_\mu)$, $\mathbf{Y}_\mu = \mathcal{P}_\vartheta^*(Y_\mu)$. Set $h = 12 - 11\vartheta$ and $D_\mu = \sum_e O_{e\mu} K_e$.

Proposition B.6 (Explicit ten-panel reconstruction). *The ten Hermitian operator-valued pullbacks determine all six system coefficients through*

$$D_\mu \otimes I_{H_R} = \frac{2\sqrt{3}}{\sqrt{\vartheta h}}(\mathbf{X}_\mu + i\mathbf{Y}_\mu), \quad K_e = \frac{I}{\sqrt{18}} + \sum_\mu O_{e\mu} D_\mu. \quad (\text{B.12})$$

Thus this finite operator-valued export determines the represented system algebra and commutant.

Proof. In the normalized coefficient frame the uniform environment vector is an eigenvector of the coefficient Gram with eigenvalue h , while each r_μ has eigenvalue ϑ . Hence $C(u) = \sqrt{h/12} I$ and $C(r_\mu) = \sqrt{\vartheta} D_\mu \otimes I_{H_R}/2$. Since $X_\mu + iY_\mu = 2|u\rangle\langle r_\mu|$, Equation (B.1) gives the first reconstruction formula; orthogonality of O gives the second. $\square$

Proposition B.7 (Robust commutant stopping test). *If the ten pullbacks have Hilbert–Schmidt errors $\epsilon_{X,\mu}$ and $\epsilon_{Y,\mu}$, then*

$$\left(\sum_e \|\hat{K}_e - K_e\|_{\text{HS}}^2 \right)^{1/2} \leq e_K := \frac{2\sqrt{3}}{\sqrt{\vartheta h}} \left[\sum_\mu (\epsilon_{X,\mu} + \epsilon_{Y,\mu})^2 \right]^{1/2}. \quad (\text{B.13})$$

For the star-closed stacked commutator map,

$$\|\hat{\mathcal{D}} - \mathcal{D}\|_{2 \rightarrow 2} \leq 2\sqrt{2} e_K. \quad (\text{B.14})$$

Consequently a measured least singular value $\hat{s} > 2\sqrt{2} e_K$ on an independently verified protected complement certifies absence of any further commutant direction there.

Proof. Equation (B.12) and the triangle inequality give the D_μ errors. Orthogonality of O preserves the sum of squared coefficient errors. For unit Hilbert–Schmidt X , each coefficient error changes a commutator by at most twice its operator norm; retaining both a coefficient and its adjoint gives the factor $2\sqrt{2}$. The last statement is the standard least-singular-value perturbation inequality. $\square$

The ten-panel theorem is a sufficient finite tomography statement, not an assertion that these coherent quadratures are automatically present in every microscopic source. If only the classical resolved record is supplied, Proposition B.5 shows that the acted-on system algebra can remain underdetermined. Conversely, once the operator-valued pullbacks are physically available on a common frame, the reconstruction is exact and feeds directly into the commutant audit of the main text.

B.3 Historical operations on the regular word carrier

The preceding reconstruction problem has a second layer. Even after a finite history algebra has been reconstructed, an invariant endomorphism of a word Hilbert space need not be an operation induced by the historical system algebra. The following module criterion separates these notions.

Let $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$ be a finite unital C^* -algebra generated by the represented historical operations on one exposed carrier, and let a finite group G act by unitaries u_g with $u_g \mathcal{A} u_g^* = \mathcal{A}$. Write $\alpha_g(a) = u_g a u_g^*$ and

$$\mathcal{A} \cong \bigoplus_{\beta} M_{n_{\beta}}(\mathbb{C}), \quad D = \dim_{\mathbb{C}} \mathcal{A} = \sum_{\beta} n_{\beta}^2,$$

with normalized regular trace

$$\tau_{\text{reg}}(a) = D^{-1} \sum_{\beta} n_{\beta} \text{Tr}(a_{\beta}).$$

On $\mathcal{W} = L^2(\mathcal{A}, \tau_{\text{reg}})$ let $L_a x = ax$, $R_b x = xb$ and $U_g x = \alpha_g(x)$.

Proposition B.8 (Word-module recognition of historical internal actions). *The operations on the regular word carrier induced by invariant historical system elements are exactly*

$$R(\mathcal{A})' \cap U(G)' = L(\mathcal{A}^G), \quad \mathcal{A}^G = \mathcal{A} \cap \{u_g : g \in G\}'. \quad (\text{B.15})$$

Equivalently, $T \in \text{End}(\mathcal{W})$ is left multiplication by a unique invariant historical element if and only if it commutes with every right multiplication and every U_g ; the coefficient is $T1$.

Proof. Every L_a commutes with every R_b . Conversely, if $[T, R_b] = 0$ for all b , then $T(b) = T(R_b 1) = R_b T1 = (T1)b$, hence $T = L_{T1}$. Moreover $U_g L_a U_g^{-1} = L_{\alpha_g(a)}$, so faithfulness of left multiplication gives $[U_g, L_a] = 0$ exactly when $a \in \mathcal{A}^G$. $\square$

Proposition B.9 (Exact module and symmetry defects). *Let $(e_{\nu})_{\nu=1}^D$ be a τ_{reg} -orthonormal basis of $\mathcal{A}$ and put $a = T1$. Define*

$$\delta_R(T)^2 = \sum_{\nu=1}^D \|[T, R_{e_{\nu}}]1\|_{\tau}^2, \quad \delta_G(T)^2 = \frac{1}{|G|} \sum_g \|[U_g, T]1\|_{\tau}^2.$$

If $E_G(a) = |G|^{-1} \sum_g \alpha_g(a)$, then

$$\|T - L_a\|_{\text{HS}(\mathcal{W})}^2 = \delta_R(T)^2, \quad (\text{B.16})$$

$$\|T - L_{E_G(a)}\|_{\text{HS}(\mathcal{W})} \leq \delta_R(T) + \sqrt{D/2} \delta_G(T). \quad (\text{B.17})$$

Thus both defects vanish exactly on the algebra in (B.15).

Proof. For each basis element, $(T - L_a)e_{\nu} = T(R_{e_{\nu}} 1) - R_{e_{\nu}} T1 = [T, R_{e_{\nu}}]1$; summation gives (B.16). Since $U_g 1 = 1$, $[U_g, T]1 = \alpha_g(a) - a$. Orthogonal group averaging gives $|G|^{-1} \sum_g \|\alpha_g(a) - a\|_{\tau}^2 = 2\|a - E_G(a)\|_{\tau}^2$. Finally $\|L_b\|_{\text{HS}(\mathcal{W})}^2 = D\|b\|_{\tau}^2$, and the triangle inequality proves (B.17). $\square$

Corollary B.10 (Word-space multiplicity firewall). *Multiplicity of the conjugation representation on $L^2(\mathcal{A}, \tau_{\text{reg}})$ need not equal multiplicity on the exposed physical carrier and cannot by itself be interpreted as an executable internal factor. Any such inference must also pass Proposition B.8. In particular, the multiplicity-birth analysis of Section 4.3 always refers to an actual reachable-observable physical carrier, not merely to the regular word representation. Appendix B.3.1 contains an explicit twelve-root example with an invariant $M_7(\mathbb{C})$ word-space block but no physical internal $M_3(\mathbb{C})$ factor.*

We next identify a class of mixed source quantities that require no independent measurement once the relevant typing operations are already historical.

Definition B.11 (Word-native linear shadow). A linear map $\mathcal{L} : \mathcal{A} \rightarrow \mathcal{A}$ is *word native* if

$$\mathcal{L}(X) = \sum_{\alpha=1}^m c_{\alpha} A_{\alpha} X B_{\alpha}, \quad (\text{B.18})$$

where $A_{\alpha}, B_{\alpha} \in \mathcal{A}$ are reconstructed historical-algebra elements and the coefficients c_{α} are fixed by the declared shadow before the target conclusion is evaluated. Finite compressions, conjugations, finite group averages and finite functional-calculus operations are included whenever their defining elements belong to $\mathcal{A}$.

Proposition B.12 (Inserted word-moment compiler). *Let $x, y, z \in \mathcal{A}$ be represented words and let $\mathcal{L}, \mathcal{M}$ be word-native maps, with $\mathcal{M}(Z) = \sum_{\beta} d_{\beta} C_{\beta} Z D_{\beta}$. Then*

$$\langle y, \mathcal{L}(x) \rangle_{\text{HS}} = \sum_{\alpha} c_{\alpha} \tau_{\text{reg}}(y^* A_{\alpha} x B_{\alpha}), \quad (\text{B.19})$$

$$\langle \mathcal{L}(x), \mathcal{M}(z) \rangle_{\text{HS}} = \sum_{\alpha, \beta} \overline{c_{\alpha}} d_{\beta} \tau_{\text{reg}}(B_{\alpha}^* x^* A_{\alpha}^* C_{\beta} z D_{\beta}). \quad (\text{B.20})$$

Hence every ordinary-shadow and shadow-shadow Gram block is a finite linear combination of ordinary Grand-Tensor word moments. If the historical word algebra is already flat, the reconstructed multiplication table reduces all inserted products and no deeper physical histories are required.

Proof. The formulas are the Hilbert–Schmidt inner product followed by substitution of (B.18). Cyclicity of τ_{reg} rewrites each summand as the inner product of two represented algebra elements. At flatness the finite word span is a $*$ -algebra, so every inserted product reduces in its known multiplication table. $\square$

Corollary B.13 (Table-native ancestry short). *Let W_r synthesize a flat historical word bank, let $S = W_r B_S$, $T = \mathcal{L} W_r B_T$, and define*

$$K^{00} = W_r^* W_r, \quad K^{0L} = W_r^* \mathcal{L} W_r, \quad K^{LL} = W_r^* \mathcal{L}^* \mathcal{L} W_r.$$

Then

$$G = B_S^* K^{00} B_S, \quad C = B_S^* K^{0L} B_T, \quad H = B_T^* K^{LL} B_T, \quad (\text{B.21})$$

and the ancestry short

$$R_{T|S} = H - C^* G^{\dagger} C \quad (\text{B.22})$$

is a deterministic function of the historical word Gram, multiplication table and the fixed word-native compiler. No independently executed mixed cross-Hankel panel is required on this branch.

Proposition B.14 (Word moments cannot manufacture an external shadow). *There are conservative extensions with identical historical word algebra, multiplication table, complete word-Gram hierarchy and resolved classical cylinder probabilities but different values of a proposed grading-changing or typed-incidence cross block whose defining shadow is not represented in the historical algebra. Therefore no function of the old word moments alone can recover such an external shadow.*

Proof. Tensor a faithful historical realization with an unread two-level spectator on which every old operation acts trivially. One extension contains no coherent cross-support link; another admits a future-visible spectator link and its typed compression. All old word products and moments agree, whereas the proposed external cross block is zero in the first extension and nonzero in the second. $\square$

B.3.1 Large word multiplicities need not be physical internal factors

Let $\Omega = \{(i, j) : 1 \leq i, j \leq 4, i \neq j\}$ and $\mathcal{H} = \mathbb{C}^\Omega$, with the simultaneous relabeling action $u_g|i, j\rangle = |g(i), g(j)\rangle$ of S_4 . Adjoin all diagonal point projections and the two fibre averages

$$(Tf)(i, j) = \frac{1}{3} \sum_{k \neq i} f(i, k), \quad (Hf)(i, j) = \frac{1}{3} \sum_{k \neq j} f(k, j).$$

Their support graph is connected, so the generated system algebra is $M_{12}(\mathbb{C})$.

Proposition B.15 (Invariant word-space inflation without an internal M_3). *For the preceding benchmark,*

$$\mathcal{A} = M_{12}(\mathbb{C}), \quad \mathcal{A}^{S_4} \cong \mathbb{C} \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}.$$

In the irreducible order $(\mathbf{1}, \text{sgn}, W, W \otimes \text{sgn}, V_2)$, the multiplicities of the physical S_4 action on $\mathcal{H}$ are $(1, 0, 2, 1, 1)$, whereas the multiplicities of the conjugation action on $L^2(M_{12})$ are $(7, 5, 19, 17, 12)$. Thus the latter commutant contains an $M_7(\mathbb{C})$ block although the word-induced invariant system algebra has no $M_3(\mathbb{C})$ corner.

Proof. The permutation character on ordered unequal pairs is $(12, 2, 0, 0, 0)$ on the conjugacy classes $1, (12), (12)(34), (123), (1234)$. Character inner products with the five irreducibles of S_4 give $(1, 0, 2, 1, 1)$ and hence the displayed physical commutant. Conjugation on $M_{12} \cong \mathcal{H} \otimes \overline{\mathcal{H}}$ has character $|\chi_{\mathcal{H}}|^2 = (144, 4, 0, 0, 0)$, whose character inner products are $(7, 5, 19, 17, 12)$. Proposition B.8 retains only left multiplication by $\mathcal{A}^{S_4}$ as word-induced invariant system actions. $\square$

B.4 Typed compilation from historical tables

Corollary B.16 (Historical typed-compilation criterion). *Let the relevant historical word hierarchy be flat. Suppose the grading, type, weak-leg and incidence source/target projections used to form the Store–Clifford and matter-incidence shadows are represented elements, or finite functional-calculus elements, of that same historical $\ast$ -algebra, and that every router or finite symmetry average entering the compression is word native in the sense of Definition B.11. Then the complete mixed ancestry Gram of those typed shadows, its Schur ancestry residual, and the alternating determinant shadow are deterministic functions of the existing Grand-Tensor word Gram and multiplication table. No independent mixed cross-Hankel source row is required for these word-native quantities. The same table computes the six naturality defects when their amplitudes and tensor-leg actions are historical; for a mixed branch, Proposition D.12 instead gives the contextual Choi-functional criterion.*

Conversely, if a required typing projector or router is external to the historical algebra, Proposition B.14 prevents its historical occurrence from being inferred from the old word moments alone.

Proof. The typed compressions are word-native linear shadows, so Proposition B.12 and Corollary B.13 reconstruct all ordinary-shadow and shadow-shadow blocks. Complete alternation on the already reconstructed typed Hom space is fixed linear postprocessing, so it introduces no additional physical source datum. The converse is exactly Proposition B.14. $\square$

C Tetrahedral representations and internal-algebra reconstruction

The representation decomposition, general router criterion, reciprocal return test, and scalar-separation argument support the finite branch classification in Section 4.

C.1 The oriented tetrahedral carrier

This appendix records the representation-theoretic decomposition used in Proposition 4.1 for the tetrahedral relational carrier supplied by [46]. The decomposition itself is used here only as finite representation-theoretic input and is independent of the probabilistic origin of the source Gram.

Let $X = \{1, 2, 3, 4\}$ and let $\Phi = \{(i, j) : i \neq j\}$ be the twelve oriented edges. Root reversal is

$$\iota(i, j) = (j, i).$$

Let $P_{\mathbb{C}} = \mathbb{C}^{\Phi}$. Its even and odd reversal subspaces are

$$P^{\pm} = \{f : f(j, i) = \pm f(i, j)\}, \quad \dim P^{\pm} = 6.$$

C.1.1 The even sector

The even sector is the permutation representation on the six unordered edges of a tetrahedron. Let χ_+ be its character. For the five conjugacy classes of S_4 represented by

$$1, \quad (12), \quad (12)(34), \quad (123), \quad (1234),$$

the number of fixed unordered edges is

$$6, \quad 2, \quad 2, \quad 0, \quad 0.$$

Comparing with the S_4 character table gives

$$P^+ \cong \mathbf{1} \oplus W \oplus V_2, \tag{C.1}$$

where W is the standard three-dimensional representation and V_2 is the two-dimensional irreducible representation.

C.1.2 The odd sector

The odd sector is the oriented edge space $\Lambda^2 \mathbb{C}^4$ after identifying $e_i \wedge e_j$ with the signed edge $i \rightarrow j$. Write

$$\mathbb{C}^4 = \mathbb{C}u_0 \oplus W, \quad u_0 = \frac{1}{2}(1, 1, 1, 1).$$

Then

$$\Lambda^2 \mathbb{C}^4 = u_0 \wedge W \oplus \Lambda^2 W.$$

The first summand is isomorphic to W . Since $\dim W = 3$,

$$\Lambda^2 W \cong W \otimes \text{sgn}.$$

Therefore

$$P^- \cong W \oplus (W \otimes \text{sgn}). \tag{C.2}$$

The boundary map sends the first summand isomorphically onto the centred endpoint space and annihilates the second. Hence the two odd triplets are canonically distinguished as endpoint and cycle sectors.

Combining (C.1) and (C.2), removing the even root mean, and adjoining the scalar type contrast gives

$$\mathbb{C}_{\text{type}}^+ \oplus W_+ \oplus (V_2)_+ \oplus W_- \oplus (W \otimes \text{sgn})_-,$$

which is (4.5).

C.2 General routers and minimal support residuals

Proposition C.1 (Connected-router full-matrix theorem). *Let*

$$\mathcal{D}_n = \text{span}\{p_1, \dots, p_n\} \cong \mathbb{C}^n$$

be the diagonal algebra of mutually orthogonal rank-one projections on $\mathbb{C}^n$, and let $u \in M_n(\mathbb{C})$. Form the undirected support graph with an edge $i - j$ whenever $p_i u p_j \neq 0$ or $p_j u p_i \neq 0$. Then

$$C^*(\mathcal{D}_n, u) = M_n(\mathbb{C}) \iff \text{the support graph of } u \text{ is connected.} \quad (\text{C.3})$$

If the graph is disconnected, every connected component is a reducing subspace and the generated algebra is a proper block-diagonal subalgebra.

Proof. For every support edge, $p_i u p_j$ is a nonzero scalar multiple of the matrix unit E_{ij} ; adjoints give E_{ji} . Products along a graph path generate all matrix units between vertices in the same connected component. Thus a connected graph generates all of $M_n(\mathbb{C})$. Conversely the sum of the p_i over any connected component is a nontrivial central reducing projection when the graph is disconnected. $\square$

Corollary C.2 (One-router source minimality). *Once a coherent n -dimensional multiplicity carrier and its diagonal minimal projections have occurred, one cyclic nonnormal router is sufficient to generate $M_n(\mathbb{C})$. Without any off-diagonal word the generated algebra remains commutative. Thus one router is source-minimal at fixed multiplicity carrier.*

Proposition C.3 (Minimal bridge and weak-router residuals). *On $T \oplus H_{\text{priv}}$ define*

$$\delta_{\text{col}}(u) := \|p_H u p_T\|_{\text{HS}}^2 + \|p_T u p_H\|_{\text{HS}}^2.$$

On a weak multiplicity plane define

$$\delta_{\text{wk}}(p, q) := \|[p, q]\|_{\text{HS}}^2.$$

Then $\delta_{\text{col}} = 0$ exactly on the reducing colour branch $M_2 \oplus \mathbb{C}$, while $\delta_{\text{col}} > 0$ gives M_3 . For rank-one projections p, q on $\mathbb{C}^2$, one has

$$\delta_{\text{wk}}(p, q) = 2c^2(1 - c^2), \quad c = |\langle t, h \rangle|, \quad (\text{C.4})$$

so $\delta_{\text{wk}} > 0$ exactly when the projections are distinct and nonorthogonal, in which case they generate M_2 .

Proof. The colour statement is Proposition 4.4. For the weak statement, choose a basis with $p = |e_1\rangle\langle e_1|$ and $h = ce_1 + \sqrt{1 - c^2}e_2$ after absorbing a phase. Direct multiplication gives (C.4). Its strict positivity is equivalent to $0 < c < 1$, which is the Burnside condition of Proposition 4.4. $\square$

C.3 Subdirect matrix algebras and scalar separation

Proposition C.4 (Finite subdirect matrix-algebra classification). *Let $\mathcal{B} \subseteq \bigoplus_{i=1}^r M_{n_i}(\mathbb{C})$ be a unital C^* -subalgebra whose coordinate projection onto every displayed factor is surjective. Then the coordinates partition into classes of equal matrix size, and $\mathcal{B}$ contains one independent full matrix algebra per class, embedded diagonally across that class up to inner automorphisms. In particular, distinct matrix sizes cannot be diagonally identified.*

Proof. Write the intrinsic decomposition $\mathcal{B} \cong \bigoplus_{\beta} M_{m_{\beta}}(\mathbb{C})$. A surjective $*$ -homomorphism to a simple factor $M_{n_i}(\mathbb{C})$ annihilates every intrinsic summand except one: the images of the intrinsic central units are orthogonal central projections summing to the identity, so exactly one is nonzero. On that summand the map is a nonzero homomorphism between simple full matrix algebras and hence an isomorphism, forcing $m_{\beta} = n_i$. Coordinates seeing the same intrinsic summand are related by automorphisms of a full complex matrix algebra, hence by inner automorphisms. Different intrinsic summands are independent. $\square$

Applying this proposition to matrix sizes $(3, 2, 1, 1)$ leaves only the two scalar factors available for diagonal identification. Their independent central supports, or the positive character-separation residual, exclude it.

C.4 Reciprocal returns through a represented mediator

The bridge need not be a primitive operation when the historical route table contains reciprocal access through one common mediator. Let G be the external finite symmetry, let V be an irreducible unitary G -module of dimension d , and let the mediator carrier be $\mathcal{E} = V \otimes \mathcal{N}$ with $n = \dim \mathcal{N}$. Suppose actual entry and reverse-exit operations have the covariant form

$$A_g = (v_g \otimes I)A : T \rightarrow \mathcal{E}, \quad B_h^* = B^*(v_h^* \otimes I) : \mathcal{E} \rightarrow H_{\text{priv}},$$

and that the actual mediator dynamics contains $D = I_V \otimes h$ with $h = h^*$. Put

$$r_{g,k} = B^*(v_g \otimes h^k)A, \quad 0 \leq k < n,$$

$$\rho_A = \text{Tr}_V(AA^*), \quad \rho_B = \text{Tr}_V(BB^*), \quad W_A = \sum_{k=0}^{n-1} h^k \rho_A h^k.$$

Proposition C.5 (Finite reciprocal-return criterion). *For the declared route bank,*

$$\mathbf{m}_{\text{ret}} := \frac{1}{|G|} \sum_{g \in G} \sum_{k=0}^{n-1} \|r_{g,k}\|_{\text{HS}}^2 = \frac{1}{d} \text{Tr}(\rho_B W_A) \geq 0. \quad (\text{C.5})$$

The following are equivalent:

- (i) *an admitted word in the route bank has a nonzero $H_{\text{priv}} \leftarrow T$ corner;*
- (ii) *some $r_{g,k}$ is nonzero;*
- (iii) $\mathbf{m}_{\text{ret}} > 0$;
- (iv) *the h -cyclic multiplicity spaces generated by $\text{Ran } \rho_A$ and $\text{Ran } \rho_B$ are not orthogonal.*

If these conditions fail, an explicit reducing projection separates T from H_{priv} for the entire declared bank, so arbitrary repeated excursions cannot create a colour bridge within that bank. If they hold, a witness has at most $n + 1$ route letters.

Proof. Schur averaging on the irreducible V factor gives

$$\frac{1}{|G|} \sum_g (v_g \otimes I) A A^* (v_g^* \otimes I) = \frac{I_V}{d} \otimes \rho_A.$$

Taking the Hilbert–Schmidt norm of $B^*(v_g \otimes h^k)A$ and summing yields (C.5). Since it is a finite sum of squared norms, positivity is equivalent to a nonzero returned word. The range of W_A is $\text{span}\{h^k \text{Ran } \rho_A : 0 \leq k < n\}$ by Cayley–Hamilton. Thus $\text{Tr}(\rho_B W_A) = 0$ exactly when the two cyclic multiplicity spaces are orthogonal. In that case the projection onto $T \oplus (V \otimes \text{Ran } W_A)$ commutes with the route generators and their adjoints while annihilating H_{priv} , excluding all longer returns. The finite horizon $k < n$ gives the stated word-length bound. $\square$

Corollary C.6 (Port-relative reachability forces colour). *If $W_A \succeq \kappa I_{\mathcal{N}}$ with $\kappa > 0$ and $B \neq 0$, then*

$$\mathbf{m}_{\text{ret}} \geq \frac{\kappa}{d} \|B\|_{\text{HS}}^2, \quad \max_{g, k < n} \|r_{g,k}\|_{\text{HS}}^2 \geq \frac{\kappa}{dn} \|B\|_{\text{HS}}^2. \quad (\text{C.6})$$

Hence the colour bridge follows from the upstream route structure and Proposition 4.4. In particular, for a multiplicity-one mediator, two nonzero reciprocal ports force a two-leg colour return and no independent direct bridge is required.

For the mediator notation of Proposition C.5, set

$$\mathcal{N}_A = \text{span}\{h^k \text{Ran } \rho_A : 0 \leq k < n\}, \quad \mathcal{N}_B = \text{span}\{h^k \text{Ran } \rho_B : 0 \leq k < n\}.$$

Cayley–Hamilton makes both spaces h -invariant, and self-adjointness of h makes them reducing.

Proposition C.7 (All-excursion return alternative). *If $\mathbf{m}_{\text{ret}} = 0$, then $\mathcal{N}_A \perp \mathcal{N}_B$ and*

$$Q = I_T \oplus 0_{H_{\text{priv}}} \oplus (I_V \otimes P_{\mathcal{N}_A})$$

commutes with every generator of the declared reciprocal route bank and its adjoint. Consequently every word w in that bank has $p_H w p_T = 0$. Thus failure of the finite return panel is an exact obstruction to arbitrarily long repeated excursions within the same bank.

Proof. The equality $\mathbf{m}_{\text{ret}} = d^{-1} \text{Tr}(\rho_B W_A) = 0$ and positivity imply $\text{Ran } \rho_B \perp \mathcal{N}_A$, hence $\mathcal{N}_B \perp \mathcal{N}_A$. Partial-trace positivity gives $\text{Ran } A \subseteq V \otimes \text{Ran } \rho_A$ and $\text{Ran } B \subseteq V \otimes \text{Ran } \rho_B$. The stated projection therefore reduces the entry, exit, mediator and adjoint generators, contains the type line and annihilates the private plane. Every generated word has zero $H_{\text{priv}} \leftarrow T$ corner. $\square$

C.5 Router failure loci and the finite alternatives

For the diagonal algebra $\mathcal{D}_n$ of Proposition C.1, define the router failure set

$$\mathfrak{F}_n = \{u \in M_n(\mathbb{C}) : C^*(\mathcal{D}_n, u) \neq M_n(\mathbb{C})\}.$$

For every nonempty proper subset $S \subset \{1, \dots, n\}$ let

$$L_S = \{u : p_i u p_j = p_j u p_i = 0 \text{ for all } i \in S, j \notin S\}.$$

Proposition C.8 (Algebraic structure of the router failure set). *One has*

$$\mathfrak{F}_n = \bigcup_{\emptyset \neq S \subsetneq \{1, \dots, n\}} L_S. \quad (\text{C.7})$$

Each L_S is a proper complex linear subspace of $M_n(\mathbb{C})$. Hence $\mathfrak{F}_n$ is a proper Zariski-closed set and its complement is Zariski open, Euclidean open, and dense.

Proof. The support graph is disconnected exactly when some nontrivial union S of connected components has no support edge to its complement, which is precisely membership in L_S . Each L_S is defined by homogeneous linear equations on matrix entries and is proper because, for example, a cyclic shift matrix has connected support. A finite union of closed algebraic subsets is closed, and a finite union of proper linear subspaces cannot fill $M_n(\mathbb{C})$. $\square$

In particular, fixing diagonal minimal projections on the admitted three- and two-dimensional multiplicity carriers gives a Zariski-open dense set of routers generating $M_3(\mathbb{C}) \oplus M_2(\mathbb{C})$. This statement is conditional on those carriers having occurred.

Proposition C.9 (Finite internal landing alternative). *The following finite implications apply to the colour/weak landing problem. Failure of a finite-depth test without flatness is not a terminal obstruction; the implications are not asserted to be mutually exclusive.*

- (A1) *If the word hierarchy is flat and $m_{0,r} < 3$, an external-trivial internal colour $M_3(\mathbb{C})$ is impossible at every later depth.*
- (A2) *If $m_{0,r} \geq 3$ but every represented type-private bridge has $\delta_{\text{col}} = 0$, the colour block is reducible and remains $M_2 \oplus \mathbb{C}$ on the displayed three-space.*
- (A3) *If the weak multiplicity plane is present but every available pair has $\delta_{\text{wk}} = 0$, the represented weak multiplicity algebra is commutative.*
- (A4) *If the complete census is $\mathbf{m}_r = (3, 0, 2, 1, 1)$, both a positive colour residual and a positive weak-router residual occur, and the two scalar characters are centrally separated, the represented internal word algebra equals the full relative commutant $M_3 \oplus M_2 \oplus \mathbb{C} \oplus \mathbb{C}$.*

On the fixed complete census $\mathbf{m}_r = (3, 0, 2, 1, 1)$ with independent displayed central supports, the intermediate algebra dimensions are exactly those of Proposition 4.6. Without scalar central separation the blockwise-surjective terminal branch may instead have dimension fourteen. Therefore every failure returns a finite algebraic witness and no failed condition is silently completed to the terminal branch.

Proof. Item (A1) is Proposition 4.3. Item (A2) is Proposition 4.4; item (A3) is Proposition 4.4; and item (A4) is Theorem 4.5. The dimension statement is Proposition 4.6. $\square$

D Determinant naturality, incidence normalization, and anomalies

The principal determinant and gauge-group proofs are in Section 5. Here we retain the mixed-occurrence alignment criterion, the unrestricted incidence envelopes, the exact naturality spectrum, and the source and anomaly bookkeeping.

D.1 Tensor alignment from mixed source occurrences

Proposition D.1 (Fusion-native sufficient criterion for same-carrier alignment). *Let H_C, H_1, H_2 be finite source-minimal Hilbert spaces with source syntheses $S_A : E_A \rightarrow H_A$ and Grams $G_A = S_A^* S_A$. Let H_{C1}, H_{12}, H_{C12} be the source-minimal carriers of the represented pair and triple occurrences, with syntheses S_{C1}, S_{12}, S_{C12} and coefficient occurrence maps*

$$m_{C,1} : E_C \otimes E_1 \rightarrow E_{C1}, \quad m_{1,2} : E_1 \otimes E_2 \rightarrow E_{12},$$

$$m_{C1,2} : E_{C1} \otimes E_2 \rightarrow E_{C12}, \quad m_{C,12} : E_C \otimes E_{12} \rightarrow E_{C12}.$$

Assume the coefficient occurrence law is associative,

$$m_{C1,2}(m_{C,1} \otimes I) = m_{C,12}(I \otimes m_{1,2}). \quad (\text{D.1})$$

For each composable pair (A, B) suppose

$$m_{A,B}^* G_{AB} m_{A,B} = G_A \otimes G_B. \quad (\text{D.2})$$

Then the rule on represented source vectors

$$\Gamma_{A,B}(S_A x \otimes S_B y) := S_{AB} m_{A,B}(x \otimes y) \quad (\text{D.3})$$

extends uniquely to an isometry $\Gamma_{A,B} : H_A \otimes H_B \rightarrow H_{AB}$. Assume in addition that the four fusion innovations vanish,

$$S_{AB}^*(I - \Gamma_{A,B} \Gamma_{A,B}^*) S_{AB} = 0 \quad \text{for } (A, B) = (C, 1), (1, 2), (C1, 2), (C, 12), \quad (\text{D.4})$$

and that the associator defect vanishes,

$$\|\Gamma_{C1,2}(\Gamma_{C,1} \otimes I) - \Gamma_{C,12}(I \otimes \Gamma_{1,2})\|_{\text{HS}}^2 = 0. \quad (\text{D.5})$$

Then the two parenthesizations define the same unitary

$$\Gamma : H_C \otimes H_1 \otimes H_2 \xrightarrow{\sim} H_{C12}. \quad (\text{D.6})$$

Consequently the three transported factor actions

$$\begin{aligned} \hat{\pi}_C(a) &= \Gamma(a \otimes I \otimes I) \Gamma^*, \\ \hat{\pi}_1(b) &= \Gamma(I \otimes b \otimes I) \Gamma^*, \\ \hat{\pi}_2(c) &= \Gamma(I \otimes I \otimes c) \Gamma^* \end{aligned} \quad (\text{D.7})$$

commute pairwise. If $\dim H_C = 3$ and $\dim H_1 = \dim H_2 = 2$, they generate $\mathcal{B}(H_{C12}) \cong M_{12}(\mathbb{C})$, and the same-carrier defect $\Delta_{\otimes}$ of Proposition 5.1 vanishes for these factor actions. With an additional reducing multiplicity space M , the generated algebra is $M_{12}(\mathbb{C}) \otimes I_M$.

Proof. Equation (D.2) says exactly that (D.3) preserves inner products on the source span, hence defines an isometry. Source minimality gives $\text{Ran } S_{AB} = H_{AB}$; therefore (D.4) implies that every $\Gamma_{A,B}$ is onto and hence unitary. The associator defect identifies the two triple unitaries. The three tensor-leg matrix algebras commute before conjugation by Γ and therefore after conjugation. Their tensor product is $\mathcal{B}(H_C \otimes H_1 \otimes H_2)$, so in dimensions $(3, 2, 2)$ the transported factors generate $M_{12}(\mathbb{C})$. The commutator defect is a sum of squared norms and consequently vanishes. Tensoring all factors with I_M gives the final statement. $\square$

The direct commutation residual applies when factor actions are supplied independently. The preceding sufficient criterion instead reconstructs commuting actions from the actual mixed occurrence packet. Neither follows from separate factor marginals, as the next witness shows.

Proposition D.2 (Factor marginals do not fix their relative tensor position). *There are two pairs of unital factor embeddings on the same six-dimensional Hilbert space such that each colour marginal is $M_3(\mathbb{C})$, each weak marginal is $M_2(\mathbb{C})$, and all intrinsic single-factor traces and Hilbert–Schmidt Grams agree, while one pair commutes and the other does not.*

Proof. On $\mathbb{C}^3 \otimes \mathbb{C}^2$ put

$$\mathcal{A} = M_3(\mathbb{C}) \otimes I_2, \quad \mathcal{B} = I_3 \otimes M_2(\mathbb{C}).$$

Choose a non-scalar Hermitian $D \in M_3(\mathbb{C})$, a Pauli matrix $X \in M_2(\mathbb{C})$, and for generic nonzero θ define

$$U = \exp(i\theta D \otimes X), \quad \mathcal{B}' = U\mathcal{B}U^*.$$

The algebras $\mathcal{B}$ and $\mathcal{B}'$ are unitarily conjugate, so every intrinsic algebraic invariant, trace, and Hilbert–Schmidt Gram computed within the weak marginal agrees. The colour marginal is unchanged. The pair $(\mathcal{A}, \mathcal{B})$ commutes. For generic θ , however, choose a colour matrix unit not commuting with D and a weak generator not commuting with X ; conjugation by U produces an element of $\mathcal{B}'$ with a nonzero commutator with that colour element. Hence the second pair is not a commuting tensor-factor realization despite identical separate marginals. $\square$

D.2 Rectangular and semi-intertwiner envelopes

Let an input carrier split as $A = \bigoplus_{s \in \mathcal{S}} A_s$, with $p_s = \dim A_s$. A recorded branch a has source $A_{s(a)}$ and target T_a of dimension q_a . Suppose its only amplitude restriction is the full rectangular envelope $\text{Hom}(A_{s(a)}, T_a)$. Use the coefficient convention in which the normalization synthesis of matrix-unit amplitudes $B_{a,i\alpha} = |f_i\rangle\langle e_\alpha|$ sends a coefficient matrix unit to $B_{a,i\alpha}^* B_{a,j\beta}$. This is the usual Choi normalization map after the fixed input-transpose identification; its rank and kernel do not depend on that convention.

Proposition D.3 (Exact rectangular normalization kernel). *If every source sector occurs in at least one branch, the normalization map is*

$$\Gamma_{\text{rect}}((X_a)_a) = \bigoplus_{s \in \mathcal{S}} \sum_{a: s(a)=s} \text{Tr}_{T_a} X_a, \quad X_a \in M_{q_a p_s}(\mathbb{C}). \quad (\text{D.8})$$

It is surjective onto $\bigoplus_s M_{p_s}(\mathbb{C})$ and has complex nullity

$$\dim_{\mathbb{C}} \text{Ker } \Gamma_{\text{rect}} = \sum_s p_s^2 \left(\sum_{a: s(a)=s} q_a^2 - 1 \right). \quad (\text{D.9})$$

It is injective exactly when every source sector has one outgoing recorded branch and that branch has one-dimensional target.

Proof. The products of amplitude matrix units are $B_{a,i\alpha}^* B_{a,j\beta} = \delta_{ij} |e_\alpha\rangle\langle e_\beta|$. This proves (D.8). Any one branch on a source sector makes its partial trace surjective onto $M_{p_s}(\mathbb{C})$. The domain dimension on that sector is $p_s^2 \sum_{a: s(a)=s} q_a^2$ and the image dimension is p_s^2 , giving the nullity. All q_a are positive integers, so vanishing of each summand is equivalent to the asserted one-branch, one-dimensional-target condition. $\square$

For the incidence port of (5.4), $p = 12$, $q = 3$; therefore the normalization kernel has dimension 1152. This is a statement about the unrestricted envelope, not the nullity at an already rank-one-supported Choi packet. Finite naturality removes the unwanted amplitude directions before normalization is used.

D.2.1 Schur compression before normalization

Proposition D.4 (Typed semi-intertwiner envelope). *Let a compact group act on a source H_s and target H_t , and let χ be a unitary character. Decompose*

$$H_s = \bigoplus_{\lambda} V_{\lambda} \otimes \mathbb{C}^{m_{\lambda}}, \quad \chi^{-1} H_t = \bigoplus_{\lambda} V_{\lambda} \otimes \mathbb{C}^{n_{\lambda}}.$$

Then the amplitude envelope

$$\mathbf{S}_{\chi} = \{K : R_t(g)KR_s(g)^* = \chi(g)K\}$$

*is isomorphic to $\bigoplus_{\lambda} I_{V_{\lambda}} \otimes M_{n_{\lambda} \times m_{\lambda}}(\mathbb{C})$ and has dimension $\sum_{\lambda} m_{\lambda} n_{\lambda}$. Every K^*L with $K, L \in \mathbf{S}_{\chi}$ lies in $R_s(G)'$. For several recorded semi-intertwiner envelopes on this source, injectivity of their joint normalization map therefore requires*

$$\sum_a \left(\sum_{\lambda} m_{\lambda} n_{a,\lambda} \right)^2 \leq \sum_{\lambda} m_{\lambda}^2. \quad (\text{D.10})$$

Proof. Twist the target by χ^{-1} and apply Schur's lemma. The character cancels in K^*L , so the normalization image lies in the source commutant, of the dimension on the right of (D.10). Injectivity requires the coefficient-domain dimension not to exceed it. $\square$

D.3 Exact naturality spectrum

Fix an oriented orthonormal basis (c_1, c_2, c_3) of C and its dual (c^1, c^2, c^3) . The unitary volume identification $J_C : C \rightarrow \Lambda^2 C^*$ is

$$J_C c_1 = c^2 \wedge c^3, \quad J_C c_2 = -c^1 \wedge c^3, \quad J_C c_3 = c^1 \wedge c^2.$$

It obeys

$$J_C^{-1} d\rho_C^{\text{out}}(X) J_C = X - (\text{Tr } X) I_C. \quad (\text{D.11})$$

This identity follows by evaluating on the three exterior basis vectors, or by differentiating the natural action on $C \otimes (\det C)^* \cong \Lambda^2 C^*$. For every off-diagonal matrix unit the trace term vanishes. Thus on

$$\mathcal{U} \cong M_3(\mathbb{C}) \otimes (W_2 \otimes W_2)^*$$

the positive defect splits into commuting tensor summands

$$H_{\text{nat}} \cong H_C \otimes I_4 + I_9 \otimes H_W, \quad H_C = \sum_{X \in \mathcal{G}_C} \text{ad}_X^* \text{ad}_X. \quad (\text{D.12})$$

Here H_W is the dual-space version of $L_+ L_- + L_- L_+$, where $L_{\pm} = F_{12/21} \otimes I + I \otimes F_{12/21}$. Transposition onto the dual space preserves the following spectral calculation.

Lemma D.5 (Colour defect spectrum). *The spectrum of H_C is*

$$\text{spec } H_C = \{0^{(1)}, 2^{(3)}, 3^{(4)}, 6^{(1)}\}. \quad (\text{D.13})$$

Proof. For the undirected edges $1 - 2$ and $2 - 3$, the two opposite matrix-unit terms contribute

$$(E_{ii} + E_{jj})A + A(E_{ii} + E_{jj}) - 2(E_{ij}AE_{ji} + E_{ji}AE_{ij}).$$

On an off-diagonal matrix unit E_{ab} , this is diagonal with eigenvalue $d_a + d_b$, where $(d_1, d_2, d_3) = (1, 2, 1)$. Thus E_{13}, E_{31} have eigenvalue 2 and $E_{12}, E_{21}, E_{23}, E_{32}$ have eigenvalue 3. On diagonal matrices the action is twice the three-vertex path Laplacian:

$$2 \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Its eigenvectors $(1, 1, 1)$, $(1, 0, -1)$, and $(1, -2, 1)$ have eigenvalues 0, 2, and 6, respectively. These nine eigenvectors give the spectrum. $\square$

Lemma D.6 (Weak defect spectrum). *The spectrum of H_W is*

$$\text{spec } H_W = \{0^{(1)}, 2^{(2)}, 4^{(1)}\}. \quad (\text{D.14})$$

Its zero eigenspace is the alternating weak covector line.

Proof. With an orthonormal basis (w_1, w_2) , both L_+ and L_- annihilate $(w_1 \otimes w_2 - w_2 \otimes w_1)/\sqrt{2}$. On the symmetric orthonormal vectors

$$w_1 \otimes w_1, \quad (w_1 \otimes w_2 + w_2 \otimes w_1)/\sqrt{2}, \quad w_2 \otimes w_2,$$

direct multiplication of $L_+L_- + L_-L_+$ gives eigenvalues 2, 4, and 2. Passing to the dual representation gives the same spectrum. $\square$

Proposition D.7 (Exact naturality spectrum). *The complete spectrum is*

$$\text{spec } H_{\text{nat}} = \{0^{(1)}, 2^{(5)}, 3^{(4)}, 4^{(7)}, 5^{(8)}, 6^{(4)}, 7^{(4)}, 8^{(2)}, 10^{(1)}\}. \quad (\text{D.15})$$

In particular, the one-dimensional kernel is the tensor product of the scalar colour matrix and alternating weak covector, and the first positive eigenvalue is exactly two.

Proof. Add the eigenvalues of (D.13) and (D.14), multiplying their multiplicities, in the tensor sum (D.12). The resulting multiplicities sum to 36. The smallest positive sum is two, which proves (5.11) without a numerical spectral assumption. $\square$

D.4 Source-independent selection and positive branch refinements

Proposition D.8 (Determinant selector, population, and branchwise purity). *On the aligned tensor-factor branch, $(H_{\text{nat}}, P_{\text{det}})$ is defined entirely by the factor actions and typed slots, before any source population. Let $J = \sum_b J_b$, $J_b \succeq 0$, and put*

$$m_b = \text{Tr } J_b, \quad d_b = \text{Tr}(H_{\text{nat}} J_b), \quad m_{D,b} = \text{Tr}(P_{\text{det}} J_b). \quad (\text{D.16})$$

For the corresponding totals, $d = \sum_b d_b$, $m_D = \sum_b m_{D,b}$, and $d_b \geq 0$. The exact spectrum gives

$$2(I - P_{\text{det}}) \preceq H_{\text{nat}} \preceq 10(I - P_{\text{det}}), \quad (\text{D.17})$$

so, branchwise and in total,

$$\mathrm{Tr}[(I - P_{\mathrm{det}})J_b] \leq d_b/2, \quad \max\{0, m_b - d_b/2\} \leq m_{D,b} \leq m_b - d_b/10. \quad (\mathrm{D.18})$$

If $d = 0$, every positive branch satisfies

$$J_b = m_b P_{\mathrm{det}}. \quad (\mathrm{D.19})$$

Hence $m > 0$ gives $J = m P_{\mathrm{det}}$ and forgetting positive labels cannot hide non-determinant contamination. If merely $d < 2m$, then $m_D > 0$, certifying occurrence without exact purity. Simultaneous unitary changes of factor coordinates preserve all these scalar statements.

Proof. The defect maps depend on the factor actions, not on J or its Kraus coordinates. Linearity and positivity give additivity and $d_b \geq 0$. Proposition D.7 places the positive spectrum in $[2, 10]$; tracing the resulting operator bounds against each J_b gives the inequalities. If $d = 0$, every $d_b = 0$, so positivity forces each J_b onto the one-dimensional kernel of H_{nat} . Its trace then gives $J_b = m_b P_{\mathrm{det}}$. Finally $m_D \geq m - d/2$, and unitary covariance of the factor actions conjugates the selector and source simultaneously. $\square$

D.5 Determinant realization, normalization, and tensor arity

For $C \cong \mathbb{C}^3$ and $W_2 \cong \mathbb{C}^2$, define

$$\mathcal{D} = (\det C \otimes \det W_2)^*, \quad \mathcal{U} = \mathrm{Hom}(C \otimes W_2 \otimes W_2, \Lambda^2 C^*).$$

For $\tau \in \mathcal{D}$, define

$$(\Theta_\tau(c \otimes w_1 \otimes w_2))(c_1, c_2) = \tau(c \wedge c_1 \wedge c_2 \otimes w_1 \wedge w_2).$$

Complete alternation in the three colour and two weak arguments gives a map $\mathfrak{A}_{\mathrm{det}} : \mathcal{U} \rightarrow \mathcal{D}$.

Proposition D.9 (Determinant-incidence line). *The maps Θ and $\mathfrak{A}_{\mathrm{det}}$ restrict to inverse isomorphisms between $\mathcal{D}$ and the fully alternating incidence line. In oriented orthonormal bases,*

$$\mathfrak{A}_{\mathrm{det}}(\Theta_\tau) = \tau, \quad \|\Theta_\tau\|_{\mathrm{HS}}^2 = 6\|\tau\|^2.$$

Thus a nonzero fully alternating incidence shadow is equivalent to a nonzero determinant seed.

Proof. Complete alternation is the orthogonal projection onto the unique $(\det C \det W_2)^{-1}$ symmetry type. Evaluation on oriented orthonormal bases gives the factor $3!2!/2 = 6$ in the Hilbert–Schmidt normalization. $\square$

D.5.1 Partial-isometry normalization and competing branches

Let $a = (w_1 \otimes w_2 - w_2 \otimes w_1)/\sqrt{2}$ and define $T_0(c \otimes \zeta) = J_C c \langle a, \zeta \rangle$. Then

$$T_0^* T_0 = I_C \otimes |a\rangle\langle a| = P_{\mathrm{alt}}, \quad T_0 T_0^* = I_{\Lambda^2 C^*}, \quad \|T_0\|_{\mathrm{HS}}^2 = 3. \quad (\mathrm{D.20})$$

For unit τ in Proposition D.9, $T_0 = \Theta_\tau/\sqrt{2}$, up to a common phase. This distinguishes partial-isometry normalization from unit Hilbert–Schmidt normalization.

Proposition D.10 (Alternating-sector isolation fixes the branch weight). *Let a normalized instrument contain a branch satisfying (5.13), with T_0 normalized by (D.20). Suppose every Kraus amplitude K of every other recorded branch satisfies $K P_{\mathrm{alt}} = 0$. Then $x = 1$. If instead k recorded branches on this sector have the same normalized one-dimensional envelope $\mathbb{C} T_0$, their effects constrain only $\sum_{a=1}^k x_a = 1$; the linear scalar normalization kernel has dimension $k - 1$.*

Proof. Compress the trace-preservation identity on both sides by P_{alt} . The singled-out branch contributes xP_{alt} and the other branches contribute zero, proving $x = 1$. For k competing ray branches, compression gives $(\sum_a x_a)P_{\text{alt}} = P_{\text{alt}}$, a single independent scalar equation. Positivity gives $x_a \geq 0$ but does not remove the scalar ambiguity in the interior of that simplex. $\square$

D.5.2 Sharp arity of the determinant character

Let a native finite carrier have defining central weights $(1, 0)$ on $C \cong \mathbb{C}^3$, $(0, 1)$ on $W_2 \cong \mathbb{C}^2$, and $(0, 0)$ on any neutral sector. The centre of $U(3) \times U(2)$ acts on a weight (a, b) by $e^{i(a\alpha + b\beta)}$.

Proposition D.11 (Sharp five-slot determinant-mode floor). *The central-weight support of the single-system endomorphism algebra contains only*

$$(0, 0), \quad \pm(1, 0), \quad \pm(0, 1), \quad \pm(1, -1),$$

and therefore does not contain the determinant-incidence weight $(-3, -2)$. On a diagonal k -fold tensor carrier this character is absent for $k < 5$. If the native carrier contains a nonzero neutral line, the bound is attained at $k = 5$.

Proof. Single-system endomorphism weights are target-minus-source differences among $(1, 0), (0, 1), (0, 0)$, giving the displayed list. In k tensor slots every vector weight (a, b) has $a, b \geq 0$ and $a + b \leq k$, so a target-minus-source weight with total negative charge -5 cannot occur for $k < 5$. At $k = 5$ the alternating line in $C^{\otimes 3} \otimes W_2^{\otimes 2}$ transforms by $\det C \det W_2$; a rank-one map from that line to a neutral line therefore has weight $(-3, -2)$. $\square$

This result concerns tensor arity, not temporal word length. It explains why longer evolution inside a fixed low-arity memory cannot substitute for the typed five-leg incidence of Theorem 5.2.

D.6 Finite functional and word-moment evaluation

Proposition D.12 (Finite functional and word-moment realization). *Fix the represented tensor-leg actions. Suppose an actual contextual test bank determines numbers $\text{Tr}(F_\ell J_D)$ and its real linear span contains H_{nat} and $I_{\mathcal{W}}$. Then this bank determines d_{nat} and $\text{Tr } J_D$ without determining the full Choi operator. More generally it suffices that those two functionals belong to the span modulo linear functionals constant on the declared feasible Choi class.*

Alternatively, if a historical amplitude bank (T_a) and all inserted matrix-factor actions are represented historical operations, then

$$d_{\text{nat}} = \sum_a \left(\sum_{X \in \mathcal{G}_C} \|\mathcal{R}_X(T_a)\|_{\text{HS}}^2 + \sum_{Y \in \mathcal{G}_W} \|\mathcal{S}_Y(T_a)\|_{\text{HS}}^2 \right) \quad (\text{D.21})$$

is a finite sum of inserted word moments. At flatness their multiplication table computes it exactly. This second route is conditional on amplitude access; the contextual route does not require a Kraus refinement.

Proof. Express each required functional as a real linear combination of the available tester functionals, adding its known constant when necessary. For the second route, use $J_D = \sum_a |T_a\rangle\langle T_a|$ and (5.10). Each squared norm expands into traces of products of the represented amplitudes and inserted actions, to which Proposition B.12 applies on the corresponding represented source algebra. A known faithful trace convention must be used consistently; converting between a physical

trace and the regular trace requires their reconstructed block weights. Finite table reduction then evaluates the products. Identification of a scalar functional does not itself implement that functional as a physical measurement operation. $\square$

D.7 Chiral conventions and the full gauge-charged audit

With $(a, b) = (-2, 3)$, the physical right-handed convention gives

$$\begin{aligned} Q &: (\mathbf{3}, \mathbf{2})_1, & u &: (\mathbf{3}, \mathbf{1})_4, & d &: (\mathbf{3}, \mathbf{1})_{-2}, \\ L &: (\mathbf{1}, \mathbf{2})_{-3}, & e &: (\mathbf{1}, \mathbf{1})_{-6}, & H &: (\mathbf{1}, \mathbf{2})_3. \end{aligned}$$

Writing all fermions as left-handed Weyl fields replaces

$$u \mapsto u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-4}, \quad d \mapsto d^c : (\bar{\mathbf{3}}, \mathbf{1})_2, \quad e \mapsto e^c : (\mathbf{1}, \mathbf{1})_6.$$

The common kernel of these representations of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_y$ is generated by

$$\left(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3} \right),$$

hence is $\mathbb{Z}_6$. This agrees with the quotient in (5.15).

Proposition D.13 (Hypercharge support can miss a gauge-charged exotic). *There are finite fermion carriers for which every target-sector anomaly and every nonzero-hypercharge inventory test is unchanged, while the full gauge-charged inventory residual of Proposition 6.2 is nonzero. Adjoin to a passing target packet a pair of weak adjoint triplets $(\mathbf{1}, \mathbf{3})_0$, one in each physical chirality. These are representations of the same $\mathbb{Z}_6$ quotient: the weak centre acts trivially in the adjoint representation, and the colour and abelian factors act trivially. The pair is invisible to $\mathrm{supp} Y$ but spans a six-dimensional nontrivial weak sector. Consequently, if these directions are included in the physical source,*

$$\mathrm{rank} P_{0Y}^{\mathrm{nt}} = 6, \quad \Delta_{\mathrm{inv}}^F = 6, \tag{D.22}$$

where P_{0Y}^{nt} is the zero-hypercharge subspace with the fully gauge-trivial part removed.

Proof. The added triplets lie in $\mathrm{Ker} Y$ but not in the gauge-trivial sector, so they are absent from $\mathrm{supp} Y$ and present in P_{gch}^F . Their total dimension is six. Their zero hypercharge makes every local anomaly term containing the abelian generator vanish; opposite physical chiralities cancel the remaining local chiral contributions. They add no weak defining doublet. Nevertheless, their irreducible gauge type differs from every target type, so the full source-minimal inventory detects all six directions as exotic. The trivial action of the quotient kernel ensures that this example does not change the represented global gauge group. $\square$

E Fermion sources, occurrence, and allocation certificates

These results make explicit the completeness and occurrence conditions used in the charged-copy theorem. They also separate physical source support from ambient multiplicity, and process-level grading change from the structure of a chosen generator.

E.1 Complete fermion and typed sources

Proposition E.1 (Finite certificate for completeness of the retained fermion source). *Let $U_F : \mathbb{C}^M \rightarrow \mathcal{H}_F^{\text{amb}}$ synthesize an independently complete physical fermion-source atlas. For an $\mathcal{A}_F$ -reducing candidate carrier $\mathcal{K}$ containing the retained source bank, define*

$$\mathbb{C}_{F|\mathcal{K}} := U_F^*(I - P_{\mathcal{K}})U_F \succeq 0. \quad (\text{E.1})$$

Then

$$\mathbb{C}_{F|\mathcal{K}} = 0 \iff \text{Ran } U_F \subseteq \mathcal{K} \iff \mathcal{A}_F \text{Ran } U_F \subseteq \mathcal{K}. \quad (\text{E.2})$$

Moreover,

$$\begin{aligned} \text{Tr } \mathbb{C}_{F|\mathcal{K}} &= \|(I - P_{\mathcal{K}})U_F\|_{\text{HS}}^2, \\ \text{rank } \mathbb{C}_{F|\mathcal{K}} &= \dim \text{Ran } U_F - \dim(\text{Ran } U_F \cap \mathcal{K}). \end{aligned} \quad (\text{E.3})$$

The zero test is exact; the trace measures source leakage and the rank counts independent source directions modulo $\mathcal{K}$. This rank need not equal the dimension of their subsequent cyclic closure.

Proof. Writing $V = (I - P_{\mathcal{K}})U_F$ gives $\mathbb{C}_{F|\mathcal{K}} = V^*V$, hence positivity, the trace identity, and $\mathbb{C}_{F|\mathcal{K}} = 0$ exactly when $\text{Ran } U_F \subseteq \mathcal{K}$. The latter inclusion is equivalent to $\mathcal{A}_F \text{Ran } U_F \subseteq \mathcal{K}$ because $\mathcal{K}$ is invariant under $\mathcal{A}_F$ and $I \in \mathcal{A}_F$. Finally, apply rank-nullity to $(I - P_{\mathcal{K}})|_{\text{Ran } U_F}$ and use $\text{rank}(V^*V) = \text{rank } V$. $\square$

Exact completeness and quantitative separation. A lower frame bound for U_F on $\text{Ran } U_F$ alone does not separate every nonzero completeness residual from zero: the source range can meet $\mathcal{K}$ at an arbitrarily small angle. A uniform positive separation requires additional geometry, for example $\mathcal{K} \subseteq \text{Ran } U_F$. Appendix E.4 proves the contained-range bound and gives an explicit counterexample without that inclusion.

Lemma E.2 (Complete multiplicity synthesis from the saturated fermion source). *For a populated charged type f , set $N_f = \mathcal{I}_{f, \epsilon_f}$ with the multiplicity inner product of Proposition 6.2. Its evaluation map*

$$\text{ev}_f : V_f \otimes N_f \longrightarrow P_{f, \epsilon_f} \mathcal{K}_F^{\min}, \quad \text{ev}_f(v \otimes T) = Tv$$

is unitary. Let $Z_F : E_{\text{sat}} \rightarrow \mathcal{K}_F^{\min}$ synthesize a terminal source-word bank from Proposition 6.1, so that $\text{Ran } Z_F = \mathcal{K}_F^{\min}$. Define

$$\begin{aligned} \widehat{Z}_f &= \text{ev}_f^* P_{f, \epsilon_f} Z_F, \\ S_f : \overline{V}_f \otimes E_{\text{sat}} &\longrightarrow N_f, \quad S_f(\overline{v} \otimes e) = (\langle v | \otimes I_{N_f}) \widehat{Z}_f e. \end{aligned} \quad (\text{E.4})$$

Then $\text{Ran } S_f = N_f$. Thus the faithful quotient of this typed source Gram is precisely the canonical charged multiplicity, even if the displayed bank has redundant columns.

Proof. For an orthonormal intertwiner basis, the identities $T_a^* T_b = \delta_{ab} I_{V_f}$ make evaluation an isometry onto the complete type block. Since Z_F is onto the saturated carrier, $\widehat{Z}_f$ is onto $V_f \otimes N_f$. Given a unit $v \in V_f$ and $n \in N_f$, choose e with $\widehat{Z}_f e = v \otimes n$. Then $S_f(\overline{v} \otimes e) = n$, proving surjectivity. The conjugate space $\overline{V}_f$ makes this contraction linear in its source coordinates. $\square$

E.2 Source-native allocation formulas

Corollary E.3 (Source-native endpoint allocation). *Under Proposition 6.4, assume $\Delta_f^{\text{anc}}(\Xi_f) = 0$ and $T_f^*T_f \succ 0$. Let $S_f : E_f \rightarrow N_f$ be the complete typed synthesis of Lemma E.2. Then every quantity in the endpoint allocation short of Theorem 6.5 is determined by S_f and the represented ancestry operator Ξ_f through the canonical coefficient (6.21). In particular,*

$$a_{f|G} = 0 \iff \text{Ran } \Xi_f = P_{f,\epsilon_f} \mathcal{K}_F^{\min} \iff n_f = 3, \quad (\text{E.5})$$

where the middle equality is understood after the canonical identification $P_{f,\epsilon_f} \mathcal{K}_F^{\min} \cong V_f \otimes N_f$: namely $\text{Ran } \Xi_f = V_f \otimes N_f$.

Proof. The compiler supplies the unique faithful $T_f : G_{\text{gen}} \rightarrow N_f$ required by Theorem 6.5. Since $\hat{\Xi}_f = I_{V_f} \otimes T_f$, one has $\text{Ran } \Xi_f = V_f \otimes \text{Ran } T_f$ under evaluation coordinates. The zero allocation residual is equivalent to $\text{Ran } T_f = N_f$, and completeness of S_f then gives $n_f = 3$ by (6.27). The converse implications are the same identities in reverse. $\square$

Corollary E.4 (Same-source coefficient formula). *Suppose the actual endpoint subsource is specified by a represented coefficient map $J_{fG} : G_{\text{gen}} \rightarrow E_f$ with $T_f = S_f J_{fG}$ and $J_{fG}^* G_f J_{fG} \succ 0$. Then*

$$\mathbb{A}_{f|G} = G_f - G_f J_{fG} (J_{fG}^* G_f J_{fG})^{-1} J_{fG}^* G_f. \quad (\text{E.6})$$

Thus the allocation residual, its trace, and its excess rank are determined by the same typed source Gram and the specified endpoint compiler, with no independent mixed Gram required. If these data are word native at a flat historical depth, the existing multiplication and Gram tables compute the residual.

Proof. Use $A_f = J_{fG}^* G_f J_{fG}$ and $C_f = J_{fG}^* G_f$ in (6.24). The historical assertion follows from Proposition B.12 and finite matrix operations on the reconstructed Gram. The map J_{fG} must specify the actual endpoint subsource. On the covariant same-carrier branch of Proposition 6.4, it is replaced by the canonical coefficient extracted from Ξ_f ; choosing an arbitrary rank-three coefficient subspace is not a source-occurrence argument. $\square$

E.3 Common-carrier source quotients

The source-short argument also applies when the two banks are interpreted as gravitational and internal variations. It is a Gram-space statement; it does not specify a variational action or its continuum limit.

Let

$$S_g : E_g \rightarrow \mathcal{H}_{\text{cyl}}, \quad S_i : E_i \rightarrow \mathcal{H}_{\text{cyl}}$$

be the source syntheses of the complete gravitational and internal physical variation banks on one same-history cylinder. Assume the gravitational bank has already been reduced to its faithful quotient and put

$$G_g = S_g^* S_g, \quad G_i = S_i^* S_i, \quad C = S_i^* S_g.$$

Proposition E.5 (Universal coupled source quotient). *Define*

$$G_{i|g} = G_i - C G_g^{-1} C^* = S_i^* (I - P_g) S_i. \quad (\text{E.7})$$

Then

$$\begin{pmatrix} G_g & C^* \\ C & G_i \end{pmatrix} = \begin{pmatrix} I & 0 \\ C G_g^{-1} & I \end{pmatrix} \begin{pmatrix} G_g & 0 \\ 0 & G_{i|g} \end{pmatrix} \begin{pmatrix} I & G_g^{-1} C^* \\ 0 & I \end{pmatrix}. \quad (\text{E.8})$$

Consequently:

- (i) the joint variation carrier is faithful exactly when $G_{\text{ig}} \succ 0$;
- (ii) the number of genuinely new internal source directions is $\text{rank } G_{\text{ig}}$;
- (iii) the source-minimal internal variation space is

$$E_{\text{ig}}^{\min} = E_{\text{i}} / \text{Ker } G_{\text{ig}}.$$

Proof. The first equality in (E.7) is the Gram of $(I - P_g)S_{\text{i}}$. Direct block multiplication gives (E.8); the remaining statements are the standard Schur-complement rank and positivity consequences. $\square$

Thus “adding the Standard Model” does not mean appending every internal variation as a new primitive. Only the component orthogonal to the already represented gravitational source counts as genuinely new.

E.4 Completeness residuals and quantitative allocation thresholds

Example E.6 (A source frame bound does not control an arbitrary transverse candidate). On $\mathbb{C}^3$ take $\mathcal{A}_F = \mathbb{C}I$, let the retained source be $B_F z = z e_1$, and let $U_F : \mathbb{C}^2 \rightarrow \mathbb{C}^3$ be $U_F(a, b) = a e_1 + b e_2$. It has frame bound one on $H_U = \text{span}\{e_1, e_2\}$. For $0 < \theta < \pi/2$, the carrier

$$\mathcal{K}_\theta = \text{span}\{e_1, \cos \theta e_2 + \sin \theta e_3\}$$

reduces $\mathcal{A}_F$ and contains the retained source, but

$$U_F^*(I - P_{\mathcal{K}_\theta})U_F = \begin{pmatrix} 0 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}. \quad (\text{E.9})$$

Thus a nonzero completeness residual can have arbitrarily small norm while the frame bound remains one. Notice that $\mathcal{K}_\theta \not\subseteq H_U$.

Proposition E.7 (Contained-source frame separation). Let $U : E \rightarrow \mathcal{H}$ have range H_U and satisfy $UU^* \succeq mP_{H_U}$ with $m > 0$. If $K \subseteq H_U$, put

$$C_{U|K} = U^*(I - P_K)U, \quad r = \dim H_U - \dim K.$$

Then

$$\text{rank } C_{U|K} = r, \quad \text{Tr } C_{U|K} \geq mr. \quad (\text{E.10})$$

If $r > 0$, every positive eigenvalue of $C_{U|K}$ is at least m . Consequently $C_{U|K} \neq 0$ implies $\|C_{U|K}\|_{\text{op}} \geq m$; equivalently, $\text{Tr } C_{U|K} < m$ forces $r = 0$ and $C_{U|K} = 0$.

Proof. Since $K \subseteq H_U$, the operator $Q = P_{H_U} - P_K$ is the orthogonal projection onto $H_U \cap K^\perp$, and $(I - P_K)U = QU$. Thus

$$C_{U|K} = (QU)^*(QU), \quad (QU)(QU)^* = QUU^*Q \succeq mQ.$$

On $\text{Ran } Q$ the latter operator is strictly positive; outside $\text{Ran } Q$ it vanishes. It therefore has exactly r positive eigenvalues, all at least m . The two finite Gram operators have the same nonzero spectrum, proving the assertions. The conclusion includes the zero-dimensional case. $\square$

For the complete typed source of Theorem 6.5, one has $H_U = N_f$ and $K = \text{Ran } T_f \subseteq N_f$. Hence an independently known bound $S_f S_f^* \succeq m_f I_{N_f}$ yields

$$a_{f|G} \geq m_f(n_f - 3), \quad a_{f|G} < m_f \implies n_f = 3. \quad (\text{E.11})$$

In particular, if a measured scalar $\hat{a}$ satisfies $|\hat{a} - a_{f|G}| \leq \varepsilon_a$ and $\hat{a} + \varepsilon_a < m_f$, the exact three-dimensional allocation follows. This implication requires the frame bound and completeness to be verified independently of the allocation measurement.

Example E.8 (Small allocation mass without a uniform source floor). *Let $N_f = \mathbb{C}^4$, take $T_f : \mathbb{C}^3 \rightarrow \mathbb{C}^4$ to be the inclusion of the first three coordinate axes, and put*

$$S_{f,t} = \text{diag}(1, 1, 1, t), \quad 0 < t \leq 1.$$

Every $S_{f,t}$ is a complete source for N_f , yet

$$\mathbb{A}_{f|G,t} = \text{diag}(0, 0, 0, t^2), \quad a_{f|G,t} = t^2, \quad n_f = 4. \quad (\text{E.12})$$

Thus small positive allocation mass cannot alone certify exactly three multiplicity directions. The least source frame bound is $m_f = t^2$, so the quantitative estimate (E.11) is sharp.

Source range and physical multiplicity. For an arbitrary synthesis $S : E \rightarrow \mathcal{H}$ and faithful $T : G_{\text{gen}} \rightarrow \mathcal{H}$ with $\text{Ran } T \subseteq \text{Ran } S$, the same proof as Theorem 6.5 gives

$$\text{rank}(S^* S) = 3 + \text{rank}(S^*(I - P_{\text{Ran } T})S).$$

This identity counts the source range. It counts the canonical charged multiplicity only when that range is its complete typed realization. Lemma E.2 provides precisely this identification from a saturated physical fermion bank. An incomplete coefficient bank, an unread Kraus index, or a source on an unrelated carrier cannot replace that identification.

E.5 Kraus-independent grading-changing occurrence

The Clifford and internal packets must occur on one carrier. Let $J = P_+ - P_-$ be the protected pre-central grading and let $\sigma_0, \dots, \sigma_3$ be Hermitian Clifford axes generating the external matrix factor. Before selecting a coherent grading-odd generator, primitive matter occurrence can be tested directly at the completely positive process level.

For a completely positive map

$$\Phi(X) = \sum_{\alpha} K_{\alpha} X K_{\alpha}^*$$

on the active carrier, let J_{Φ} be its Choi matrix. On Hilbert–Schmidt operator space define

$$\Theta_J(X) = JXJ, \quad \mathbf{P}_{\text{odd}} = \frac{1}{2}(I - \Theta_J).$$

Proposition E.9 (Kraus-independent odd-Choi occurrence). *The quantity*

$$\mu_J(\Phi) := \text{Tr}(\mathbf{P}_{\text{odd}} J_{\Phi}) \quad (\text{E.13})$$

is independent of the Kraus representation and satisfies

$$\begin{aligned}
\mu_J(\Phi) &= \sum_{\alpha} \|(K_{\alpha})_{\text{odd}}\|_{\text{HS}}^2 \\
&= \text{Tr}[P_- \Phi(P_+)] + \text{Tr}[P_+ \Phi(P_-)] \\
&= \sum_{\alpha} (\|P_- K_{\alpha} P_+\|_{\text{HS}}^2 + \|P_+ K_{\alpha} P_-\|_{\text{HS}}^2),
\end{aligned} \tag{E.14}$$

where $(K_{\alpha})_{\text{odd}} = (K_{\alpha} - JK_{\alpha}J)/2$. Moreover,

$$\mu_J(\Phi) = 0 \iff \text{Ran } J_{\Phi} \text{ is supported on grading-even operators.} \tag{E.15}$$

Equivalently, every Kraus direction commutes with J . If $d_{\pm} = \text{Tr } P_{\pm} > 0$ and $\rho_{\pm} = P_{\pm}/d_{\pm}$, then

$$\mu_J(\Phi) = d_+ \text{Tr}[P_- \Phi(\rho_+)] + d_- \text{Tr}[P_+ \Phi(\rho_-)]. \tag{E.16}$$

Thus primitive grading change is decidable from two positive cross-sign population experiments without choosing Kraus phases.

Proof. Vectorization gives $J_{\Phi} = \sum_{\alpha} |K_{\alpha}\rangle\rangle\langle\langle K_{\alpha}|$. Taking the trace against the orthogonal projector $\mathbf{P}_{\text{odd}}$ gives the first line of (E.14) and proves Kraus-coordinate independence. For one Kraus operator,

$$\text{Tr}(P_- K P_+ K^*) = \|P_- K P_+\|_{\text{HS}}^2, \quad \text{Tr}(P_+ K P_- K^*) = \|P_+ K P_-\|_{\text{HS}}^2.$$

The two off-diagonal corners are Hilbert–Schmidt orthogonal and sum to the odd part of K , proving the remaining identities. Since every summand is nonnegative, the mass vanishes exactly when the Choi support lies in the even operator subspace. Substituting $P_{\pm} = d_{\pm}\rho_{\pm}$ gives (E.16). $\square$

Corollary E.10 (Primitive active no-matter or occurrence alternative). *Let Σ be a finite alphabet of actual autonomous CP letters and put*

$$\mathbf{m}_{\text{odd}}(\Sigma) := \sum_{a \in \Sigma} \mu_J(\Phi_a). \tag{E.17}$$

Assume the autonomous word grammar is generated by these letters together with grading-even passive writers and initializations. Then

$$\mathbf{m}_{\text{odd}}(\Sigma) = 0 \iff \text{every finite autonomous resolved word is grading even.} \tag{E.18}$$

If $\mathbf{m}_{\text{odd}}(\Sigma) > 0$, at least one represented primitive letter has nonzero cross-sign support. This is a process-level occurrence statement; identifying a particular one-route weak factor may require a separate selector when the corresponding flip Choi support has rank greater than one.

Proof. A finite sum of nonnegative odd masses vanishes exactly when every primitive Choi support is grading even by Proposition E.9. Products of grading-even Kraus directions remain grading even, and the stated writers and initializations do not change parity, proving the forward implication. The converse follows already from words of length one. Strict positivity of the sum forces at least one positive primitive summand. $\square$

E.6 Active grading change and unread spectators

Let $Z = Z^* = Z^{-1}$ be a grading on a future-visible active carrier $\mathcal{H}_{\text{act}}$, with $P_{\pm} = (I \pm Z)/2$. For a completely positive map Φ on that carrier define its active cross-sign mass by

$$\mu_Z^{\text{act}}(\Phi) = \text{Tr}[P_- \Phi(P_+)] + \text{Tr}[P_+ \Phi(P_-)]. \quad (\text{E.19})$$

This quantity asks only whether the represented active map transfers positive weight between the two grading sectors.

Proposition E.11 (Spectator-lift invariance of active grading change). *Let $\mathcal{S}$ be an unread spectator, let σ be a spectator state, and let Ψ be any trace-preserving completely positive map on $\mathcal{S}$. For the product lift*

$$\tilde{\Phi} = \Phi \otimes \Psi$$

and active preparations $P_{\pm} \otimes \sigma$ with active Reads $P_{\mp} \otimes I_{\mathcal{S}}$, one has

$$\begin{aligned} \text{Tr}[(P_- \otimes I)\tilde{\Phi}(P_+ \otimes \sigma)] + \text{Tr}[(P_+ \otimes I)\tilde{\Phi}(P_- \otimes \sigma)] \\ = \mu_Z^{\text{act}}(\Phi). \end{aligned} \quad (\text{E.20})$$

Thus active grading-changing occurrence is invariant under arbitrary unread trace-preserving spectator dynamics. By contrast, the total grading-odd mass of a chosen microscopic dilation need not be operationally identifiable; an explicit continuous family with identical visible dynamics is given in Appendix E.6.1.

Proof. Trace preservation gives $\text{Tr} \Psi(\sigma) = 1$. Hence

$$\text{Tr}[(P_- \otimes I)(\Phi \otimes \Psi)(P_+ \otimes \sigma)] = \text{Tr}[P_- \Phi(P_+)] \text{Tr} \Psi(\sigma) = \text{Tr}[P_- \Phi(P_+)],$$

and the same argument applies to the reverse sign. Adding the two identities gives (E.20). $\square$

Hidden provenance is not active matter. A passive Read of a future-sealed hidden sector may establish microscopic provenance, but it does not make that sector part of the active matter packet. For hidden structure to contribute to a typed matter incidence it must return to the active carrier through a represented router. Such a return map is additional physical loading data; it cannot be supplied by an unread spectator or inferred from unchanged visible record marginals.

E.6.1 A microscopic odd-mass ambiguity

Proposition E.12 (Microscopic odd mass is not fixed by visible dynamics). *Fix a visible carrier of dimension $d \geq 1$ and take its visible primitive map to be the identity channel. Let an unread spectator be $\mathcal{S} = \mathbb{C}^2$ with grading $Z_{\mathcal{S}} = \sigma_z$. For $0 \leq \theta \leq \pi/2$ put*

$$U_{\theta} = \cos \theta I + i \sin \theta \sigma_x, \quad \Psi_{\theta} = \text{Ad}_{U_{\theta}}.$$

Every lifted map $\text{id} \otimes \Psi_{\theta}$ induces the same visible identity channel, and every visible multitime word is unchanged after the spectator is discarded. Nevertheless, for any visible grading Z_{act} and the total grading $\tilde{Z} = Z_{\text{act}} \otimes Z_{\mathcal{S}}$, the odd Choi mass of the lift is

$$\mu_{\tilde{Z}}(\text{id} \otimes \Psi_{\theta}) = 2d \sin^2 \theta. \quad (\text{E.21})$$

Thus operationally indistinguishable lifts can have continuously different microscopic odd mass; by enlarging the unread spectator multiplicity this quantity can be increased without changing the visible process.

Proof. Discarding a trace-preserving spectator leaves the visible identity channel at every word depth. Relative to Z_S ,

$$(U_\theta)_{\text{ev}} = \cos \theta I, \quad (U_\theta)_{\text{odd}} = i \sin \theta \sigma_x.$$

The unique Kraus operator of the lifted unitary channel is $I_d \otimes U_\theta$. Its odd part relative to $\tilde{Z}$ is $I_d \otimes (U_\theta)_{\text{odd}}$, hence

$$\|I_d \otimes (U_\theta)_{\text{odd}}\|_{\text{HS}}^2 = d \|i \sin \theta \sigma_x\|_{\text{HS}}^2 = 2d \sin^2 \theta.$$

Proposition E.9 identifies this squared norm with the odd Choi mass. The visible active mass, by contrast, remains zero by Proposition E.11. $\square$

E.6.2 The narrower Clifford-occurrence margin

The quantity in (E.14) is the active cross-sign mass of (E.19); hence Proposition E.11 shows that this primitive occurrence test is unchanged by unread trace-preserving spectator lifts. The Clifford margin below is a narrower test: it asks whether the represented external Clifford action itself changes the same grading.

Definition E.13 (Clifford matter margin). Define

$$p_{\text{Cl}} = \frac{1}{2 \dim \mathcal{H}} \sum_{\mu=0}^3 \|P_- \sigma_\mu P_+\|_{\text{HS}}^2. \quad (\text{E.22})$$

Proposition E.14 (Clifford occurrence test). *One has $p_{\text{Cl}} > 0$ exactly when at least one represented Clifford axis changes the protected grading. If $p_{\text{Cl}} = 0$, the grading reduces the external Clifford action and its residual action lies in the normal multiplicity commutant.*

Proof. Every summand in (E.22) is nonnegative. The sum vanishes exactly when $P_- \sigma_\mu P_+ = 0$ for every μ , and self-adjointness then also gives $P_+ \sigma_\mu P_- = 0$. Thus J commutes with the generated external algebra. $\square$

E.7 Weak gauge-cyclic support and a rank-one counterexample

For completeness, the range step in Proposition 6.7 can be checked directly. For any $x \in W_2 \otimes M_H$,

$$\langle x, \bar{J}_H x \rangle = \int_{\text{SU}(W_2)} \|J_H^{1/2}(g^* \otimes I)x\|^2 dg.$$

The integrand is continuous and nonnegative and Haar measure has full support. The integral therefore vanishes exactly when every integrand vanishes. Since $\text{Ran } J_H^{1/2} = \text{Ran } J_H$ in finite dimension,

$$\text{Ker } \bar{J}_H = \bigcap_g ((g \otimes I) \text{Ran } J_H)^\perp = (\mathcal{K}_H^{\text{occ}})^\perp.$$

Taking orthogonal complements gives the claimed gauge-cyclic range.

Let (w_1, w_2) and (m_1, m_2) be orthonormal bases of W_2 and M_H , and put

$$\psi = \frac{w_1 \otimes m_1 + w_2 \otimes m_2}{\sqrt{2}}, \quad J_H = |\psi\rangle\langle\psi|.$$

Then $\text{rank } J_H = 1$ but

$$B_H = \frac{1}{2} I_{M_H}, \quad \bar{J}_H = \frac{1}{4} I_{W_2 \otimes M_H}, \quad \mathfrak{p}_H = \frac{1}{2}.$$

Proposition 6.7 therefore gives $\mathcal{K}_H^{\text{occ}} = W_2 \otimes M_H$ and $g_H^{\text{occ}} = 2$. The number of positive coefficient vectors is not the number of reached weak-doublet copies; the latter is the multiplicity of the gauge-cyclic support.

E.8 One-generation incidence and component structure

The commutant language also gives a compact check of how Yukawa incidence reduces residual scalar symmetries inside one generation.

Let

$$\mathcal{F}_{15} = Q \oplus u \oplus d \oplus L \oplus e$$

with the modules of (5.18). Ignoring generation multiplicity for the moment, the gauge-generated algebra is

$$\mathcal{A}_{\text{gauge}} = M_6(\mathbb{C}) \oplus M_3(\mathbb{C}) \oplus M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}, \quad (\text{E.23})$$

because the five modules are irreducible and pairwise inequivalent. Hence

$$\mathcal{A}'_{\text{gauge}} \cong \mathbb{C}^5.$$

Proposition E.15 (Incidence reduces the one-generation commutant). *Assume the three Higgs-contracted incidence histories*

$$Q \leftrightarrow u, \quad Q \leftrightarrow d, \quad L \leftrightarrow e$$

are nonzero. Then

$$\mathcal{A}_{\text{gauge+inc}} = M_{12}(\mathbb{C}) \oplus M_3(\mathbb{C}), \quad \mathcal{A}'_{\text{gauge+inc}} \cong \mathbb{C}^2.$$

The two residual central directions are the quark and lepton connected components. Adding a neutral singlet and a nonzero L -neutral incidence changes the second matrix block to $M_4(\mathbb{C})$ while leaving the residual commutant $\mathbb{C}^2$.

Proof. A nonzero block between two full matrix summands, together with multiplication by endpoint matrix units, generates every rank-one map between the summands. Products and adjoints therefore generate the full matrix algebra on each connected component of the incidence graph. The three displayed edges have components $\{Q, u, d\}$ and $\{L, e\}$, of dimensions 12 and 3. $\square$

This audit is intentionally distinct from the generation multiplicity theorem. It describes the commutant inside one fixed generation packet. The rank-three generation carrier of Proposition 4.1 is an additional multiplicity factor reconstructed before numerical flavour breaking is specified.

For the component ancestry hypotheses of Proposition 6.6, the corresponding explicit three-copy decomposition is

$$\mathcal{H}_{\text{mat}} \cong (V_Q \oplus V_u \oplus V_d) \otimes G_q \oplus (V_L \oplus V_e [\oplus V_\nu]) \otimes G_\ell, \quad \dim G_q = \dim G_\ell = 3.$$

The neutral-singlet term requires its own represented ancestry hypotheses; its multiplicity is not fixed by the charged anomaly equations.

F Stability estimates and compact-Mosco proofs

This appendix gives the incidence/flavour decomposition and coefficient perturbation estimates, followed by the complete varying-space spectral argument. The cofinal kernel theorem in the main text does not require the compactness hypotheses used in the latter argument.

F.1 Type-central connectivity and conditional flavour rigidity

Proposition F.1 (Type-central incidence graph Laplacian). *Let P_v be the represented orthogonal type projections, put $h_v = \text{rank } P_v$, and let typed incidences satisfy $Y_e = P_{t(e)}Y_eP_{s(e)}$. Set $w_e = \|Y_e\|_{\text{HS}}^2$, let B be the oriented graph incidence matrix, and define*

$$H = \text{diag}(h_v), \quad \mathcal{L}_{\text{cen}} = H^{-1/2}B^* \text{diag}(2w_e)BH^{-1/2}.$$

For the type-central operator $X_z = \sum_v z_v P_v$,

$$\sum_e (\| [X_z, Y_e] \|_{\text{HS}}^2 + \| [X_z, Y_e^*] \|_{\text{HS}}^2) = 2 \sum_e w_e |z_{t(e)} - z_{s(e)}|^2. \quad (\text{F.1})$$

If the positive-weight incidence graph has connected components $C_1, \dots, C_c$, then

$$\text{Ker } \mathcal{L}_{\text{cen}} = H^{1/2} \text{span}\{\mathbf{1}_{C_1}, \dots, \mathbf{1}_{C_c}\}, \quad (\text{F.2})$$

and this kernel corresponds exactly to the block-scalar centre $\text{span}\{P_{C_a} : 1 \leq a \leq c\}$, where $P_C = \sum_{v \in C} P_v$. The Hilbert–Schmidt projection of X_z onto that centre replaces the coefficients on each component by the weighted average

$$\bar{z}_C = \frac{\sum_{v \in C} h_v z_v}{\sum_{v \in C} h_v}.$$

Consequently the first positive eigenvalue of $\mathcal{L}_{\text{cen}}$ is the exact type-central rigidity gap of the connected-component centre.

Proof. On a typed edge $e : s \rightarrow t$, $[X_z, Y_e] = (z_t - z_s)Y_e$ and $[X_z, Y_e^*] = (z_s - z_t)Y_e^*$. Summing the squared norms proves (F.1). The energy vanishes exactly when z is constant on every positive-weight connected component, proving (F.2). Orthogonality of the type projections and $\|P_v\|_{\text{HS}}^2 = h_v$ gives the weighted component-average projector and the normalized spectral-gap statement. $\square$

Corollary F.2 (Incidence–flavour factorization of the Howe certificate). *Assume on a structural residue branch that each represented incidence factors as $Y_e = D_e \otimes F_e$ on its typed module and generation coefficient spaces. On the block-diagonal multiplicity test space, decompose the tested complement of the protected algebra into orthogonal type-central contrasts and noncentral flavour directions. Put $s = s(e)$, $t = t(e)$ and $\kappa_e = \|D_e\|_{\text{HS}}^2$. Retaining both the incidence and its adjoint, define*

$$\begin{aligned} (\mathcal{D}_{\text{cen}} z)_e &= \sqrt{\kappa_e}((z_t - z_s)F_e, (z_s - z_t)F_e^*), \\ (\mathcal{D}_{\text{fl}} X)_e &= \sqrt{\kappa_e}(X_t F_e - F_e X_s, X_s F_e^* - F_e^* X_t). \end{aligned}$$

Then the relative Howe Gram restricted to this test space has block form

$$A_{\text{cen}} = \mathcal{D}_{\text{cen}}^* \mathcal{D}_{\text{cen}}, \quad B = \mathcal{D}_{\text{cen}}^* \mathcal{D}_{\text{fl}}, \quad C_{\text{fl}} = \mathcal{D}_{\text{fl}}^* \mathcal{D}_{\text{fl}} + C_{\text{extra}},$$

where $C_{\text{extra}} \succeq 0$ contains the non-incidence flavour commutators. When $A_{\text{cen}} \succ 0$ on the declared type-central contrast space,

$$S_{\text{fl}|\text{cen}} = C_{\text{extra}} + \mathcal{D}_{\text{fl}}^*(I - P_{\text{cen}})\mathcal{D}_{\text{fl}}, \quad P_{\text{cen}} = \mathcal{D}_{\text{cen}}(\mathcal{D}_{\text{cen}}^* \mathcal{D}_{\text{cen}})^{-1} \mathcal{D}_{\text{cen}}^*. \quad (\text{F.3})$$

On this test space, exact rigidity separates into incidence connectivity and conditional flavour rigidity. In particular, if deleting a displayed incidence edge would create an additional type-central commutant direction excluded from the proposed protected algebra, then $\mathbb{G}_{\text{Howe}} \succ 0$ already forces that edge to occur.

Proof. For a type-central contrast, the commutator with Y_e is the scalar endpoint difference multiplied by Y_e ; for a noncentral multiplicity variation it is the coefficient defect $X_{t(e)}F_e - F_eX_{s(e)}$. The adjoint incidence contributes $X_{s(e)}F_e^* - F_e^*X_{t(e)}$ independently on complex operator space. Summing both Hilbert–Schmidt pairings gives the three blocks of the restricted Gram. The displayed Schur complement is the standard identity $C - B^*A^{-1}B$ with $P_{\text{cen}} = \mathcal{D}_{\text{cen}}A^{-1}\mathcal{D}_{\text{cen}}^*$. The final claim follows from Theorem 7.1: a missing edge would leave an additional exact central zero mode outside the proposed protected kernel. $\square$

F.2 Perturbed coefficient spaces and one-sided rank certification

Proposition F.3 (Stable dominant coefficient space under incidence error). *Let an exact incidence Y have operator–Schmidt rank r , smallest nonzero Schmidt coefficient $s > 0$, and largest coefficient L . Suppose $\|\tilde{Y} - Y\|_{\text{HS}} \leq \varepsilon$ and let $H_Y, H_{\tilde{Y}}$ be the slice Grams of Proposition 3.6. Then*

$$|\tilde{\sigma}_k - \sigma_k| \leq \varepsilon. \quad (\text{F.4})$$

Hence $\varepsilon < s/2$ separates an r -dimensional dominant coefficient cluster. More quantitatively, put

$$\eta = (2L + \varepsilon)\varepsilon.$$

If $\eta < s^2/2$, and

$$P = \mathbf{1}_{(0, \infty)}(H_Y), \quad \tilde{P} = \mathbf{1}_{[s^2/2, \infty)}(H_{\tilde{Y}}),$$

then $\text{rank } \tilde{P} = r$ and

$$\|\tilde{P} - P\|_{\text{op}} \leq \frac{\eta}{s^2 - \eta}. \quad (\text{F.5})$$

Thus a separated operator–Schmidt block yields a noise-stable coefficient subspace even though literal exact slice rank need not be stable.

Proof. The slice map is linear in Y and $\|\mathcal{S}_{\tilde{Y}} - \mathcal{S}_Y\|_{\text{op}} \leq \|\tilde{Y} - Y\|_{\text{HS}}$, so singular-value perturbation gives (F.4). Also

$$\|H_{\tilde{Y}} - H_Y\|_{\text{op}} \leq (2L + \varepsilon)\varepsilon = \eta.$$

The exact positive spectrum of H_Y lies in $[s^2, \infty)$, while the remaining spectrum is zero. Weyl’s eigenvalue bound therefore gives exactly r eigenvalues of $H_{\tilde{Y}}$ above $s^2/2$ when $\eta < s^2/2$. With $Q = I - P$, one has $QH_Y = 0$ and $H_{\tilde{Y}}|_{\text{Ran } \tilde{P}} \succeq (s^2 - \eta)I$, whence

$$Q\tilde{P} = Q(H_{\tilde{Y}} - H_Y)\tilde{P}(H_{\tilde{Y}}|_{\text{Ran } \tilde{P}})^{-1}.$$

Taking norms and using equality of the two projector ranks proves (F.5). $\square$

Corollary F.4 (One-sided operator–Schmidt rank certificate). *If an observed incidence $\hat{Y}$ obeys $\|\hat{Y} - Y\|_{\text{HS}} \leq \varepsilon$, then*

$$\sigma_k(Y) \geq (\hat{\sigma}_k - \varepsilon)_+, \quad \text{rank } Y \geq \#\{k : \hat{\sigma}_k > \varepsilon\}. \quad (\text{F.6})$$

Without an independent upper-rank premise, a coefficient below the error floor cannot certify exact absence of that Schmidt direction.

Proof. Apply (F.4). The final caveat is sharp because an orthogonal Schmidt term of arbitrarily small norm can be added whenever the ambient tensor spaces admit one more coefficient direction. $\square$

F.3 Compact-Mosco resolvent convergence

Proposition F.5 (Compact Mosco convergence implies norm-resolvent convergence). *If $q_X \rightarrow q_\infty$ in the sense of Definition 7.5 and the family is asymptotically compact in the sense of Definition 7.8, then*

$$\|R_X - R_\infty\|_{\text{op}} \rightarrow 0. \quad (\text{F.7})$$

Consequently, if $0 \leq \lambda_{1,X} \leq \lambda_{2,X} \leq \dots$ denote the discrete screened eigenvalues of $\mathcal{L}_X$, repeated with multiplicity, then for every fixed index in the limiting discrete spectrum,

$$\lambda_{k,X} \rightarrow \lambda_{k,\infty}. \quad (\text{F.8})$$

If an isolated spectral cluster of $\mathcal{L}_\infty$ is separated by a positive gap, the corresponding transported Riesz projections converge in operator norm and their ranks are eventually constant.

We establish the resolvent conclusion through its variational formulation and a collective-compactness argument, and then treat the low spectrum.

Retain the embeddings $J_X : \mathcal{H}_X \rightarrow \mathcal{H}$ and the closed nonnegative forms q_X of Definition 7.5 and Definition 7.8. For $f \in \mathcal{H}$ let

$$u_X = (\mathcal{L}_X + I)^{-1} J_X^* f. \quad (\text{F.9})$$

Equivalently, u_X is the unique minimizer on $D(q_X)$ of

$$\mathcal{F}_{X,f}(u) = q_X(u) + \|u\|^2 - 2 \operatorname{Re} \langle J_X u, f \rangle_{\mathcal{H}}. \quad (\text{F.10})$$

Writing $q_X(\cdot, \cdot)$ for the sesquilinear form obtained by polarizing q_X , the Euler identity is

$$q_X(u_X, v) + \langle u_X, v \rangle = \langle f, J_X v \rangle_{\mathcal{H}}, \quad v \in D(q_X). \quad (\text{F.11})$$

Taking $v = u_X$ gives

$$\|u_X\|^2 + q_X(u_X) \leq \|f\| \|u_X\| \leq \|f\|^2, \quad \|u_X\| \leq \|f\|. \quad (\text{F.12})$$

Lemma F.6 (Mosco convergence gives strong transported resolvents). *Under Definition 7.5, for every fixed $f \in \mathcal{H}$,*

$$J_X(\mathcal{L}_X + I)^{-1} J_X^* f \rightarrow J_\infty(\mathcal{L}_\infty + I)^{-1} J_\infty^* f \quad (\text{F.13})$$

strongly in $\mathcal{H}$.

Proof. The estimate (F.12) makes $(J_X u_X)$ bounded. Every weakly convergent subsequence has, by the Mosco liminf inequality, a limit $J_\infty u$ satisfying

$$q_\infty(u) + \|u\|^2 - 2 \operatorname{Re} \langle J_\infty u, f \rangle \leq \liminf_X \mathcal{F}_{X,f}(u_X).$$

For every $v \in D(q_\infty)$ choose a Mosco recovery sequence v_X . Minimality of u_X gives $\mathcal{F}_{X,f}(u_X) \leq \mathcal{F}_{X,f}(v_X)$; passing to the limit shows that u minimizes $\mathcal{F}_{\infty,f}$. Strict convexity identifies $u = (\mathcal{L}_\infty + I)^{-1} J_\infty^* f$ and therefore identifies every weak cluster point. Comparing the minimum values and using the liminf inequality gives convergence of $\|u_X\|^2 + q_X(u_X)$ to the limiting value. Weak convergence together with convergence of the Hilbert norms yields strong convergence of $J_X u_X$. Thus the whole sequence converges. $\square$

Lemma F.7 (Asymptotic compactness gives collective compactness). *Assume Definition 7.8. If (f_X) is bounded in $\mathcal{H}$, then every sequence*

$$J_X(\mathcal{L}_X + I)^{-1} J_X^* f_X$$

has a strongly convergent subsequence. Hence the transported resolvents (R_X) form a collectively compact family. The limiting resolvent R_∞ is compact.

Proof. Apply (F.12) with $f = f_X$. The corresponding resolvent vectors have uniformly bounded norms and form energies, so asymptotic compactness gives a strongly convergent subsequence. The same argument for the limiting form, applied to an arbitrary bounded sequence of right-hand sides, proves compactness of R_∞ . $\square$

Lemma F.8 (Collectively compact self-adjoint convergence is norm convergence). *Let T_X be bounded self-adjoint operators on a Hilbert space. If $T_X \rightarrow T$ strongly and (T_X) is collectively compact, then T is compact and*

$$\|T_X - T\|_{\text{op}} \rightarrow 0. \quad (\text{F.14})$$

Proof. Let B_1 be the closed unit ball and let

$$K = \overline{\bigcup_X T_X(B_1)}.$$

Collective compactness means that K is compact. For every fixed $x \in B_1$, strong convergence gives $T_X x \rightarrow Tx$, while every $T_X x$ belongs to K ; hence $Tx \in K$. Thus $T(B_1) \subseteq K$, so T is compact. This argument uses strong convergence only at a fixed vector and avoids any uniformity claim on moving vectors.

If (F.14) failed, there would be a cofinal sequence of indices, unit vectors x_X , and $\varepsilon > 0$ with $\|(T_X - T)x_X\| \geq \varepsilon$. Passing to a subsequence, weak compactness of the unit ball gives $x_X \rightharpoonup x$, collective compactness gives $T_X x_X \rightarrow y$, and compactness of T gives $Tx_X \rightarrow Tx$. For every fixed z , self-adjointness and strong convergence at the fixed vector z give

$$\langle y, z \rangle = \lim_X \langle T_X x_X, z \rangle = \lim_X \langle x_X, T_X z \rangle = \langle x, Tz \rangle = \langle Tx, z \rangle.$$

Thus $y = Tx$, so $(T_X - T)x_X \rightarrow 0$, contradicting the choice of the sequence. $\square$

The strong-resolvent lemma, the collective-compactness lemma, and the self-adjoint norm-convergence lemma prove $\|R_X - R_\infty\|_{\text{op}} \rightarrow 0$. The next lemma completes the spectral claims of Proposition F.5.

F.4 Low spectrum and the alternative protected-kernel criterion

Let $\mu_{1,X} \geq \mu_{2,X} \geq \dots > 0$ denote the nonzero eigenvalues of the compact positive operator R_X , repeated with multiplicity. On the screened space they are related to the eigenvalues of $\mathcal{L}_X$ by

$$\mu_{k,X} = \frac{1}{1 + \lambda_{k,X}}. \quad (\text{F.15})$$

Lemma F.9 (Low-spectrum convergence). *If $\|R_X - R_\infty\| \rightarrow 0$, then for every fixed k ,*

$$\mu_{k,X} \rightarrow \mu_{k,\infty}, \quad \lambda_{k,X} \rightarrow \lambda_{k,\infty} \quad (\text{F.16})$$

whenever the limiting eigenvalue is finite. If Σ is an isolated finite spectral cluster of R_∞ and Γ is a positively oriented contour separating it from the remaining spectrum, then the transported Riesz projections

$$\Pi_X(\Gamma) = \frac{1}{2\pi i} \int_{\Gamma} (z - R_X)^{-1} dz \quad (\text{F.17})$$

converge to $\Pi_\infty(\Gamma)$ in operator norm. Their ranks are eventually equal.

Proof. For compact self-adjoint operators the min-max principle gives $|\mu_{k,X} - \mu_{k,\infty}| \leq \|R_X - R_\infty\|$ for each fixed k . Equation (F.15) gives the second convergence. For the projection statement, norm convergence and the resolvent identity imply uniform convergence of $(z - R_X)^{-1}$ on Γ ; integrate around the contour. If two projections differ in operator norm by less than one, their ranges have the same finite dimension, proving eventual rank stability. $\square$

Proposition F.10 (Protected-kernel locking). *Suppose each $\mathcal{M}_X$ is an m -dimensional subspace of $\text{Ker } \mathcal{L}_X$ and $\text{Ker } \mathcal{L}_\infty = \mathcal{M}_\infty$ has dimension m . Under compact Mosco convergence, for all sufficiently late X ,*

$$\text{Ker } \mathcal{L}_X = \mathcal{M}_X,$$

the transported kernel projections converge in operator norm, and

$$\lambda_{m+1}(\mathcal{L}_X) \rightarrow \lambda_{m+1}(\mathcal{L}_\infty) > 0.$$

Proof. The eigenvalue 0 of $\mathcal{L}_\infty$ corresponds to the eigenvalue 1 of R_∞ , of multiplicity m . Compactness of the limiting resolvent and finite-dimensionality of the kernel separate this eigenvalue from the rest of the spectrum. Lemma F.9 therefore gives a nearby spectral cluster of rank exactly m for all sufficiently late X . Because $\mathcal{M}_X \subseteq \text{Ker } \mathcal{L}_X$ already supplies m exact copies of the eigenvalue 1 of R_X , that entire cluster is the kernel and no extra zero mode can occur. The same lemma gives projection and next-eigenvalue convergence. $\square$

This is a distinct spectral criterion: it assumes an exact limiting kernel and uses compact-Mosco convergence to lock the late finite kernels. By contrast, Theorem 7.7 derives limiting kernel exactness from protected transport and a uniform finite complement gap, without compactness.

F.5 Why protected transport is indispensable

Proposition F.11 (A uniformly gapped protected mode can escape). *Fixed protected dimension, transported Mosco convergence, and a uniform finite-cutoff complement gap do not by themselves produce a nonzero protected limiting kernel. On $\ell^2(\mathbb{N})$ let e_n be the standard basis, $\mathcal{M}_n = \mathbb{C}e_n$, $P_n = |e_n\rangle\langle e_n|$, and*

$$q_n[x] = \|(I - P_n)x\|^2. \quad (\text{F.18})$$

Then $\text{Ker } q_n = \mathcal{M}_n$ and the complementary gap equals one at every cutoff, but q_n Mosco-converges to $q_\infty[x] = \|x\|^2$, whose kernel is zero. Thus the finite protected-basis transport in Lemma 7.6 is a genuine hypothesis rather than a consequence of dimension and gap information.

Proof. For every fixed $x \in \ell^2(\mathbb{N})$, $\langle e_n, x \rangle \rightarrow 0$, so $q_n[x] \rightarrow \|x\|^2$; the same observation gives the Mosco liminf inequality and constant recovery sequences. The finite kernels are the moving lines $\mathbb{C}e_n$, which converge weakly to zero and have no nonzero transported limit. $\square$

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